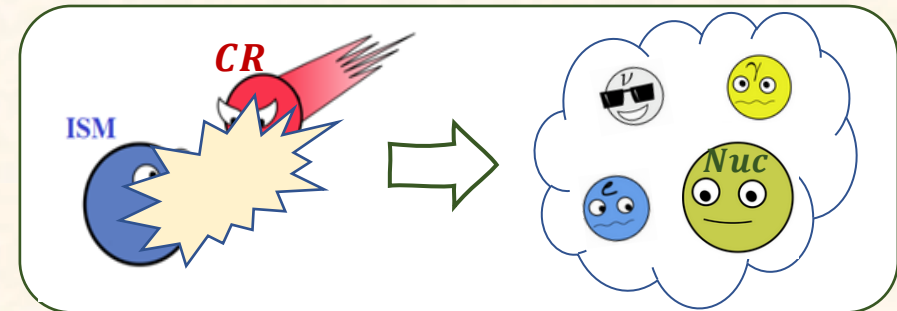
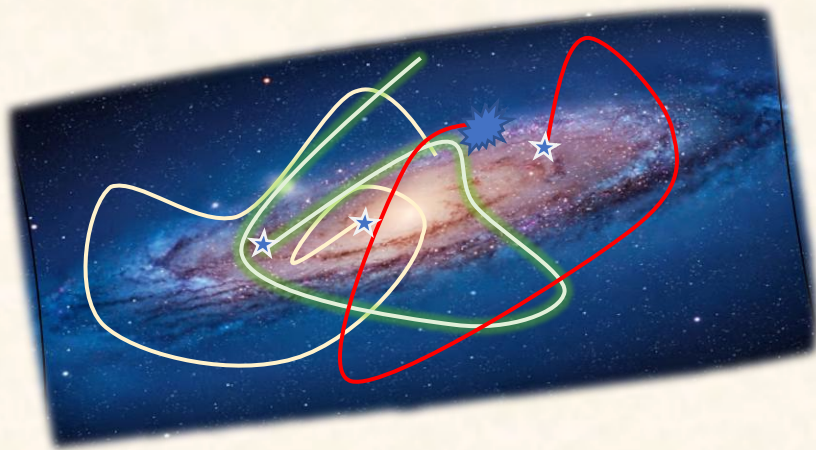


# Cosmic-ray interactions with the interstellar medium



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**Palestinian Advanced School - Cairo**

# Outline

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## Cosmic-Ray Interactions with the Interstellar Medium

Including **interactions** of cosmic rays with their environment

- Energy losses
- Inelastic interactions and secondary production

How to determine the **diffusion coefficient** and **propagation volume**

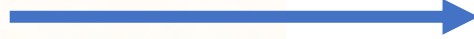
**Current open puzzles**

Production of other secondary particles

- Neutrinos and gamma rays
- Antiprotons and positrons

# The cosmic-ray diffusion equation

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$



$$\frac{\partial n}{\partial t} = \nabla \cdot (D \nabla n) + Q(\mathbf{r}, t) - \frac{n}{\tau_{\text{loss}}}$$

(This is already quite Ok to describe protons!)

Diffusion is not 1D

Source terms

Losses (sinks)

Diffusion coefficient might  
be non-uniform

- $n(\mathbf{r}, t)$ : CR density at position  $\mathbf{r}$  and time  $t$
- $D$ : diffusion coefficient [ $\text{cm}^2/\text{s}$ ], describes how fast CRs spread
- $Q(\mathbf{r}, t)$ : source term (e.g., supernova remnants)
- $\tau_{\text{loss}}$ : timescale for inelastic interactions

So far, we discussed the interaction of plasma waves with cosmic rays. However, **these particles interact with the interstellar medium in different ways, losing energy and “breaking” cosmic rays.** Also, cosmic rays may be unstable and decay

The result of these interactions are changes in their spectra and production of secondary radiations (from radio waves to gamma rays), as well as lighter cosmic rays

# The cosmic-ray diffusion equation

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- $\tau_{\text{loss}}$ : timescale for inelastic interactions

Energy losses

$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{\text{int}} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{\text{dec}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{int}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{dec}}}$$

Interactions with ISM (Particle collisions) and decays

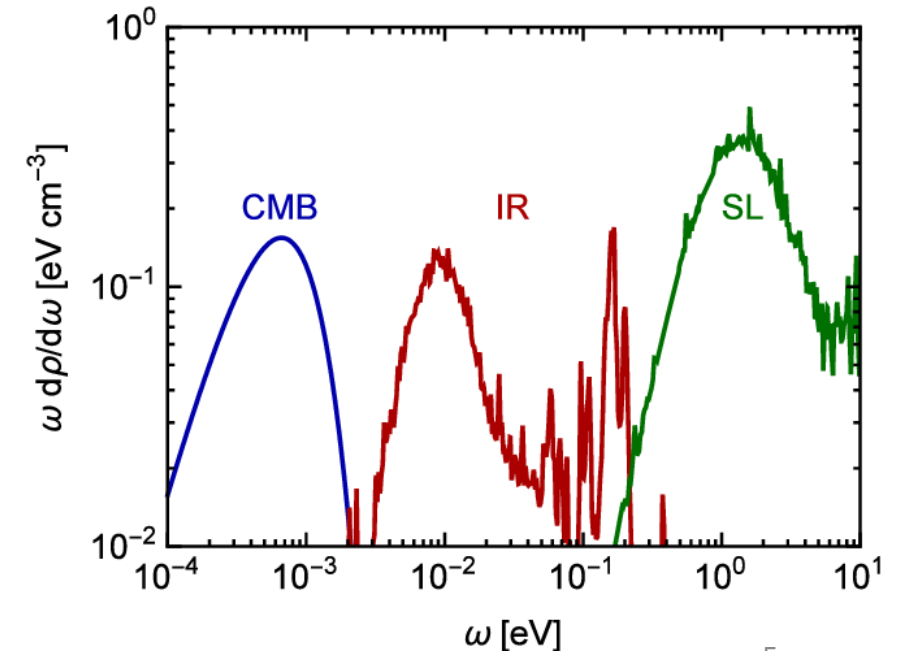
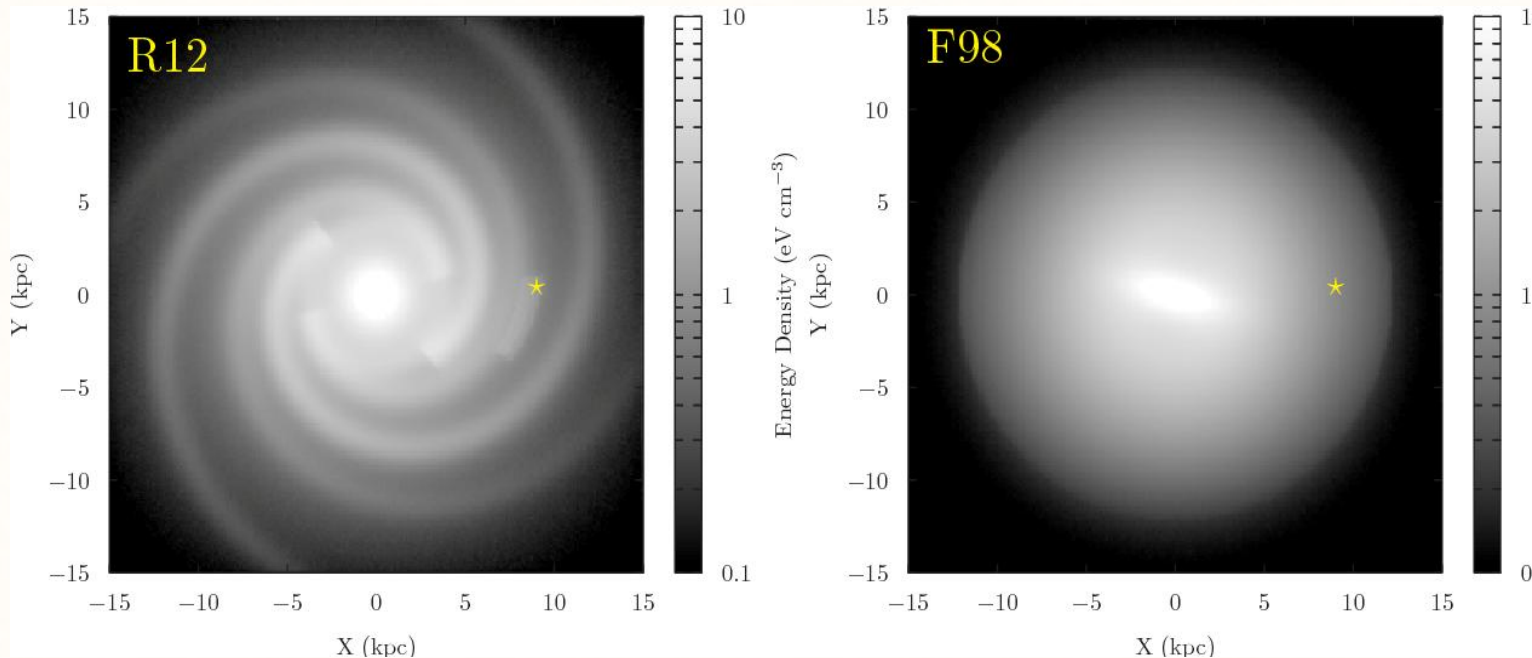
# Targets for the interaction terms

Energy losses

$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{int} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{dec}} - \frac{n_j}{\tau_{loss}^{int}} - \frac{n_j}{\tau_{loss}^{dec}}$$

Interactions with ISM (Particle collisions) and decays

## Interstellar radiation fields





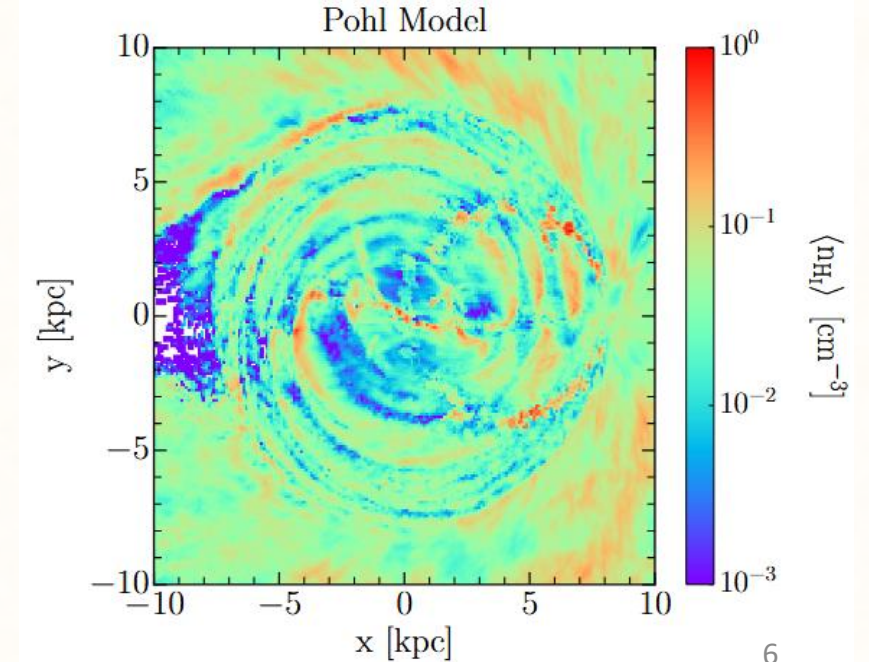
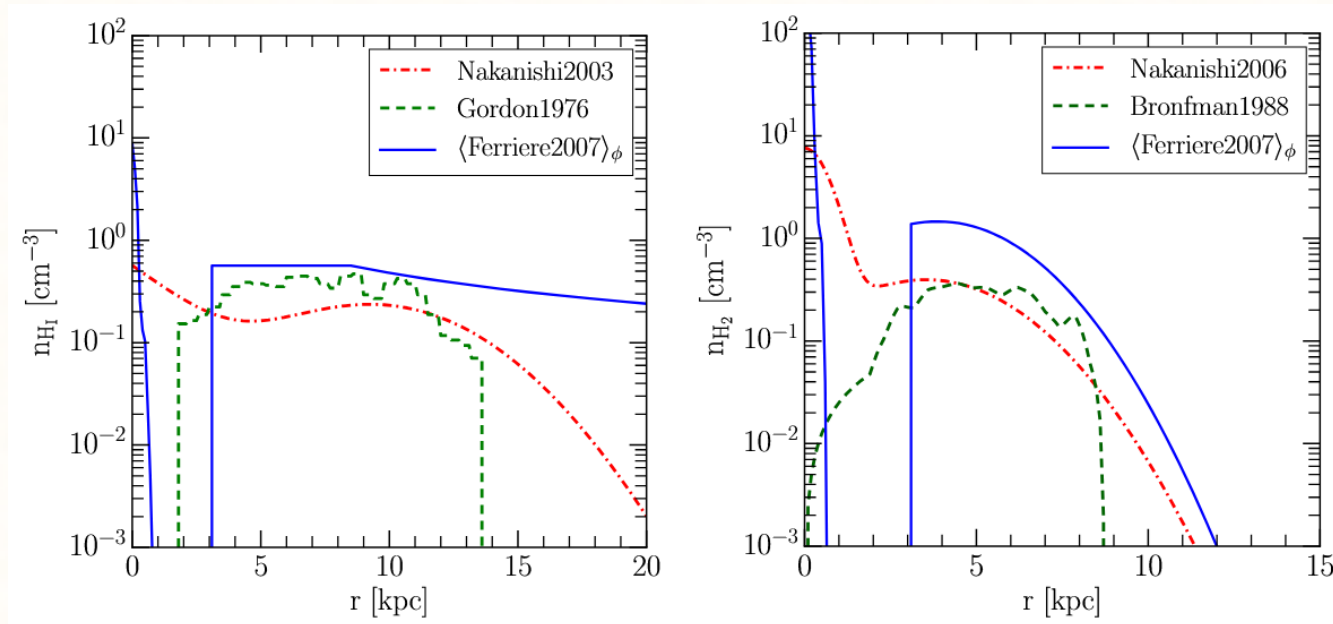
# Targets for the interaction terms

Energy losses

$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{int} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{dec}} - \frac{n_j}{\tau_{loss}^{int}} - \frac{n_j}{\tau_{loss}^{dec}}$$

Interactions with ISM (Particle collisions) and decays

**Interstellar gas** (mainly H and He – Atomic, Molecular and Ionized)



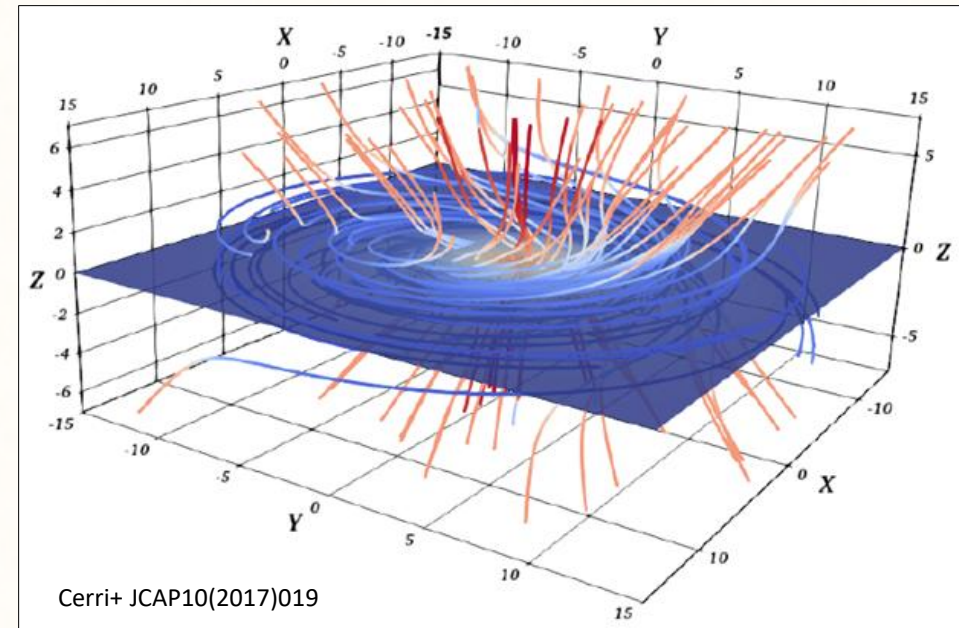
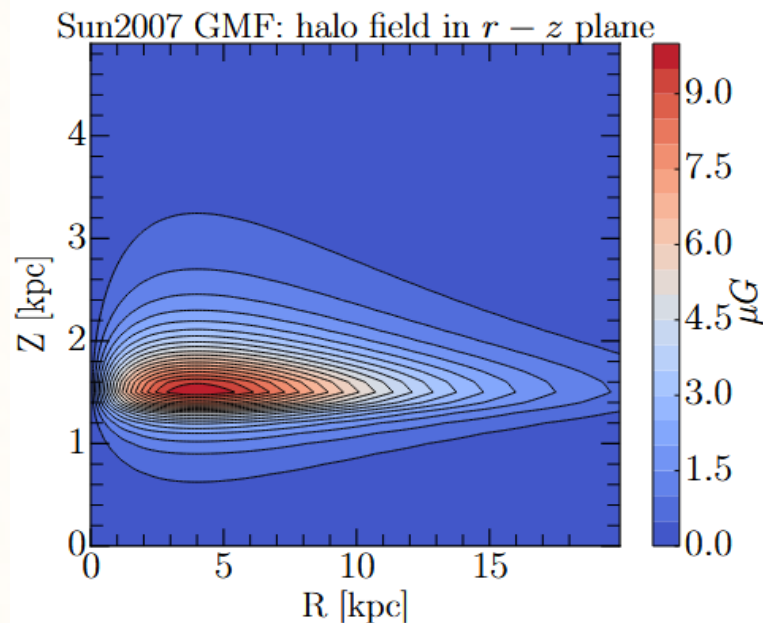
# Targets for the interaction terms

Energy losses

$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{\text{int}} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{\text{dec}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{int}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{dec}}}$$

Interactions with ISM (Particle collisions) and decays

## Galactic magnetic field (Ordered field + random field)



$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{int} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{dec}} - \frac{n_j}{\tau_{loss}^{int}} - \frac{n_j}{\tau_{loss}^{dec}}$$

The energy loss term,  $b(\mathbf{x}, E) \equiv \frac{\partial E}{\partial t}$  describes a continuum loss of energies as CRs interact with their medium during their propagation.

The main processes affecting the energy of CRs are:

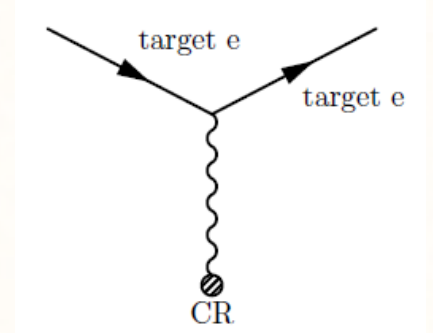
- ❖ **Bremsstrahlung** interaction
- ❖ **Ionization** of neutral gas
- ❖ **Coulomb** interactions with matter
- ❖ **Inverse Compton** interaction with radiation
- ❖ **Synchrotron** emission or curvature radiation
- ❖ **Hadronic interactions**
- ❖ **Adiabatic** energy losses

$$b \equiv \frac{\partial E}{\partial t} = \left( \frac{\partial E}{\partial t} \right)_{Coul, Ioniz} + \left( \frac{\partial E}{\partial t} \right)_{Bremss} + \left( \frac{\partial E}{\partial t} \right)_{IC, Sync} + \left( \frac{\partial E}{\partial t} \right)_{Adi} + \left( \frac{\partial E}{\partial t} \right)_{had}$$

(Not for electrons!)



# Electromagnetic collisions with gas



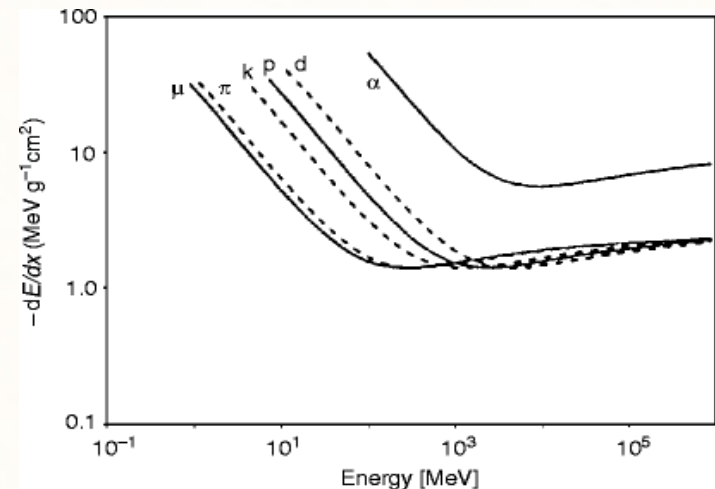
- When crossing a medium with density of electrons  $n_e$ , an electron undergoes an energy loss per unit length  $\sim \sigma_T n_e m_e F(E/m_e)$ , where  $F(x)$  depends on the properties of the medium
- For a non-relativistic simplification, this energy loss is given by

$$-\frac{dE}{dx} = \int 2\pi b n_e \Delta E db \simeq 2\pi n_e \frac{e^4}{v^2 m_e} \ln \frac{b_{\max}}{b_{\min}} \simeq \frac{\sigma_T n_e m_e}{v^2} \ln \frac{b_{\max}}{b_{\min}}$$

- Including quantum effects and relativistic corrections

**Bethe-Bloch general formula**

$$-\frac{dE}{dx} = \frac{3}{2} \frac{\sigma_T m_e n_e}{v^2} \left[ \log \left( \frac{E}{m_e} \right) + \kappa - \beta^2 \right]$$



# Coulomb losses (Numerical approximations)

- For nuclei, one can use

$$x_m \sim 0.0286 \left( \frac{T_e}{2 \times 10^6 \text{ K}} \right)^{1/2}$$

$$- \left( \frac{dE}{dt} \right) \sim 3.1 \times 10^{-16} Z^2 \left( \frac{n_e}{\text{cm}^{-3}} \right) \left( \frac{\beta^2}{x_m^3 + \beta^3} \right) \text{ GeV s}^{-1}$$

- For electrons:

$$- \left( \frac{dE}{dt} \right) = \frac{3}{4} \sigma_T c m_e c^2 n_e \left[ \ln \left( \frac{E m_e c^2}{4 \pi r_e \hbar^2 c^2 n_e} \right) - \frac{3}{4} \right]$$

$$- \left( \frac{dE}{dt} \right) \sim 7.64 \times 10^{-18} \left( \frac{n_e}{\text{cm}^{-3}} \right) \left[ \ln \left( \frac{E}{m_e c^2} \right) - \ln \left( \frac{n_e}{\text{cm}^{-3}} \right) + 73.57 \right] \text{ GeV s}^{-1}$$

# Ionization losses (Numerical approximations)

$$\left(\frac{dE}{dt}\right)_{\text{ion}} = -\frac{4\pi r_e^2 m_e c^3 Z^2}{\beta} n_{\text{target}} \left[ \ln \left( \frac{2m_e c^2 \beta^2 \gamma^2}{I} \right) - \beta^2 \right]$$

- For nuclei

$$\left|\frac{dE}{dt}\right|_{\text{ion}} = \frac{3\sigma_T c w_{e0}}{4\beta} Z^2 n_g z_{\text{target}} \left[ \ln \left( \frac{2w_{e0}(\gamma^2 - 1) Q_{\text{max}}}{I_{\text{pot}}^2} \right) - 2\beta^2 \right]$$

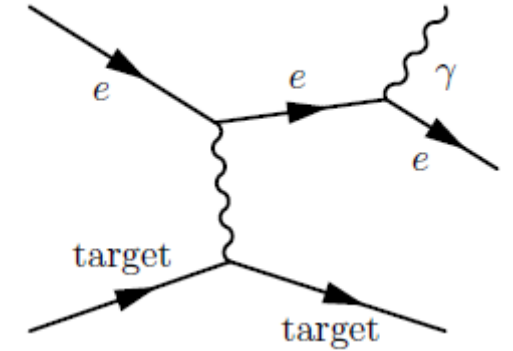
- For electrons:

$$\left|\frac{dE}{dt}\right|_{\text{ion}} = 7.64 \times 10^{-18} \frac{n_H}{\beta^2} \left[ \ln \left( \frac{\gamma^4 m_e^2}{2I^2(1 + \gamma)} \right) - \left( \frac{2}{\gamma} - \frac{1}{\gamma^2} \right) \ln 2 + \frac{1}{\gamma^2} + \frac{1}{8} \left( 1 - \frac{1}{\gamma} \right)^2 \right]$$

# Bremsstrahlung (braking radiation)

$$e + N \rightarrow e' + \gamma + N$$

- This is a similar interaction to the one previously described, but here, the electric field of the target deflects the electron trajectory and make it radiate a photon consequently.



$$\sigma \sim Z^2 \frac{\alpha r_e^2}{k} F(E, k)$$

$$-\frac{dE}{dt} \simeq \sigma_{\text{Brems}} n E$$

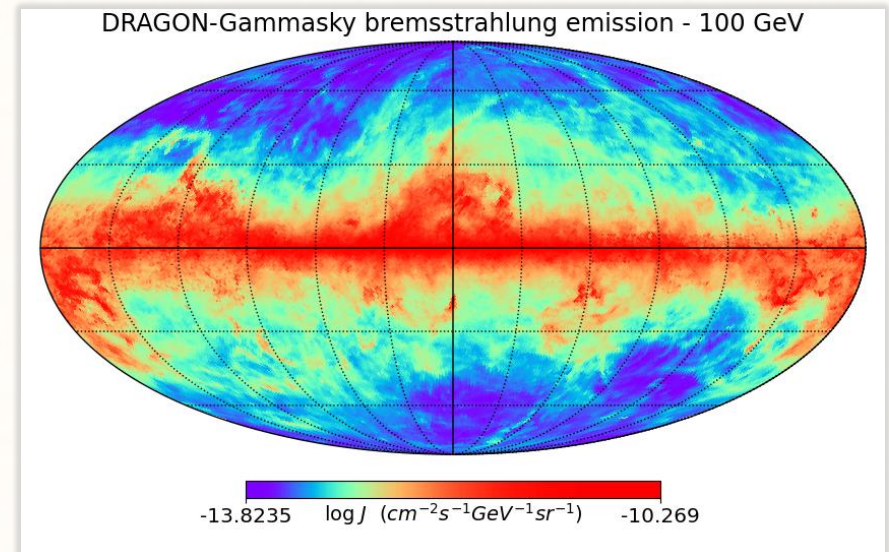
$$\sigma_{\text{brems}}(E) \approx \frac{4\alpha r_e^2 Z^2}{3} \left[ \ln \left( \frac{2E}{m_e c^2} \right) - \frac{1}{3} \right] \quad (\text{When screening is not important})$$

Interaction with ionized gas and weakly-shielded neutral gas

$$-\left(\frac{dE}{dt}\right)_{\text{ws}} = 3.55 \cdot 10^{-17} \left( \frac{E}{\text{GeV}} \right) \left[ \ln(2\gamma) - \frac{1}{3} \right] \left[ 2 \left( \frac{n_{\text{H}}}{\text{cm}^{-3}} \right) + 6 \left( \frac{n_{\text{He}}}{\text{cm}^{-3}} \right) \right] \text{GeV s}^{-1}$$

Interaction with strongly shielded neutral gas

$$-\left(\frac{dE}{dt}\right)_{\text{ss}} = 8 \times 10^{-16} \text{GeV s}^{-1} \left( \frac{E}{\text{GeV}} \right) \left( \frac{n_{\text{H}}}{\text{cm}^{-3}} + 2.7 \frac{n_{\text{He}}}{\text{cm}^{-3}} \right)$$



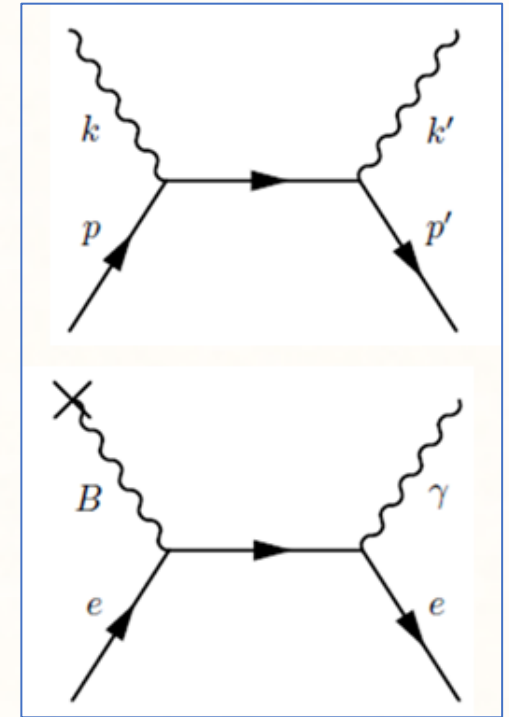
# EM radiative interactions

- In classical electrodynamics, the average radiated power of a charged particle interacting with an EM wave is:

$$\langle P \rangle = \frac{2}{3} q^2 \langle a^2 \rangle = \frac{2}{3} \frac{q^4}{m^2} \frac{E_0^2}{2}$$

- In scattering theory, the cross-section is the ratio of the radiated power to incident flux:

$$\sigma = \frac{\langle P \rangle}{|\langle S \rangle|} = \frac{8\pi}{3} \frac{q^4}{m^2} \equiv \sigma_T, \text{ where } \langle S \rangle \Leftrightarrow u \cdot c \text{ (Energy density times } c \text{)}$$



$$\langle P \rangle_{\text{scatt}} = \sigma_T \tilde{u} c \quad \langle \tilde{u} \rangle = u \gamma^2 \left( 1 + \frac{\beta^2}{3} \right)$$

$$-\frac{dE}{dt} = \langle P \rangle_{\text{scattered}} - \langle P \rangle_{\text{incoming}}$$

$$\langle P \rangle_{\text{inc}} = \sigma_T u c$$

$$-\frac{dE}{dt} = \sigma_T u \left[ \gamma^2 \left( 1 + \frac{\beta^2}{3} \right) - 1 \right] = \frac{4}{3} \gamma^2 \beta^2 u \sigma_T \simeq \frac{4}{3} \gamma^2 u \sigma_T c$$



# Synchrotron losses

$$B + e \rightarrow B + e' + \gamma'$$

Synchrotron losses are **radiative losses** due to the acceleration (change in direction) of the electrons when passing through a magnetic field

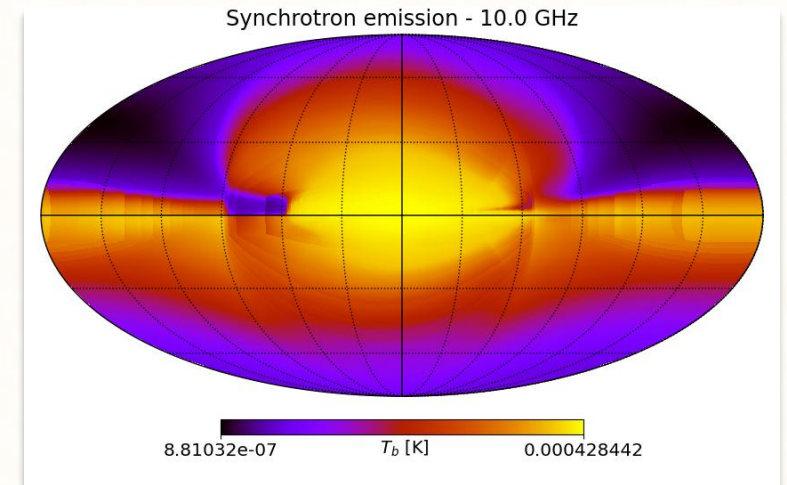
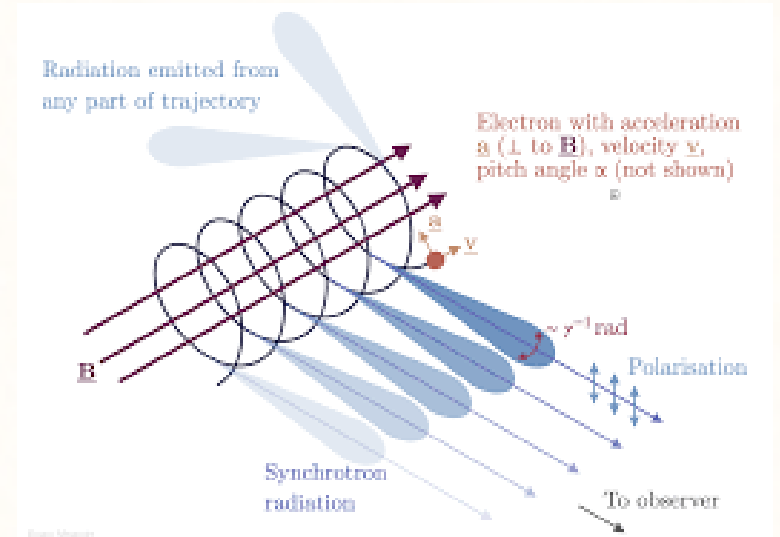
$$P_{\text{syn}} \propto \gamma^2 B^2$$

These radiations allow us to study the magnetic field intensity!

$$u_B = \frac{B^2}{2\mu_0}$$

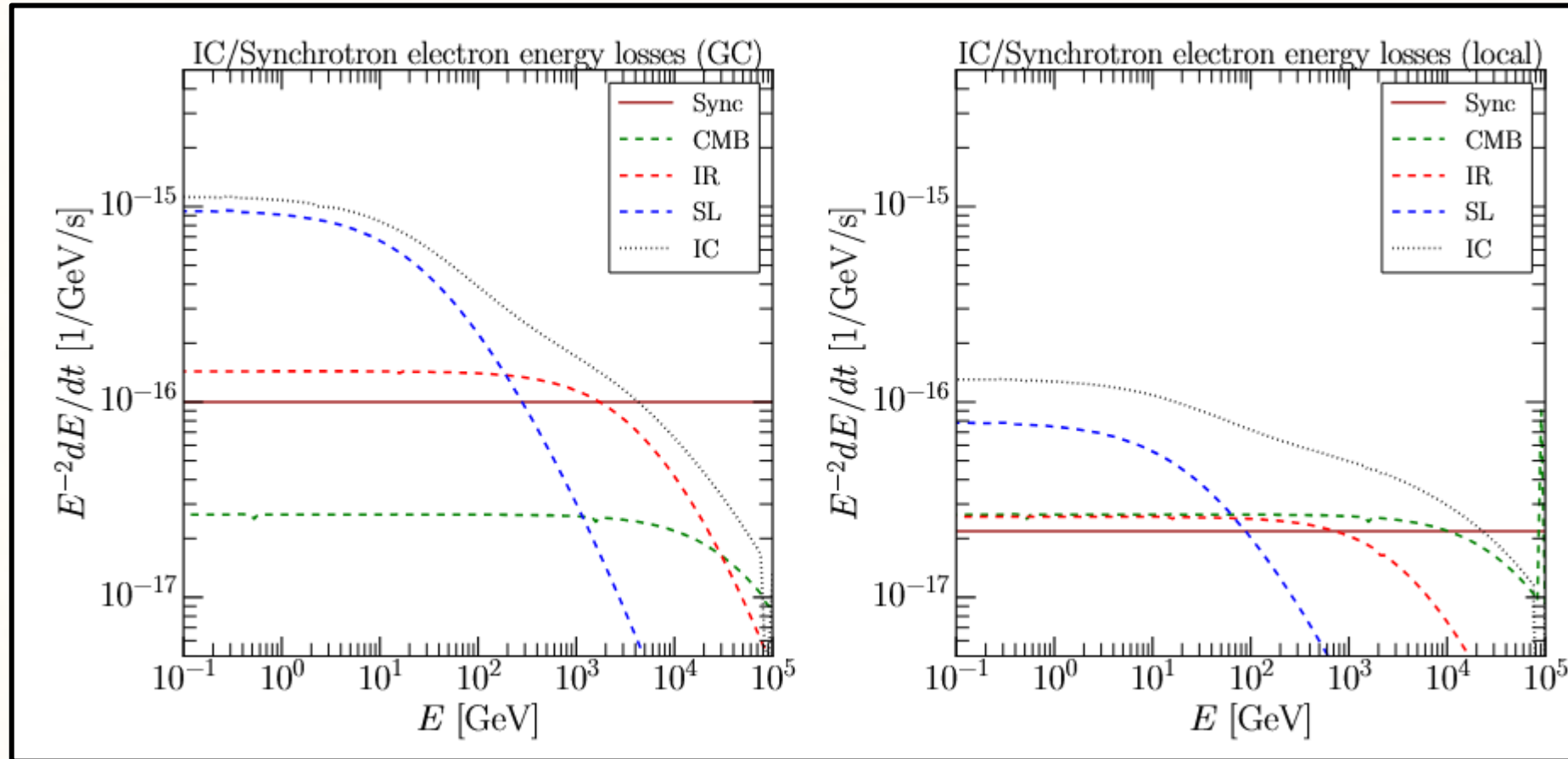
$$u_B (\text{eV}/\text{cm}^3) \approx 0.025 B_{\mu\text{G}}^2$$

$$-\left(\frac{dE}{dt}\right) = \frac{4}{3}\sigma_T c U_B \gamma^2 \beta^2 \sim 2.53 \times 10^{-18} \left(\frac{B}{\mu\text{G}}\right)^2 \left(\frac{E}{\text{GeV}}\right)^2 \text{ GeV s}^{-1}$$



# Synchrotron vs Inverse Compton

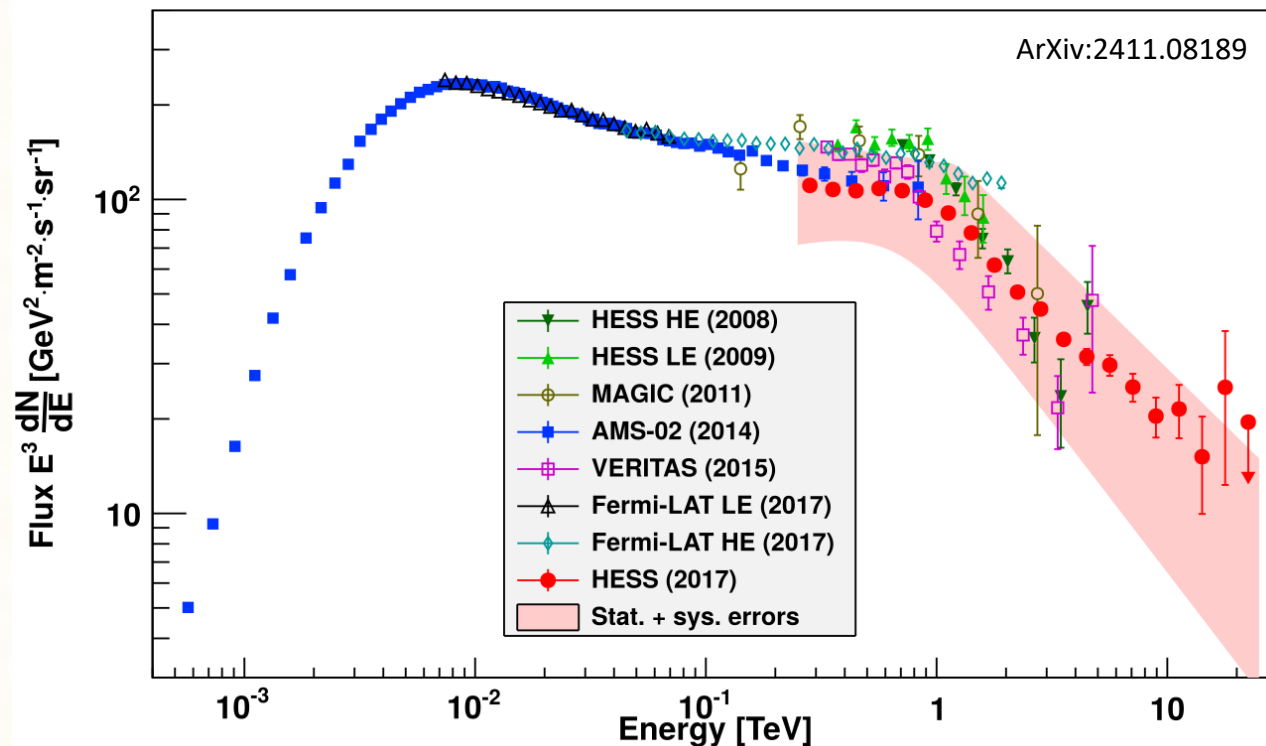
$$b(E) \equiv \frac{dE}{dt} = -\frac{4}{3}\sigma_T c [f_i^{KN} U_i + U_B] \left( \frac{E}{m_e c^2} \right)^2$$



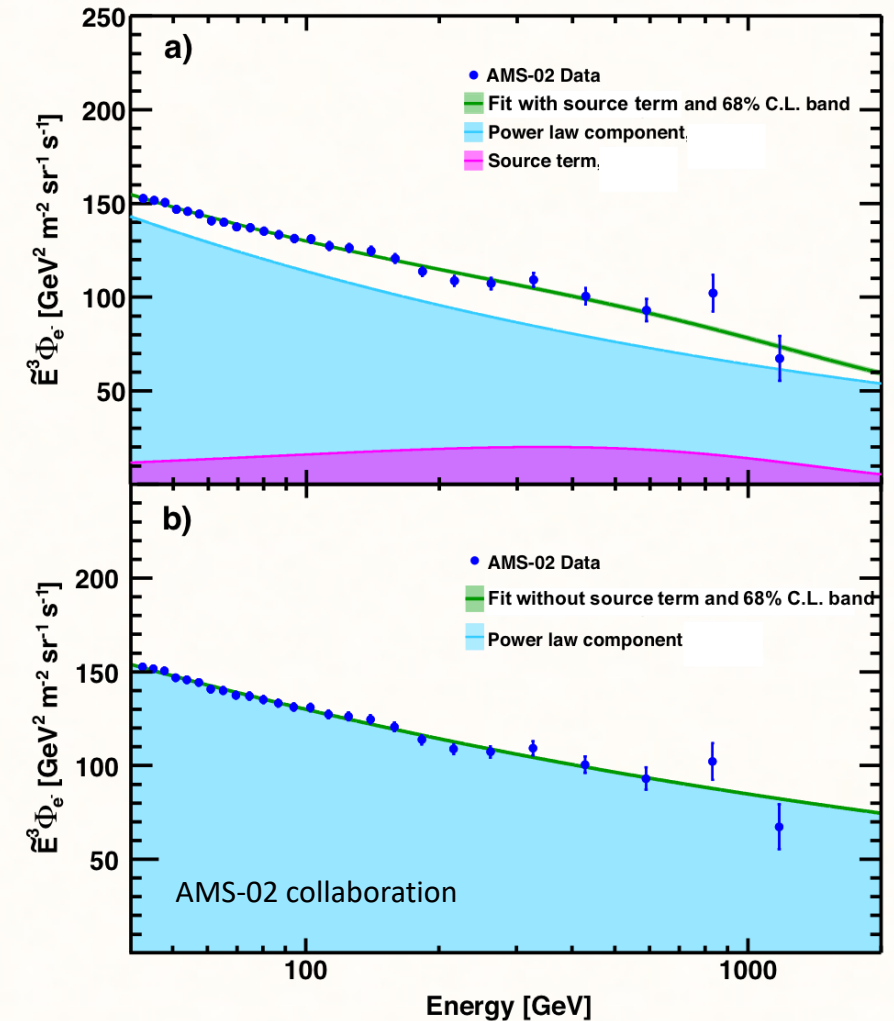
# $e^+e^-$ propagation horizon

Since high-energy electrons lose energy so quickly, they must come from nearby

$$\bar{r} = 2\sqrt{D\tau_{loss}} ; \tau_{loss} \equiv \frac{E}{b_{loss}}$$

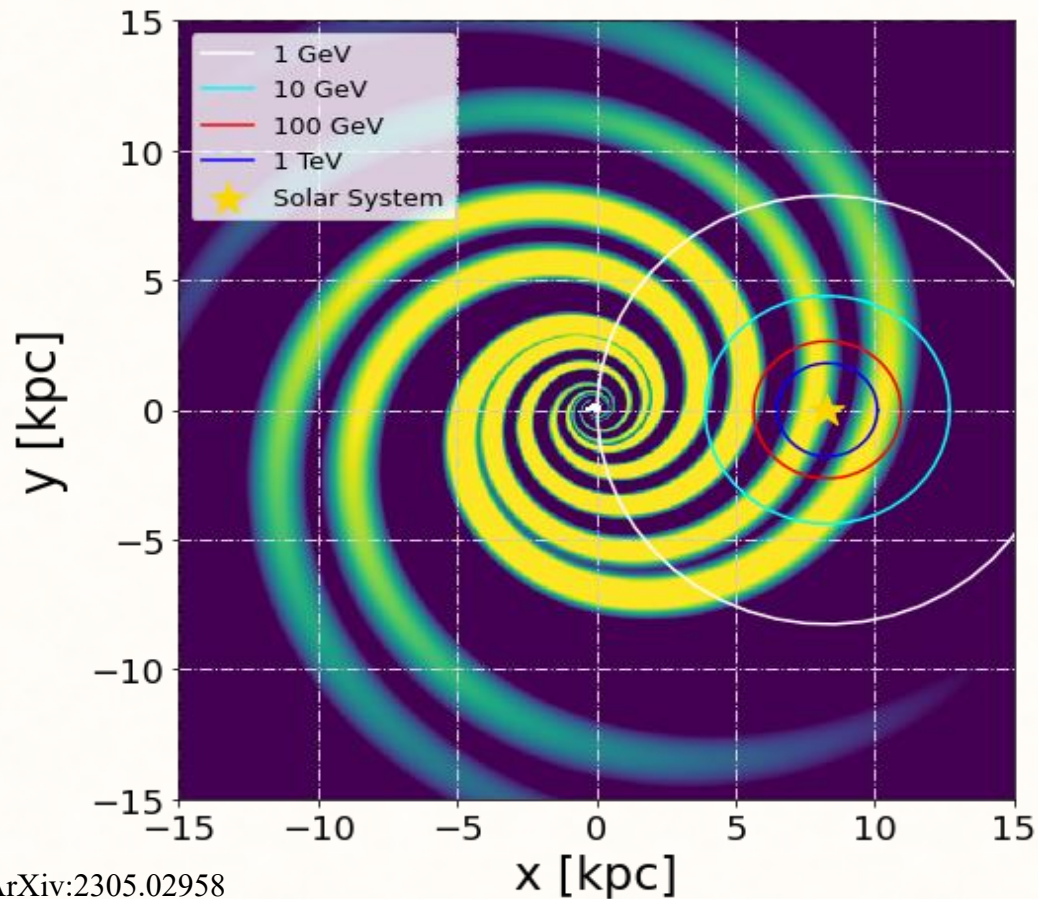


GeV-TeV  $e^-$  are dominated by the emission from local sources!

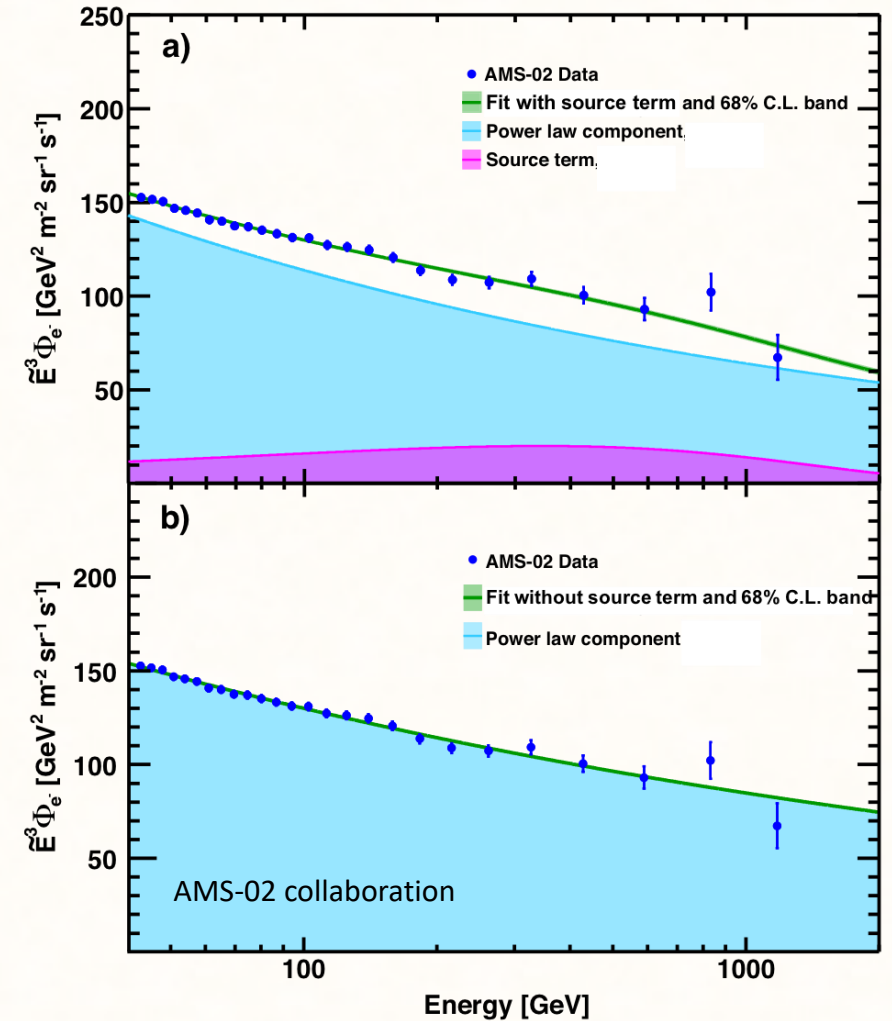


# $e^+e^-$ propagation horizon

$$\bar{r} = 2\sqrt{D\tau_{loss}} ; \quad \tau_{loss} \equiv \frac{E}{b_{loss}}$$



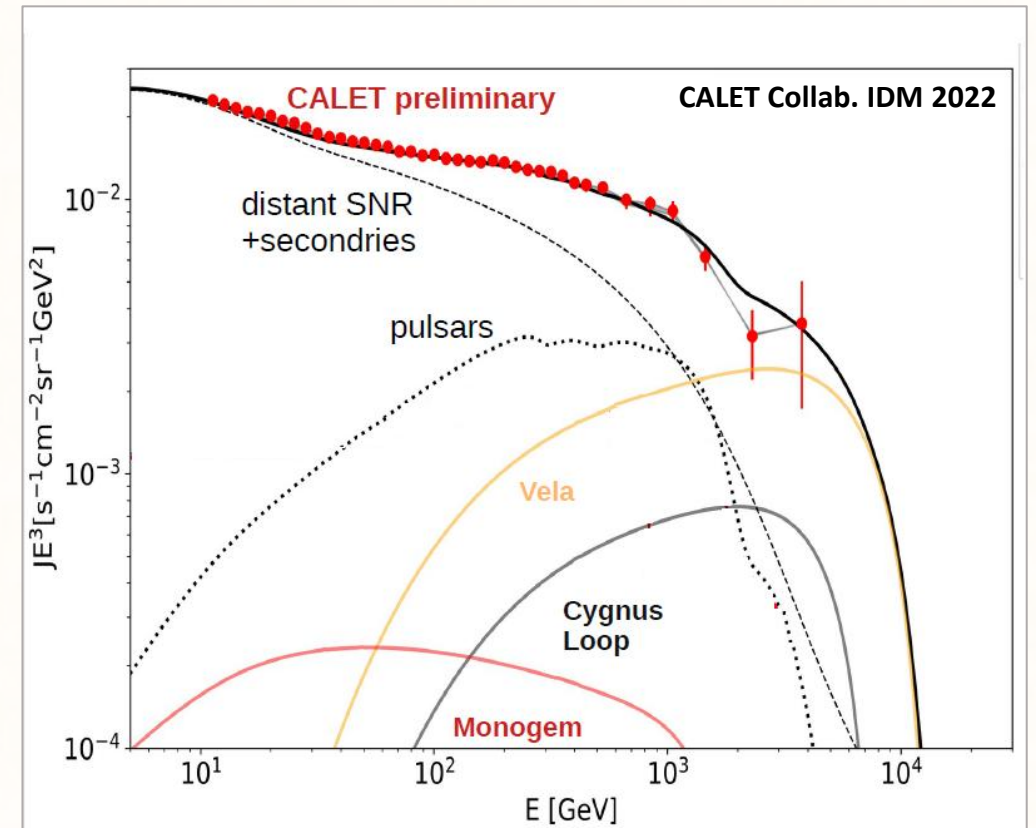
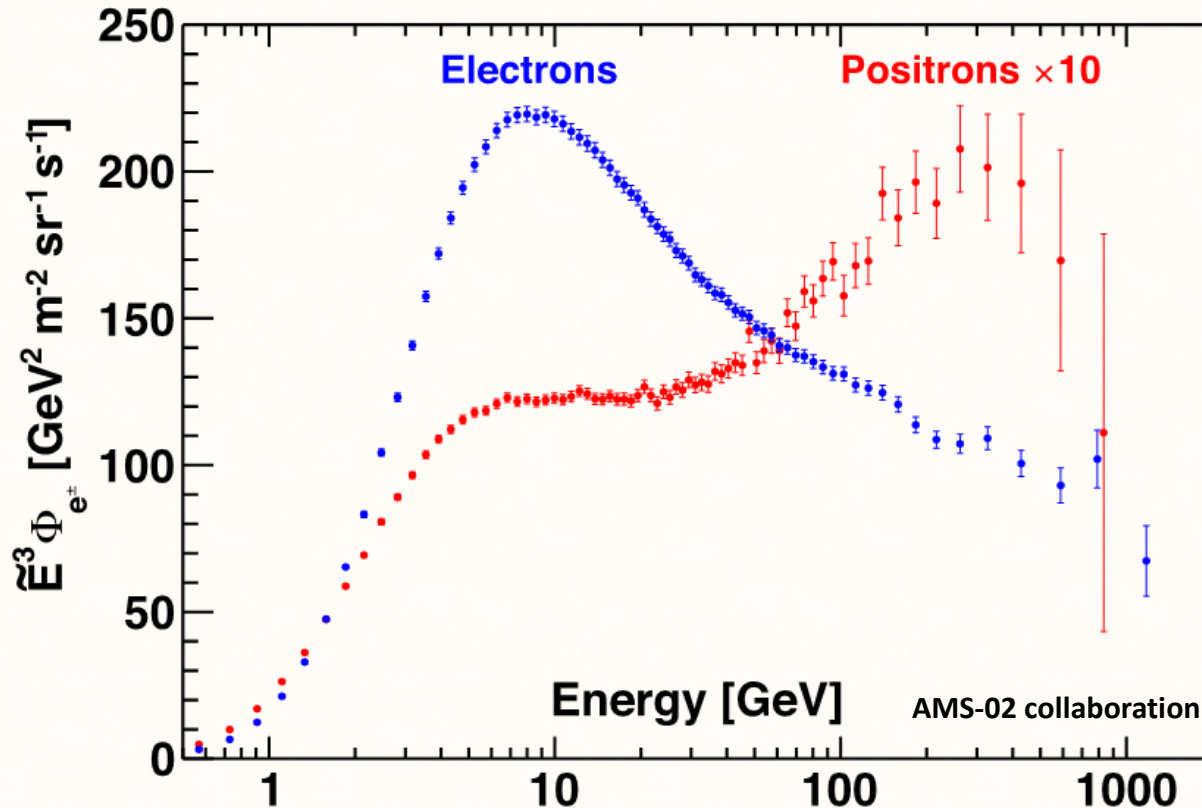
GeV-TeV  $e^-$  are dominated by the emission from local sources!



# $e^+e^-$ propagation horizon

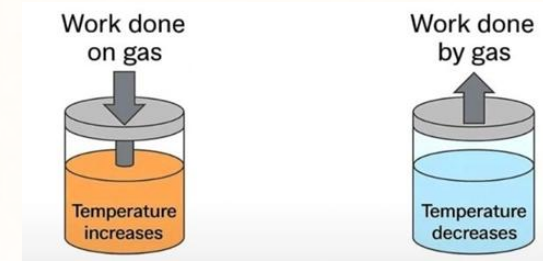
GeV-TeV  $e^-$  are dominated by the emission from local sources!

Nearby pulsars and supernova remnants explain the high-energy electron and positron spectra





# Adiabatic losses



This term can be understood from a macroscopic perspective, using thermodynamics. If an ensemble of particles are trapped in an expanding plasma, the total energy of the ensemble gets reduced:

$$dE = -P dV$$

A microscopic intuition can be given if you think of head- and tail-on collisions: If the plasma is moving out, more tail-on collisions are expected

Therefore, when cosmic rays are propagating in an expanding galactic wind with speed  $u$ , they'll suffer an energy loss proportional to the direction of the flow (namely its divergence)

$\nabla \cdot \mathbf{u} > 0 \rightarrow$  fluid elements expand  $\rightarrow$  **adiabatic energy loss**

$\nabla \cdot \mathbf{u} < 0 \rightarrow$  fluid compresses  $\rightarrow$  **adiabatic energy gain**

$$\frac{dE}{dt} = -\frac{1}{3}E (\nabla \cdot \mathbf{u})$$

$$t_{\text{ad}} \sim \frac{R}{u}$$

# Non-annihilating hadronic interactions

At high energies, the interactions of nuclei with gas become relevant (given that other losses are not important). A typical process when these interactions occur, is that of **meson production, which redistributes the energy of the colliding cosmic rays**

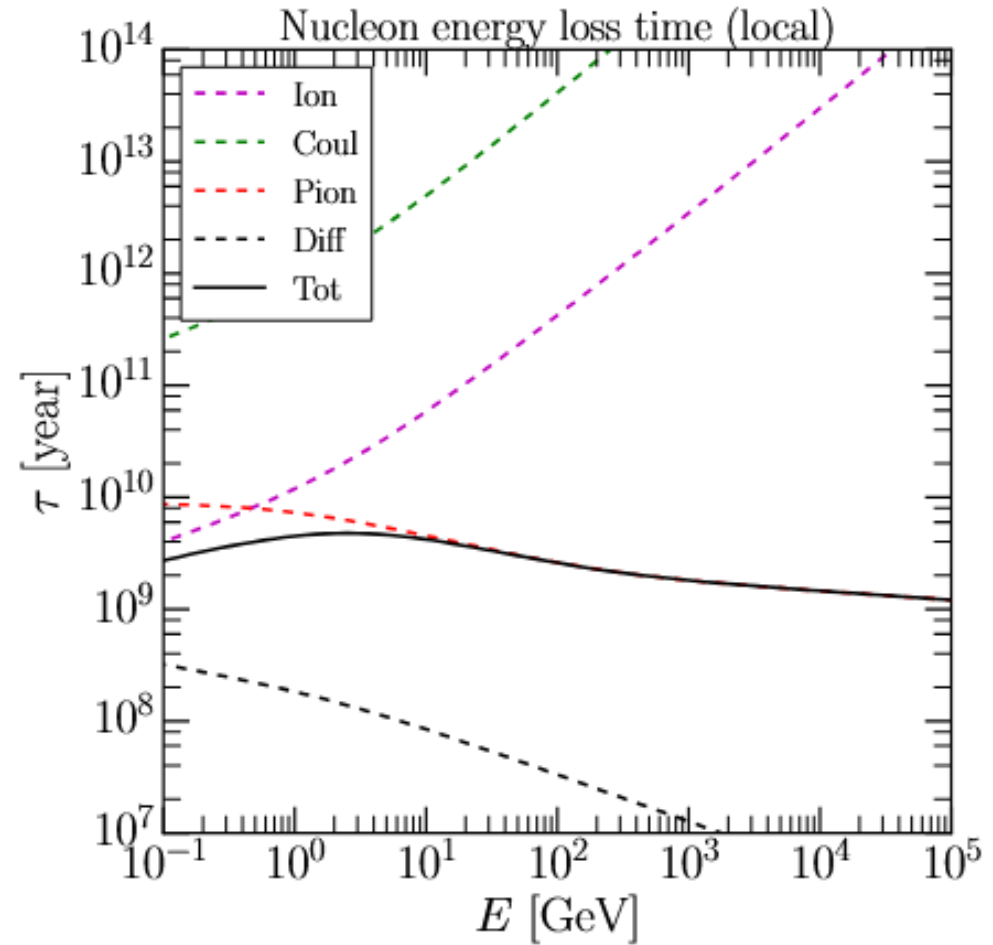
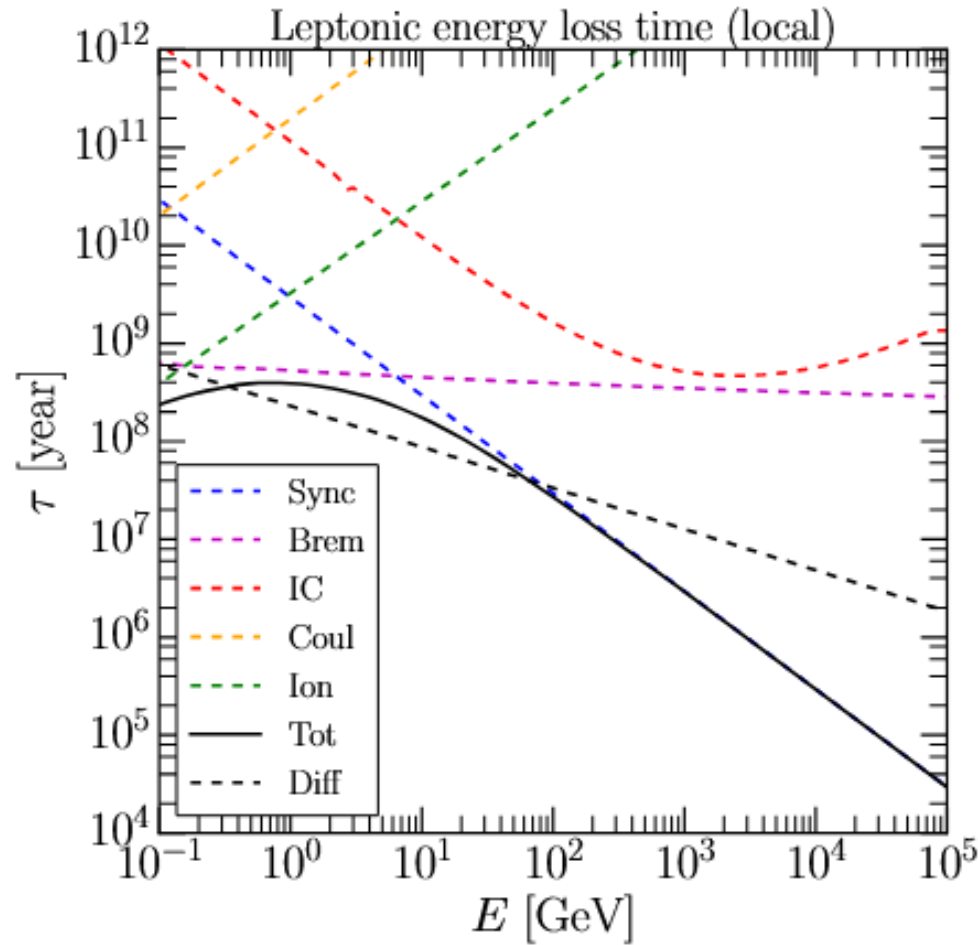
$$p + p \rightarrow p + p + \pi^0, \quad p + p + \pi^+, \text{ etc.}$$

The main product is pions, and a fraction of the kinetic energy of the original cosmic ray is carried away by them. Sometimes, this is taken as another component of production of cosmic rays, instead (depending on how the interaction loss term is defined).

The **inelastic cross-section** for hadron–gas collisions rises above the pion-production threshold and slowly increases with energy.  $\tau_{pion} \sim 1 / (n_{gas} \sigma_{pion} c)$

$$-\left(\frac{dE}{dt}\right)_p = 3.85 \times 10^{-16} \left(\frac{n_{gas}}{\text{cm}^{-3}}\right) \left(\frac{E}{\text{GeV}}\right)^{1.28} \times \left(\frac{E}{\text{GeV}} + 200\right)^{-0.2} \text{ GeV s}^{-1}$$

# Comparison of loss terms



$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{int} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{dec}} - \frac{n_j}{\tau_{loss}^{int}} - \frac{n_j}{\tau_{loss}^{dec}}$$

Decay and interaction terms are crucial to obtain information about cosmic-ray diffusion in the Galaxy. These can be divided in production and losses:

❖ **Production:**

- Interactions:  $\Gamma_{i \rightarrow j}^{Int} \propto n_{gas} \sigma_{i \rightarrow j}$
- Decays:  $\tau_{i \rightarrow j}^{dec} = \gamma \tau_{1/2}^i B_{i \rightarrow j}$

❖ **Losses:**

- Hadronic Interactions
- EM interactions
- Decays:  $\tau_{loss}^{dec} = \gamma \tau_{1/2}^j$

- The term of production of secondary CRs is crucial given the information they carry
- **$^{10}\text{Be}$  is a key unstable isotope.** Others, like  $^{26}\text{Al}$ ,  $^{36}\text{Cl}$  or  $^{54}\text{Mn}$  are important too
- Loss terms are very important at high energies for nuclei as well

# Loss terms -- $\frac{n_j}{\tau_{loss}^{int}}$

## ☐ Electrons:

- No inelastic interactions (they don't have inner structure)
- However, triplet pair production
- For positrons also the in-flight annihilation process

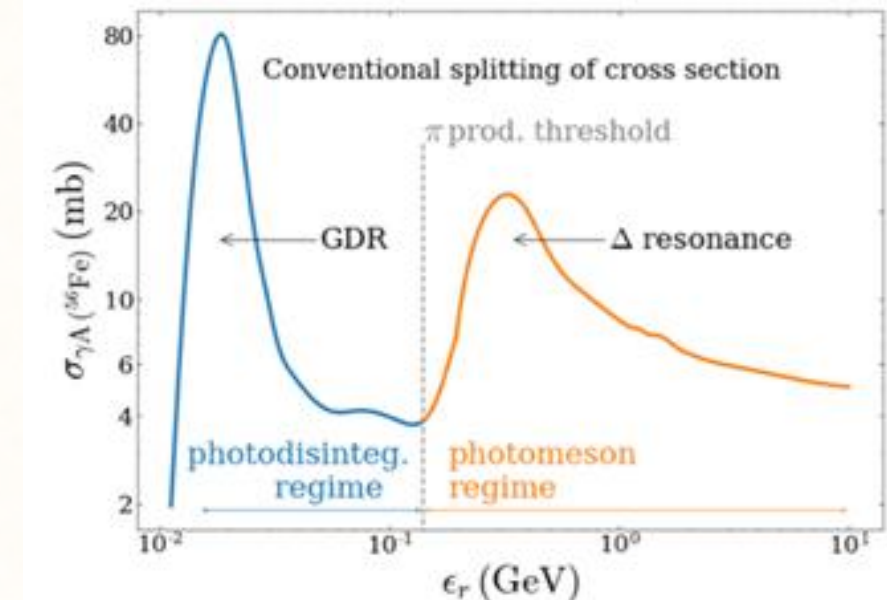
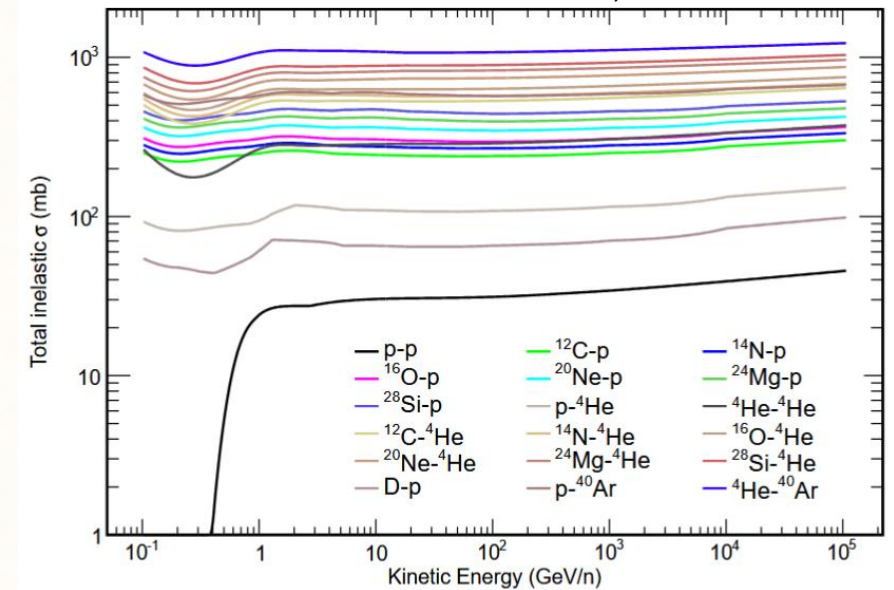
## ☐ Protons and nuclei:

Inelastic cross sections that break them up:

proton(nucleus)-gas interactions

photonuclear disintegration

$$\tau_{loss}^{int} \propto (n_{gas} \sigma_{ine} c)^{-1} \quad (\sim 10^8 \text{ yr for p-p})$$





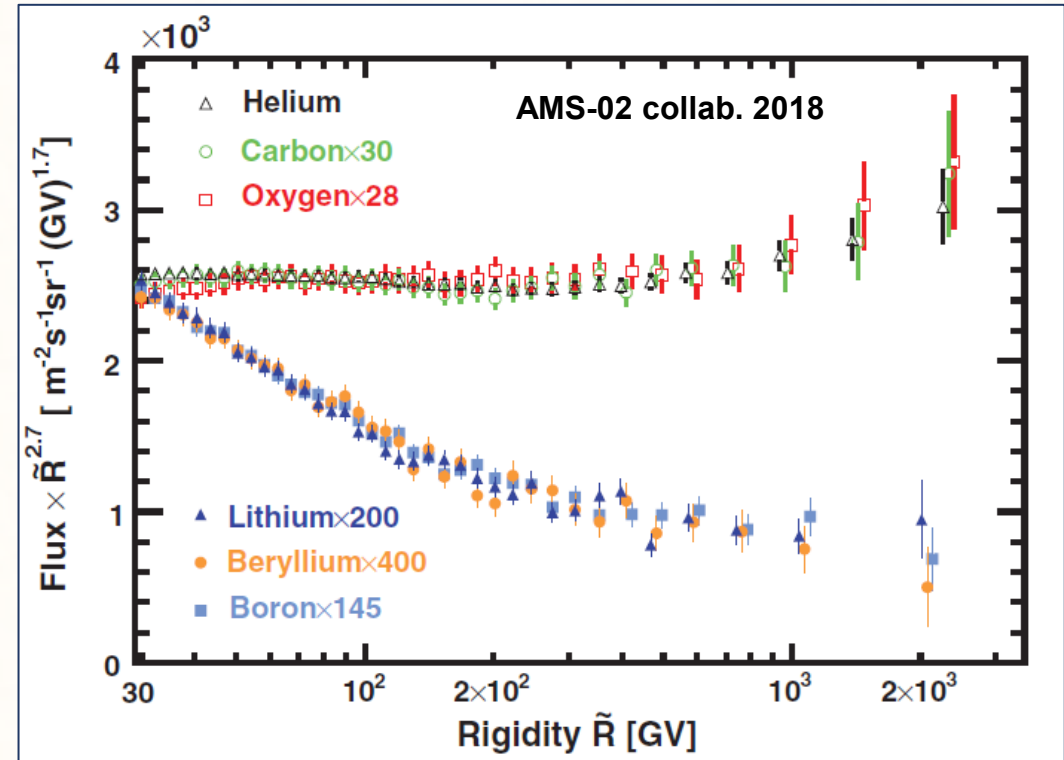
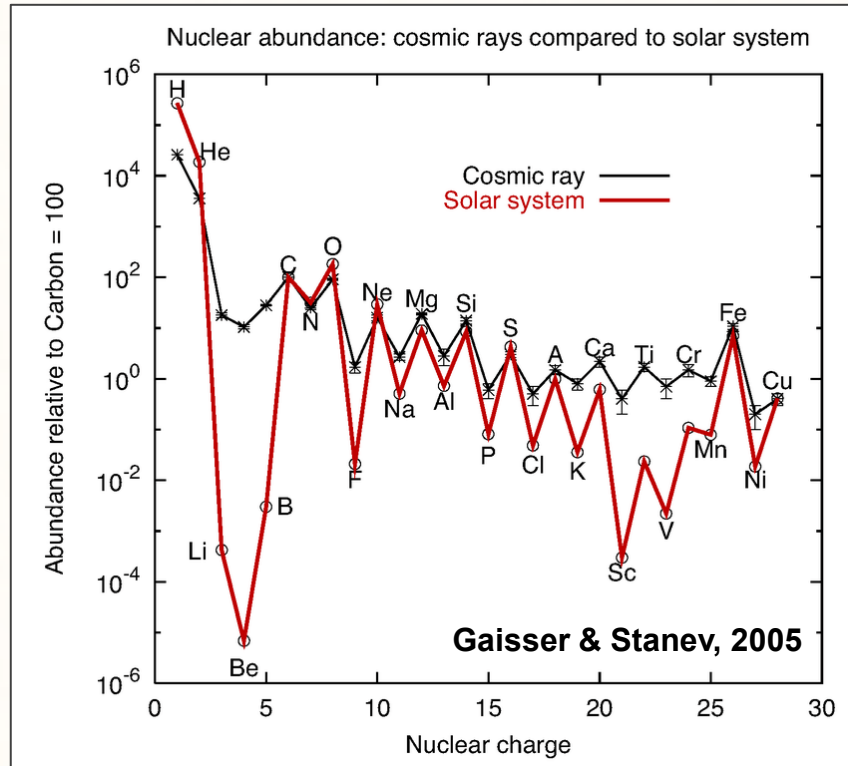
# Production of secondary cosmic rays

Cosmic rays are usually divided in primary and secondary

$$\Gamma_{i \rightarrow j}^{Int} \propto n_{gas} \sigma_{i \rightarrow j}$$

The more grammage primary CRs traverse, the more is the amount of secondary produced.

$$X = \int \rho dl$$



# Production of secondary cosmic rays

**Grammage** is the total column density of interstellar matter traversed by a cosmic ray during its propagation in the Galaxy

$$X \approx \rho_{\text{ISM}} v t_{\text{conf}}$$

If a cosmic ray traveled **straight through the Galactic disk once**, it would see only  
 $\sim 0.003 \text{ g cm}^{-2}$

But observations indicate

$$\sim 5\text{--}10 \text{ g cm}^{-2}$$

So cosmic rays effectively **wander through the disk thousands of times** before escaping

The interaction length in **grammage units** is  $\lambda = \frac{m_{\text{ISM}}}{\sigma}$   
or (not in units of grammage)  $\lambda = 1/(n_{\text{ISM}}\sigma)$

$$P_{\text{interaction}} \sim \frac{X}{\lambda}$$

# Production of secondary cosmic rays

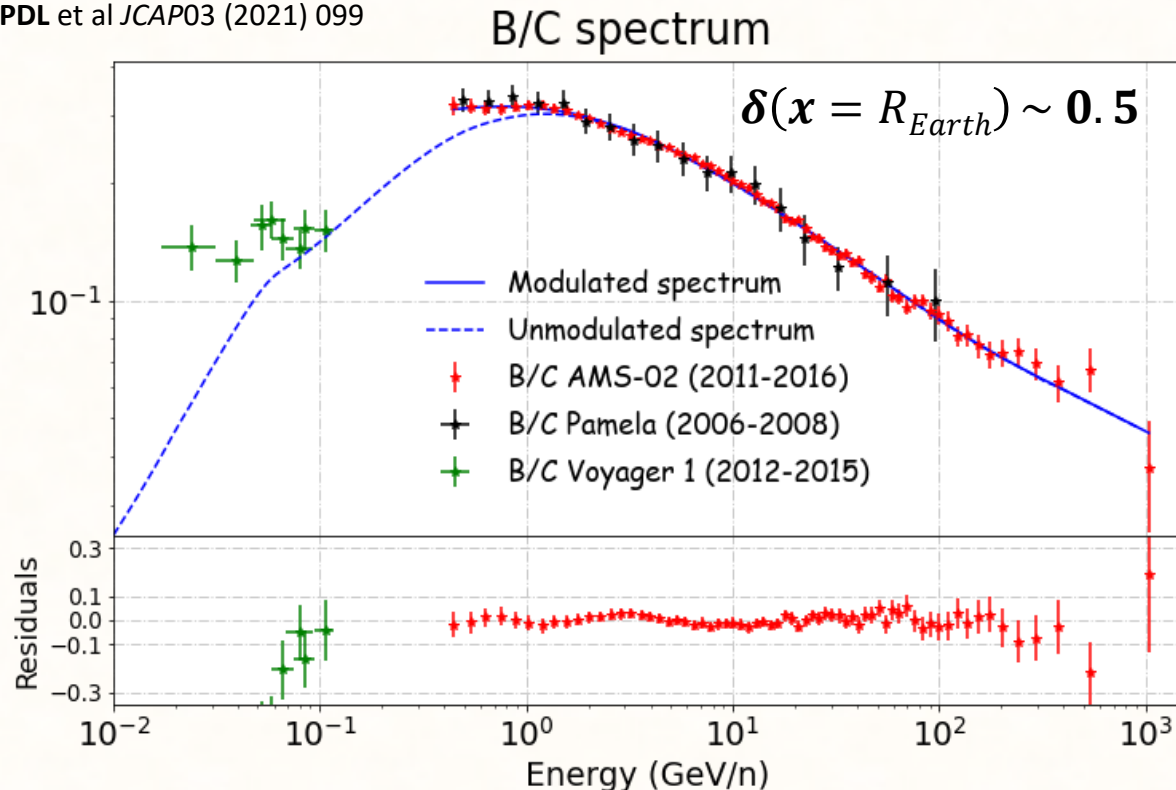
$$J_{sec} \propto X J_{pr} \propto \tau^{\text{conf}} J_{pr}$$

Secondary to primary ratios are directly sensitive to grammage or to confinement time!!

$$J_{pr}(E) \sim \frac{Q_{inj}(E)}{D(E)} \propto E^{-(\alpha_{inj} + \delta)}$$

$$J_{sec}(E) \sim \frac{\sigma(E) J_{pr}(E)}{D(E)} \propto E^{-(\alpha_{inj} + 2\delta)}$$

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They provide a direct tool to determine the **Diffusion coefficient** ( $D \propto 1/\tau^{\text{conf}}$ )

$$\frac{J_{sec}}{J_{pr}} \sim \sigma(E)/D(E) \propto E^{-\delta}$$

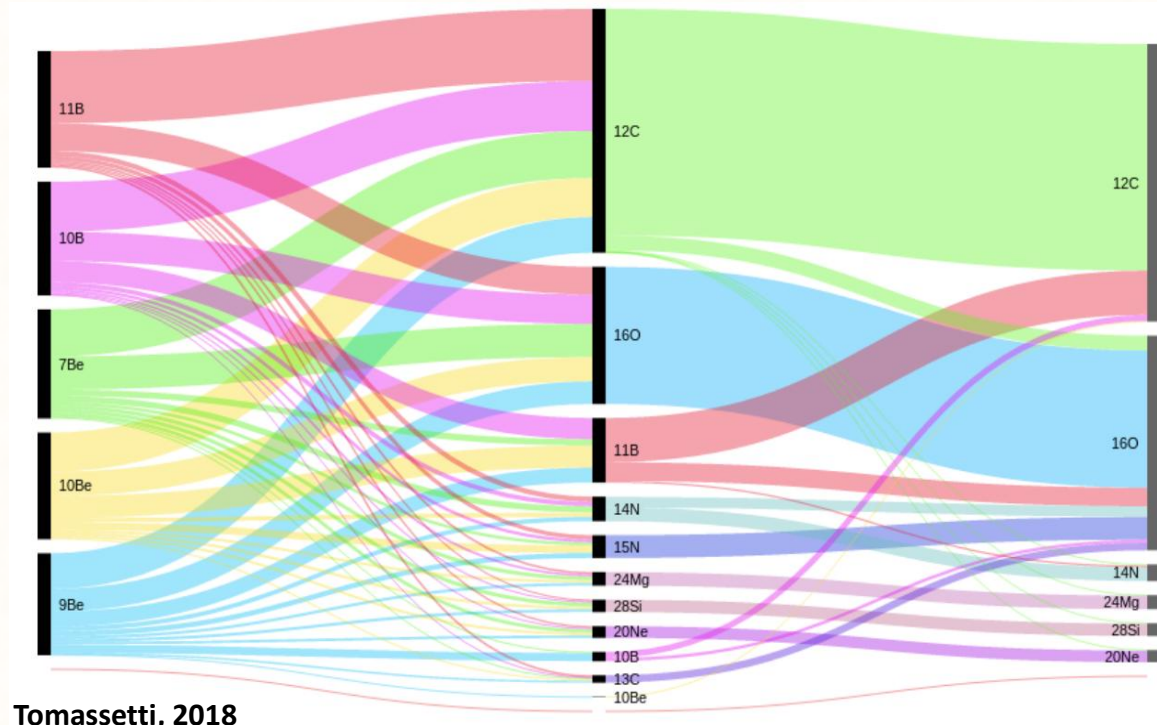
$$D = D_0 \beta^\eta \left( \frac{R}{R_0} \right)^\delta F(\vec{r}, z)$$

Different parameterizations for  $D$  can be adopted, but they'll follow the general form of a power-law in rigidity/energy with breaks

## Cross sections → Secondary CRs

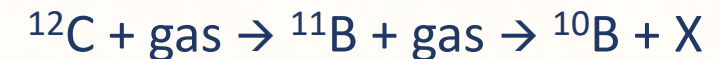
$$Q_{sec}(E) \propto \sum^{pr} J_{pr}(E) \sigma_{pr \rightarrow sec}(E)$$

### The Complexity of the XS network



### Production of B, Be and Li

- Main spallation channels: O and C
- Secondary channels (N, Ne, Mg, Si & Fe) are very important for Li and Be (< 50%)
- **Tertiary channels** and **ghost nuclei** also matter are very important :



Genolini+ 2019 ArXiv:1803.04686

Tomassetti, 2018 ArXiv:1707.06917

# Cross sections parametrizations

Need to have precise measurements !!

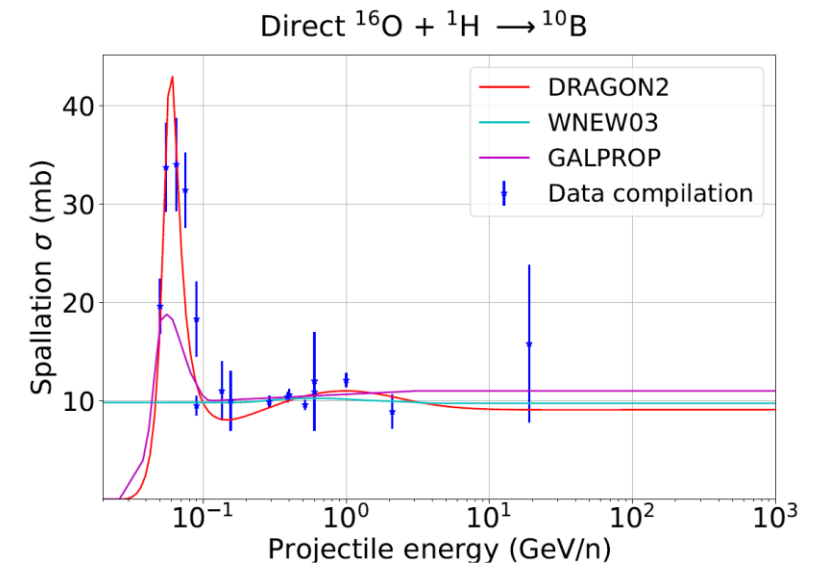
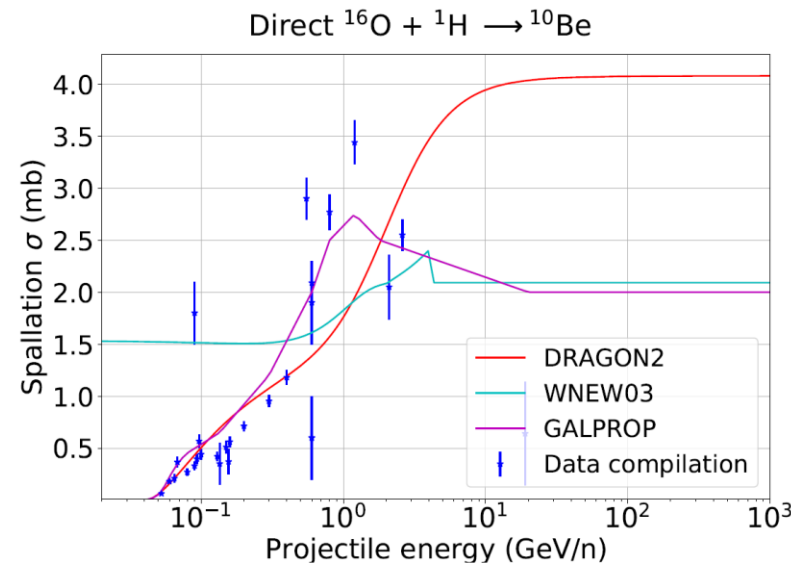
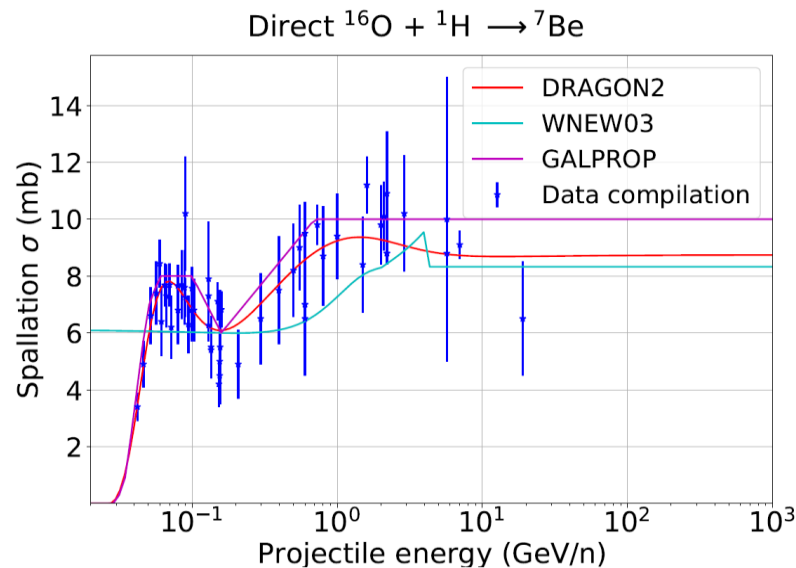
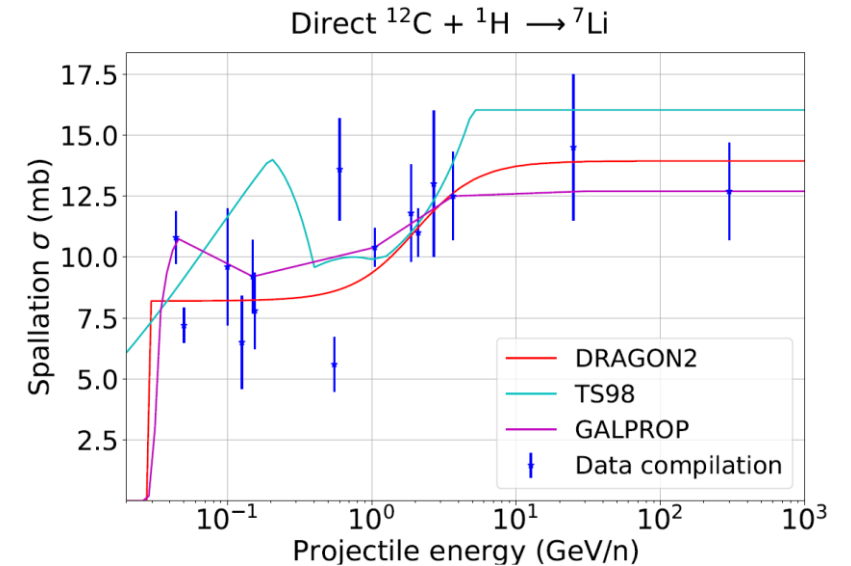
PDL et al JCAP03 (2021) 099

Phenomenological parametrization (Evoli+ ArXiv:1904.10220)

$$\sigma_{j+H \rightarrow i} = \sigma_0 \frac{\Gamma^2 (T - E_{th})^2}{(T^2 - M^2)^2 + \Gamma^2 M^2} + \sigma_1 \left(1 - \frac{E_{th}}{T}\right)^\xi \left(1 + \frac{\Delta}{1 + (T_h/T)^2}\right)$$

See recent review (Maurin+ arXiv:2503.16173):

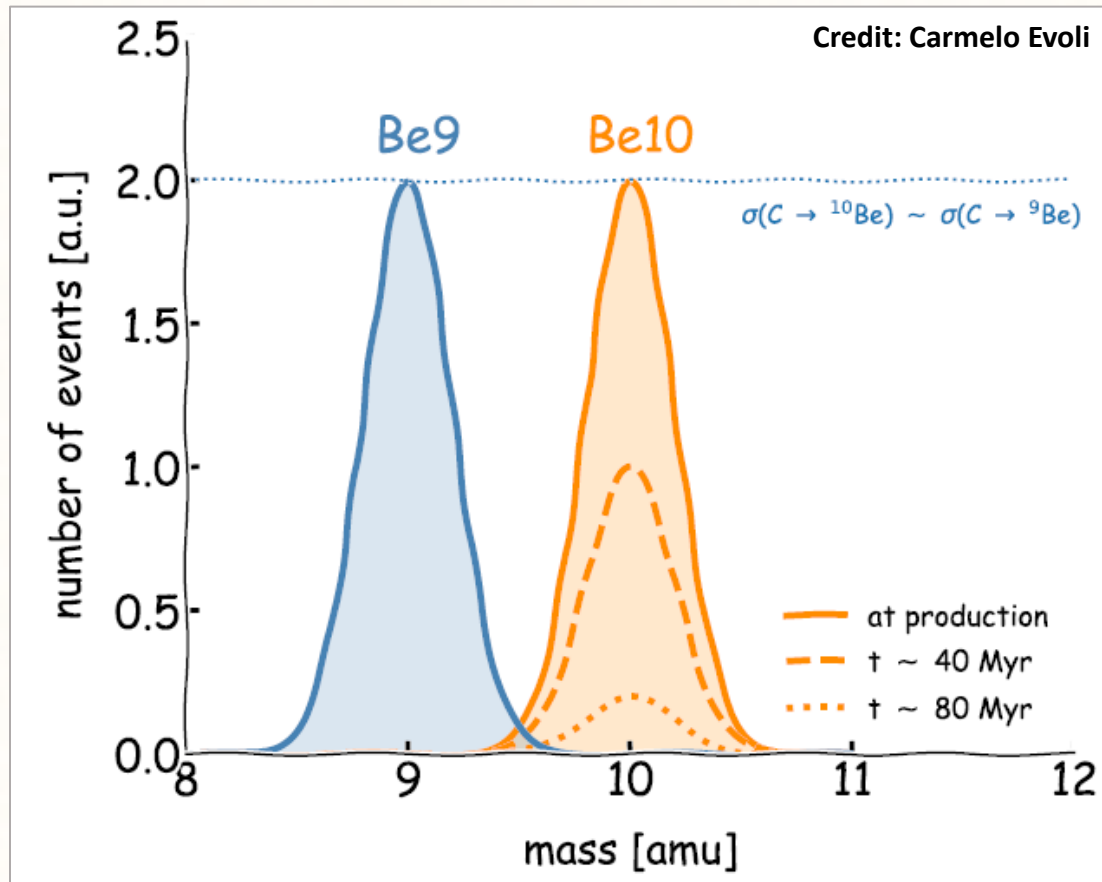
“Precision cross-sections [...] programme for the next decade”





# How to determine the propagation volume?

$^{10}\text{Be}$  is an unstable isotope that decays into  $^{10}\text{B}$  with a half-life  $\sim 1.5$  Myr



$$N_{sec}^* \propto N_{\text{prim}} \sigma \frac{\tau_{dec}}{\sqrt{\tau_{dec} D}} \propto \frac{\sqrt{\tau_{dec} \tau_{conf}}}{H}$$

$$N_{sec} \propto N_{\text{prim}} \sigma \frac{H}{D}$$

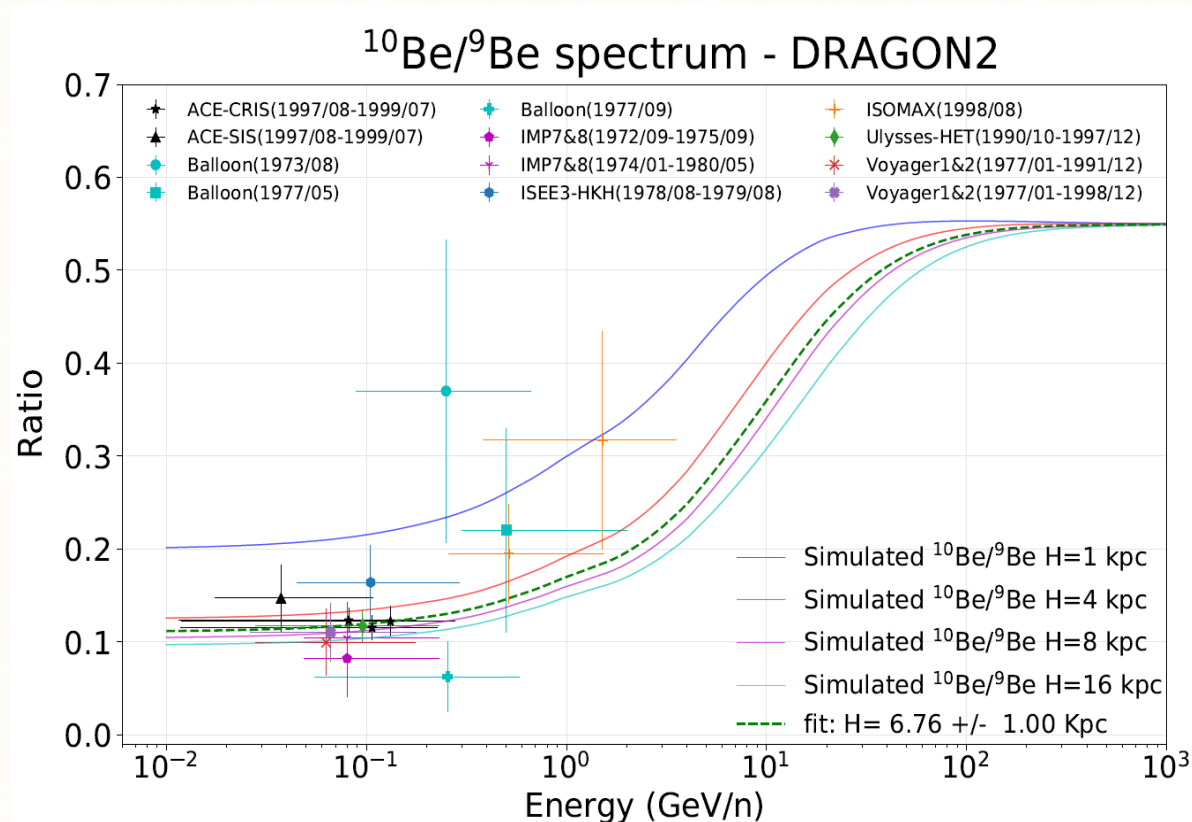
$$\frac{N_{sec}}{N_{\text{prim}}} \propto X \propto \frac{H}{D}$$

$$\frac{N_{sec}^*}{N_{sec}} \propto \frac{\sqrt{D}}{H}$$

$^{10}\text{Be}$  to  $^9\text{Be}$  ratio extremely useful to break this degeneracy!

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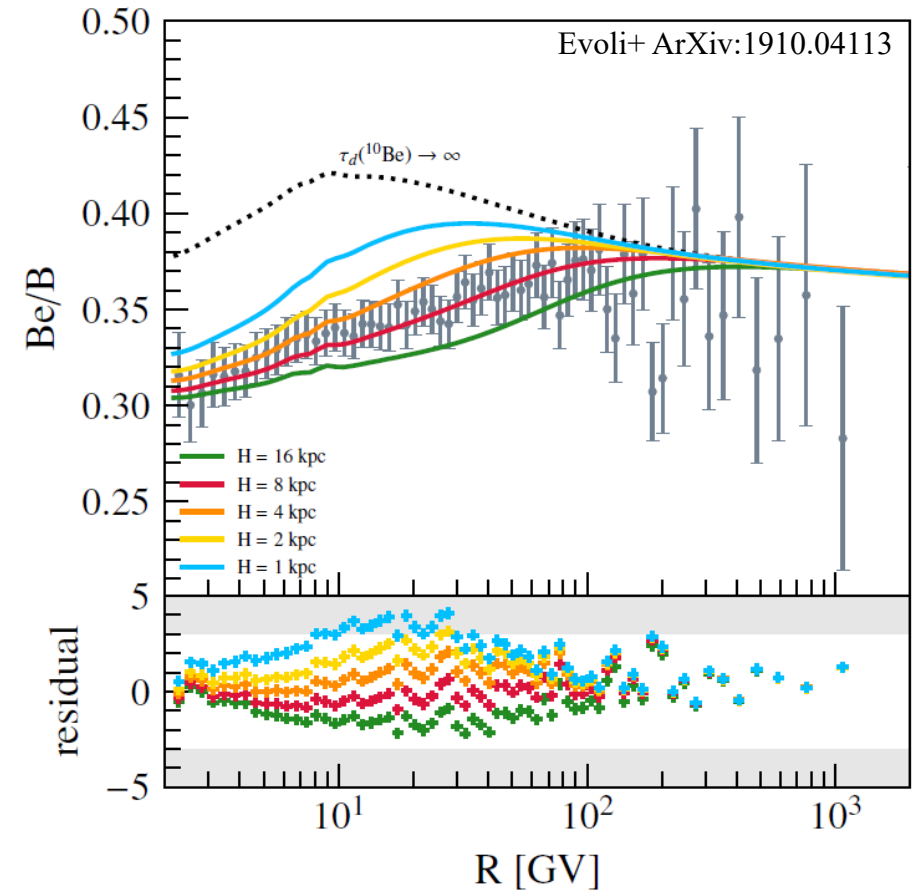
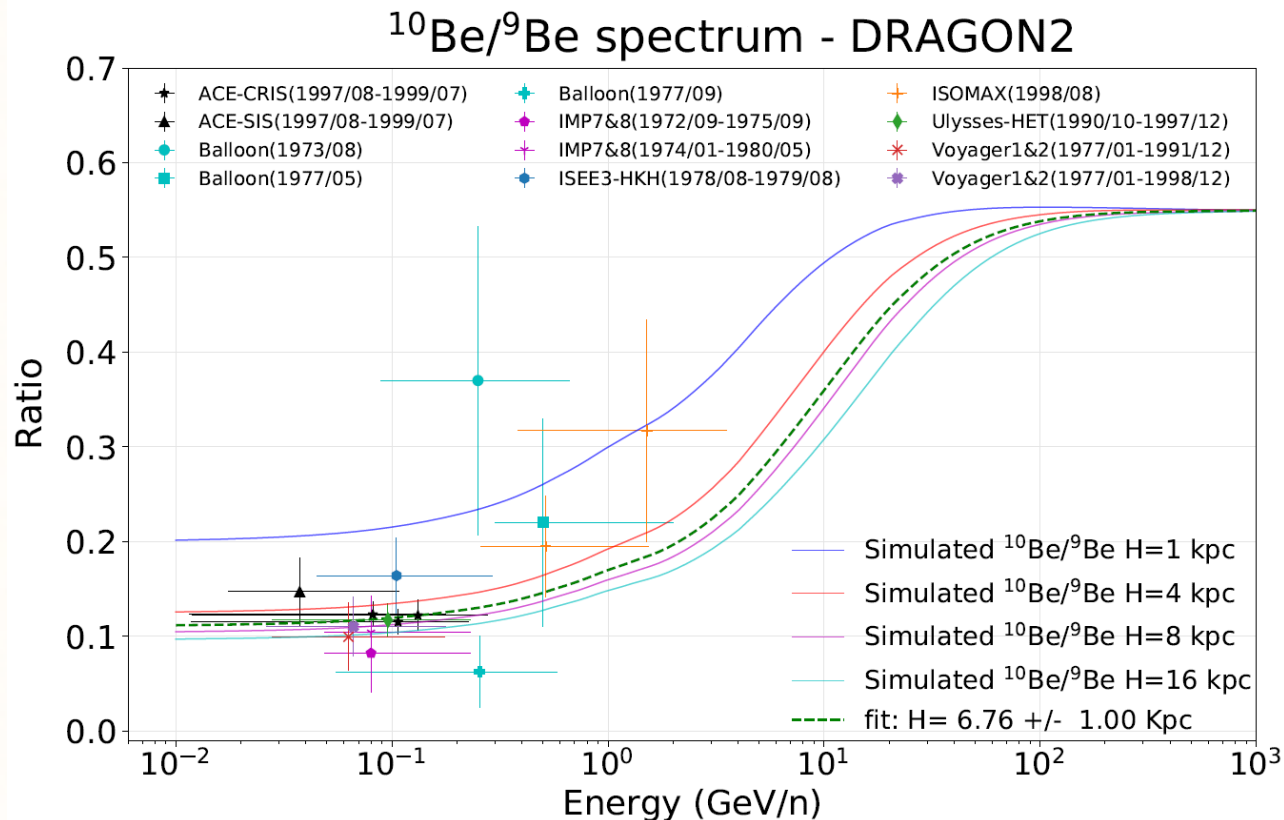
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# How to determine the propagation volume?

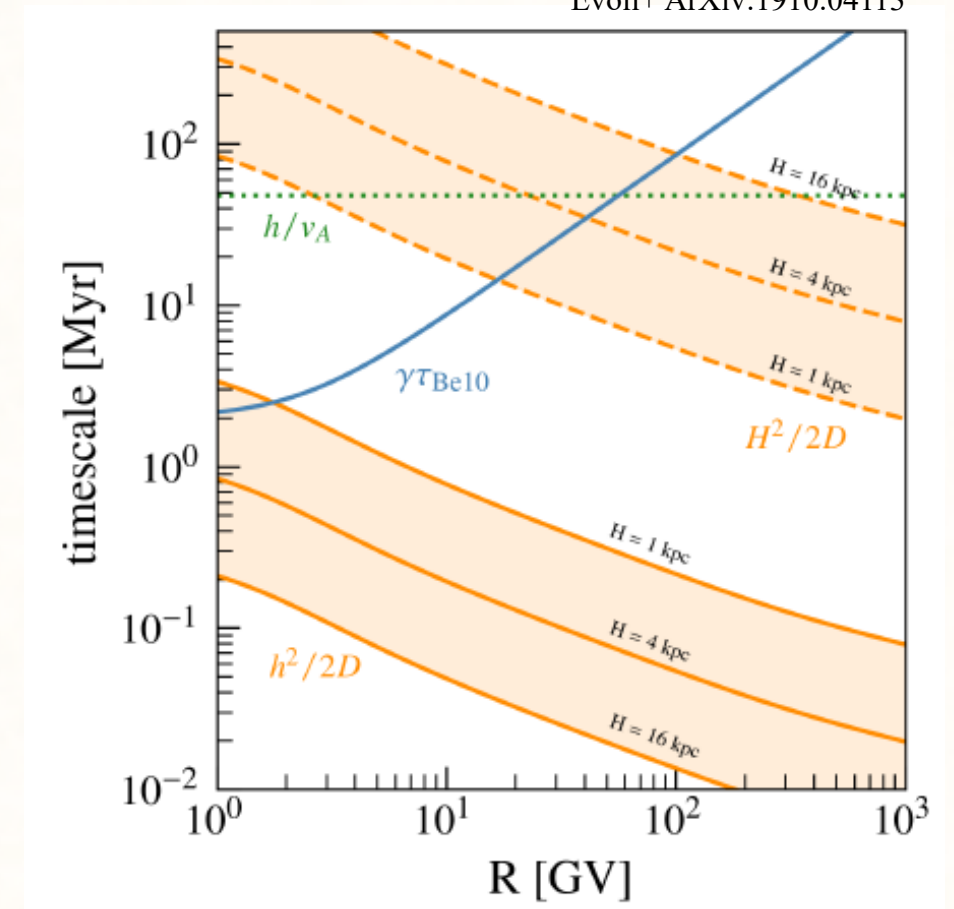
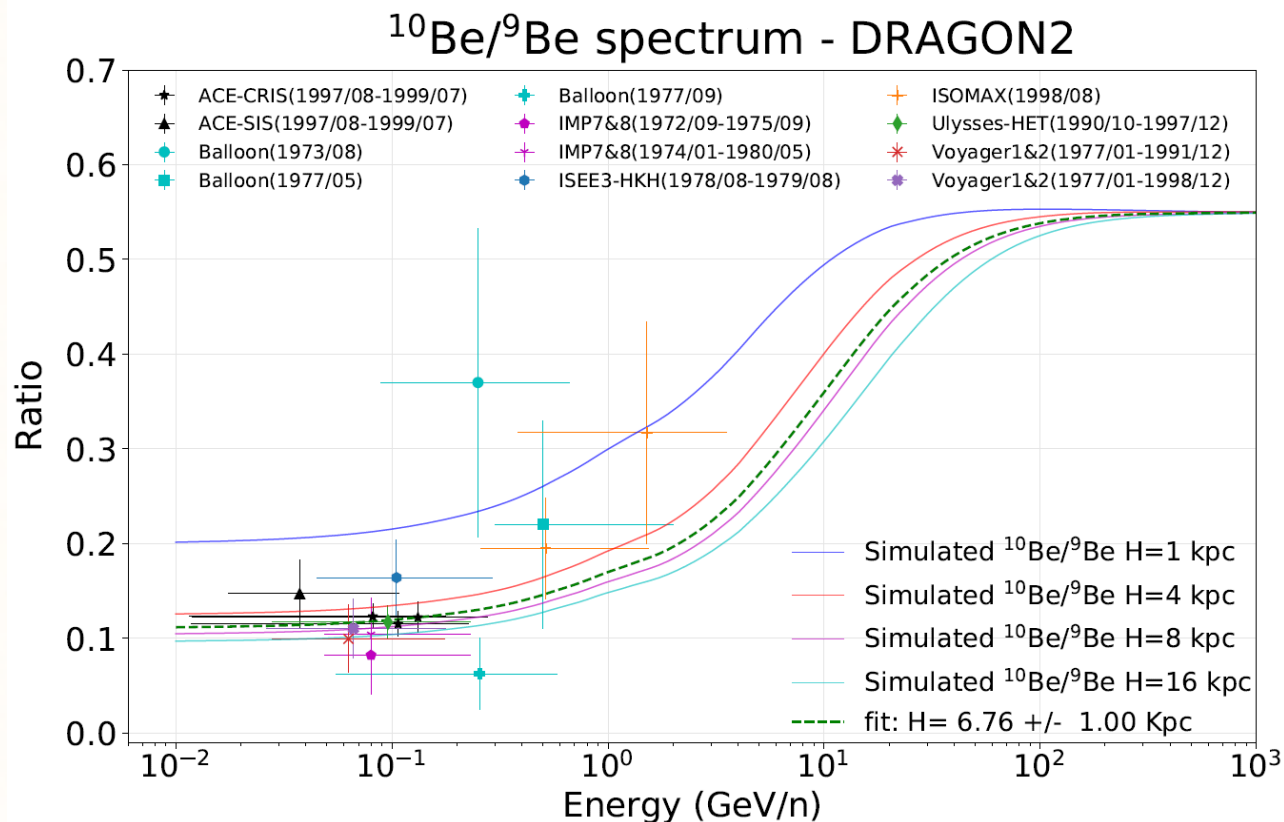
$^{10}\text{Be}$  is an unstable isotope that decays into  $^{10}\text{B}$  with a half-life  $\sim 1.5$  Myr



# How to determine the propagation volume?

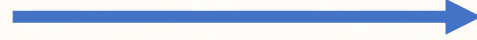
$^{10}\text{Be}$  is an unstable isotope that decays into  $^{10}\text{B}$  with a half-life  $\sim 1.5$  Myr

Evoli+ ArXiv:1910.04113



# The cosmic-ray diffusion equation

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$



$$\frac{\partial n}{\partial t} = \nabla \cdot (D \nabla n) + Q(\mathbf{r}, t) - \frac{n}{\tau_{\text{loss}}}$$

$$\frac{\partial n_j}{\partial t} = \nabla \cdot (D \nabla n_j) + Q_j(\mathbf{r}, t) + \frac{\partial}{\partial E} (b n_j) + \sum_i \Gamma_{i \rightarrow j}^{\text{int}} n_j + \sum_i \frac{n_j}{\tau_{i \rightarrow j}^{\text{dec}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{int}}} - \frac{n_j}{\tau_{\text{loss}}^{\text{dec}}}$$

$$\begin{aligned} \vec{\nabla} \cdot (-D \nabla N_i - \vec{v}_\omega N_i) + \frac{\partial}{\partial p} \left[ p^2 D_{pp} \frac{\partial}{\partial p} \left( \frac{N_i}{p^2} \right) \right] = Q_i + \frac{\partial}{\partial p} \left[ \dot{p} N_i - \frac{p}{3} (\vec{\nabla} \cdot \vec{v}_\omega N_i) \right] \\ - \frac{N_i}{\tau_i^f} + \sum \Gamma_{j \rightarrow i}^s(N_j) - \frac{N_i}{\tau_i^r} + \sum \frac{N_j}{\tau_{j \rightarrow i}^r} \end{aligned}$$



# The cosmic-ray diffusion equation

$$\vec{\nabla} \cdot (-D \nabla N_i - \vec{v}_\omega N_i) + \frac{\partial}{\partial p} \left[ p^2 D_{pp} \frac{\partial}{\partial p} \left( \frac{N_i}{p^2} \right) \right] = Q_i + \frac{\partial}{\partial p} \left[ \dot{p} N_i - \frac{p}{3} (\vec{\nabla} \cdot \vec{v}_\omega N_i) \right] - \frac{N_i}{\tau_i^f} - \sum \Gamma_{j \rightarrow i}^s(N_j) - \frac{N_i}{\tau_i^r} + \sum \frac{N_j}{\tau_{j \rightarrow i}^r}$$

$$D = D_0 \beta^\eta \left( \frac{R}{R_0} \right)^\delta F(\vec{r}, z)$$

Different parameterizations for D can be adopted, but they'll follow the general form of a power-law in rigidity/energy with breaks

$$J_{pr}(E) \sim \frac{Q_{inj}(E)}{D(E)} \propto E^{-(\alpha_{inj} + \delta)}$$

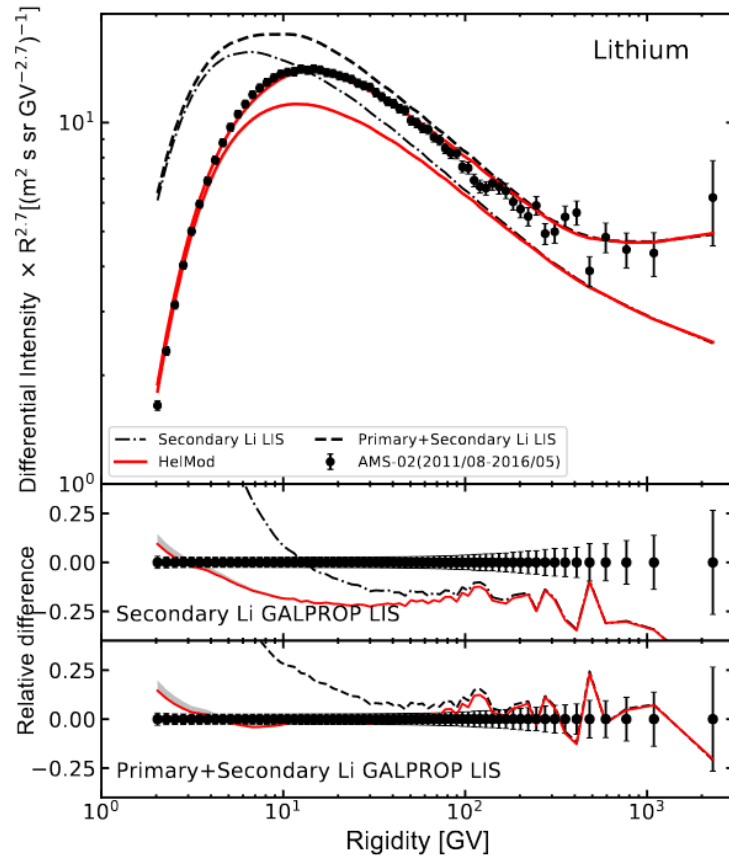
$$J_{sec}(E) \sim \frac{\sigma(E) J_{pr}(E)}{D(E)} \propto E^{-(\alpha_{inj} + 2\delta)}$$

$$\frac{J_{sec}}{J_{pr}} \sim \sigma(E)/D(E) \propto E^{-\delta}$$

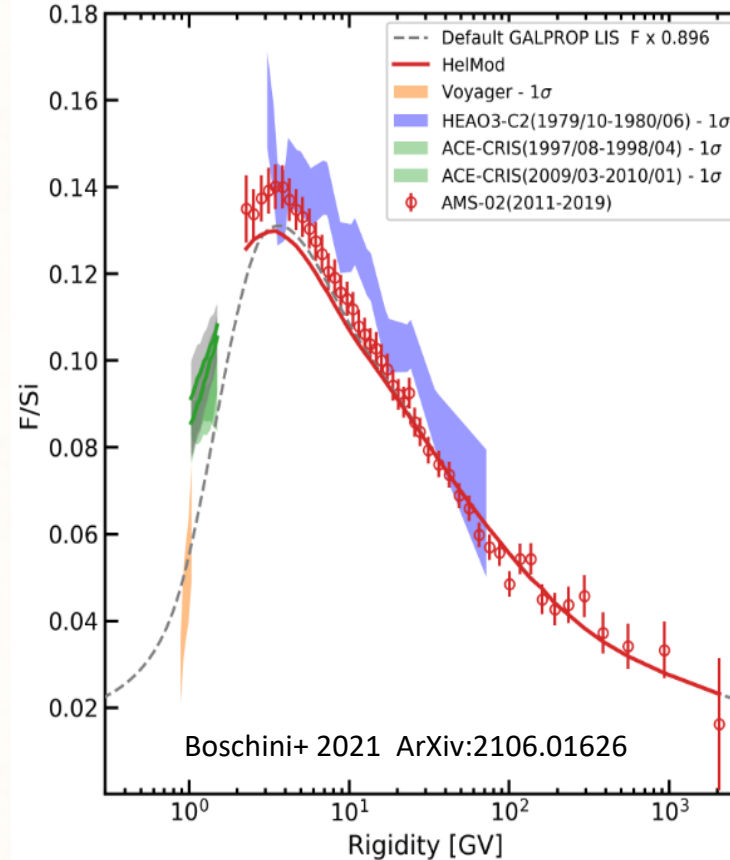
$$\Gamma_{j \rightarrow i}^s = \beta_j c n_t \sigma_{j \rightarrow i} N_j$$

$n_t$  Gas density  
 $\sigma_{j \rightarrow i}$  Cross section  
 $N_j$  Flux of primary CR

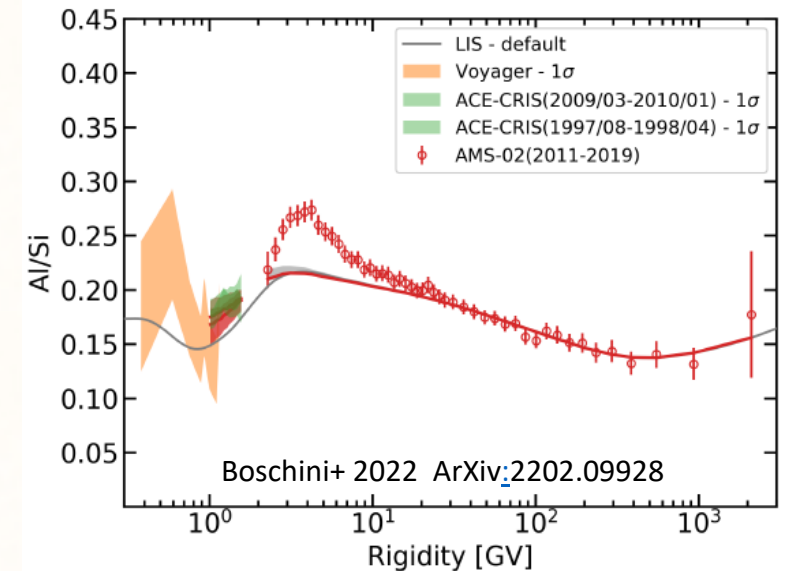
# Current issues: Extra sources?



Boschini+ 2020 ArXiv:1911.03108

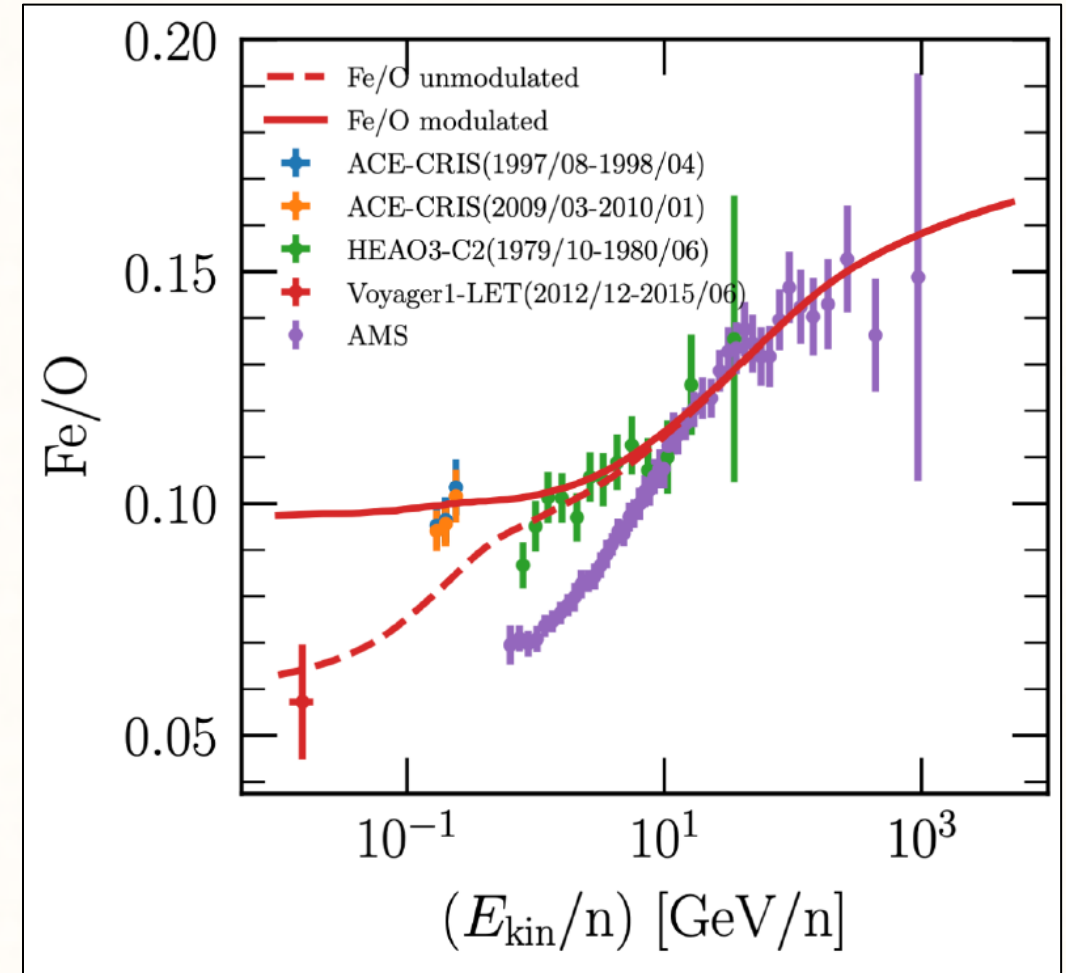
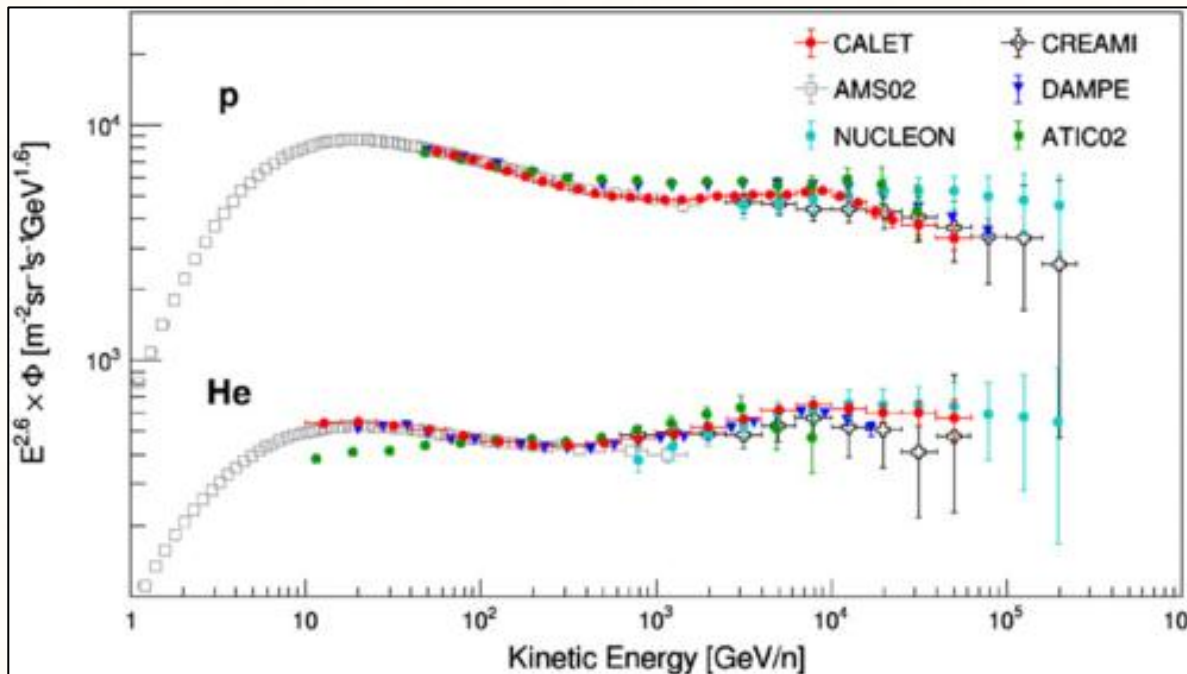


A number of potential excesses have been claimed but ignored the effect of **cross sections uncertainties: Main source of systematic uncertainty**



# Current issues: Anomalies in Primary CRs

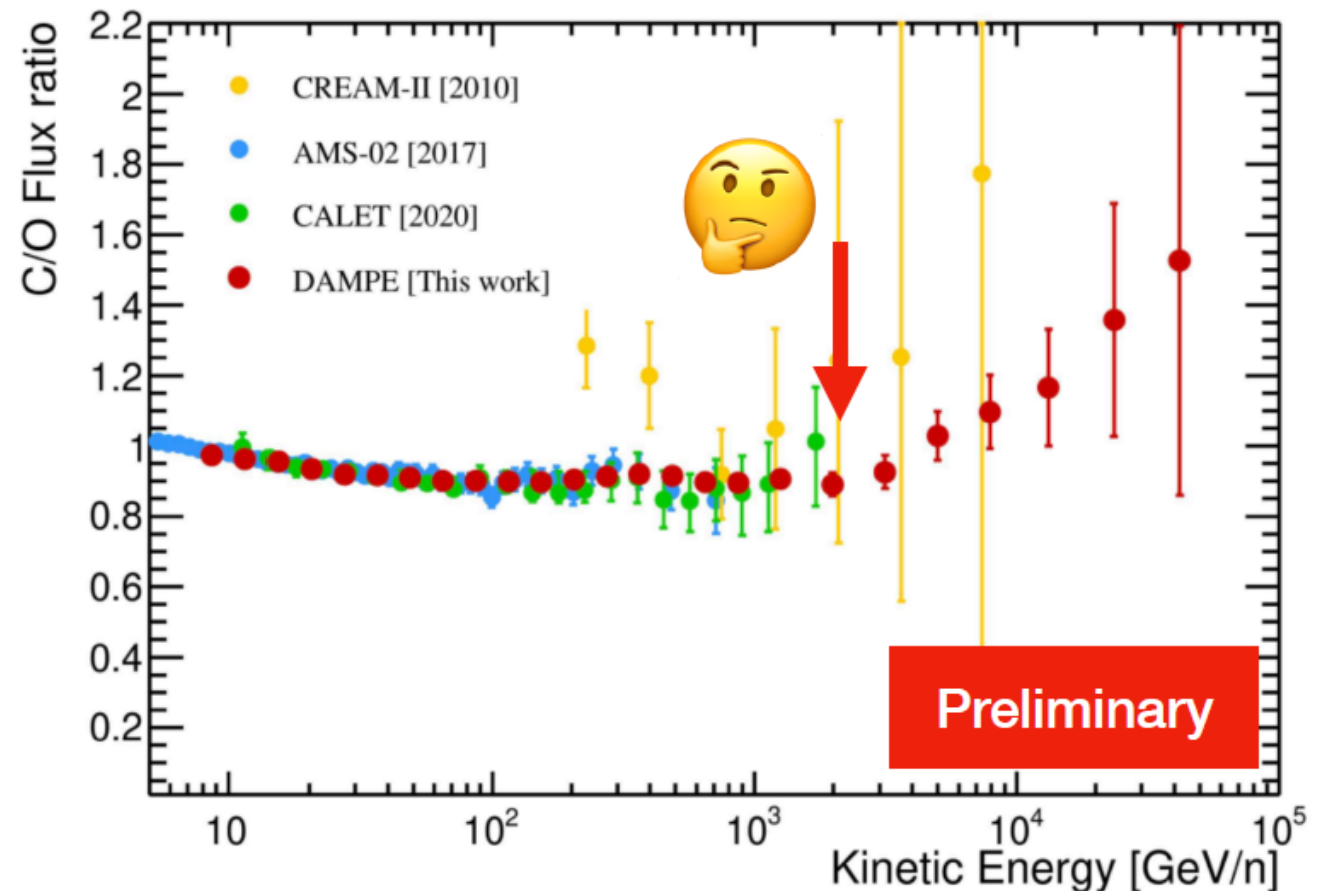
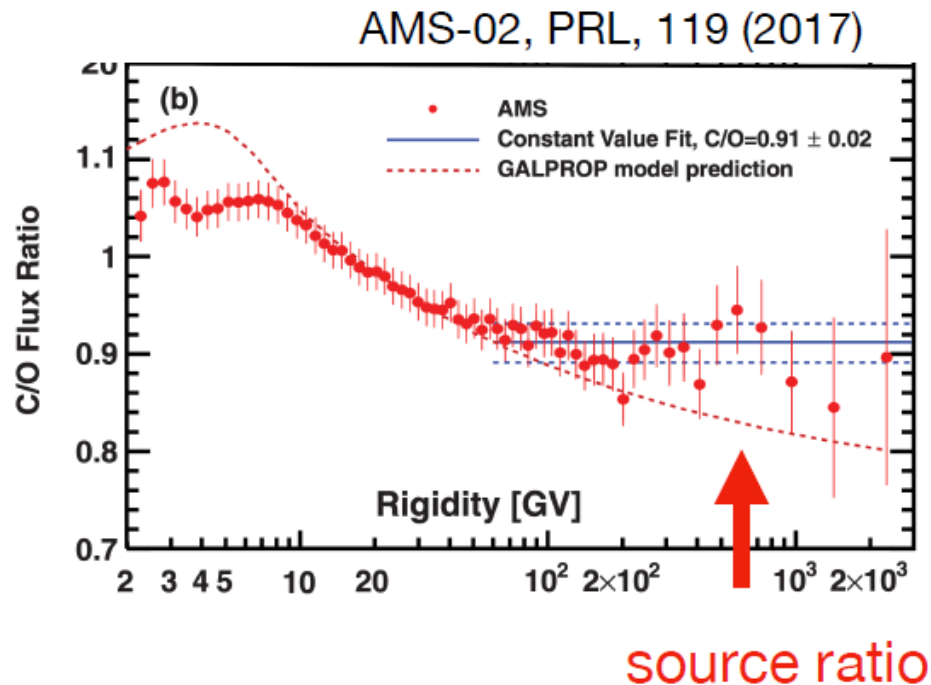
- **Protons and helium** spectra behave differently, while their rigidity dependence should be identical
- **Iron** seems to significantly deviate from our predictions at low energies



# Current issues: Anomalies in Primary CRs

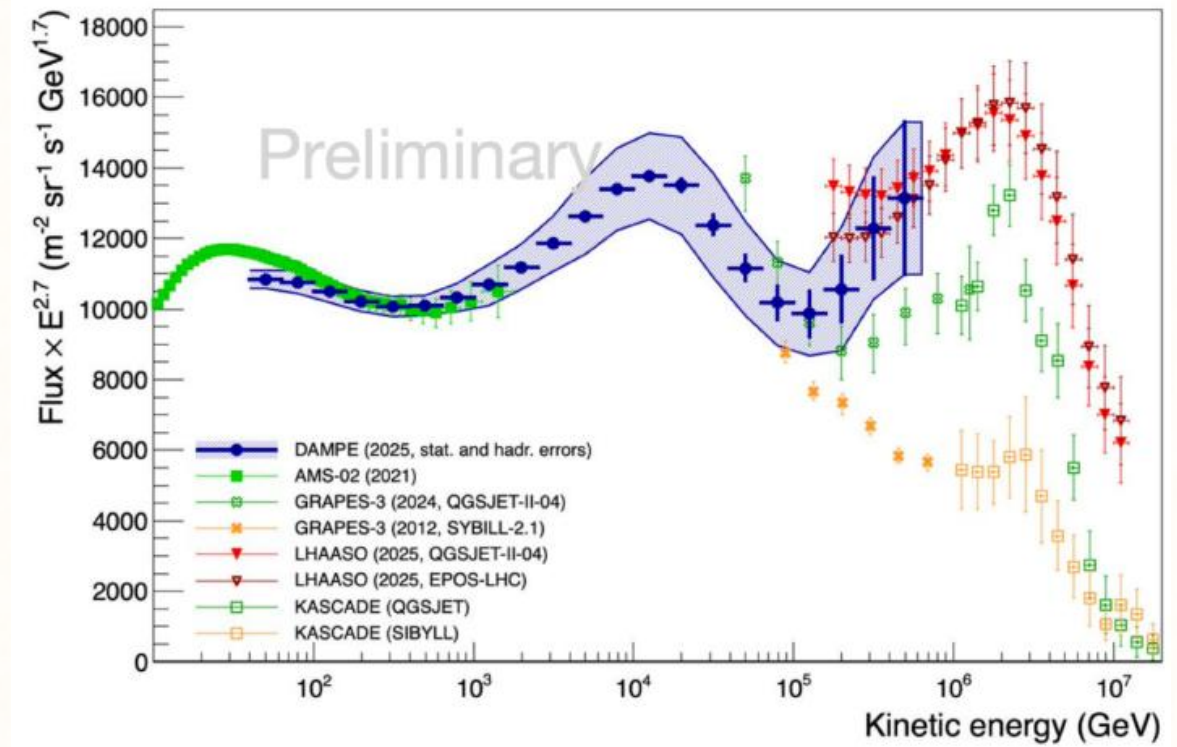
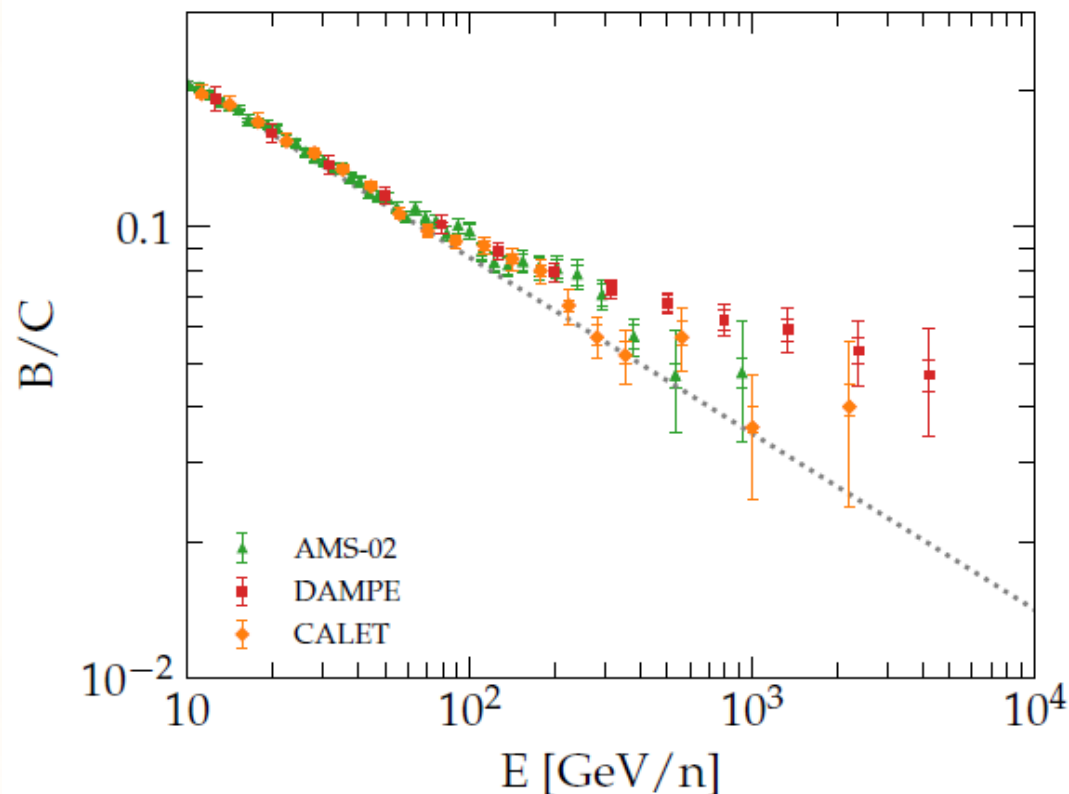
- However, **C and O** do not behave as p and He

$$\frac{N_C}{N_O} = \frac{Q_C(R)}{Q_O(R)} = \frac{q_C}{q_O} \mathcal{R}^{-\gamma_C + \gamma_O}$$



# Current issues: Many breaks!

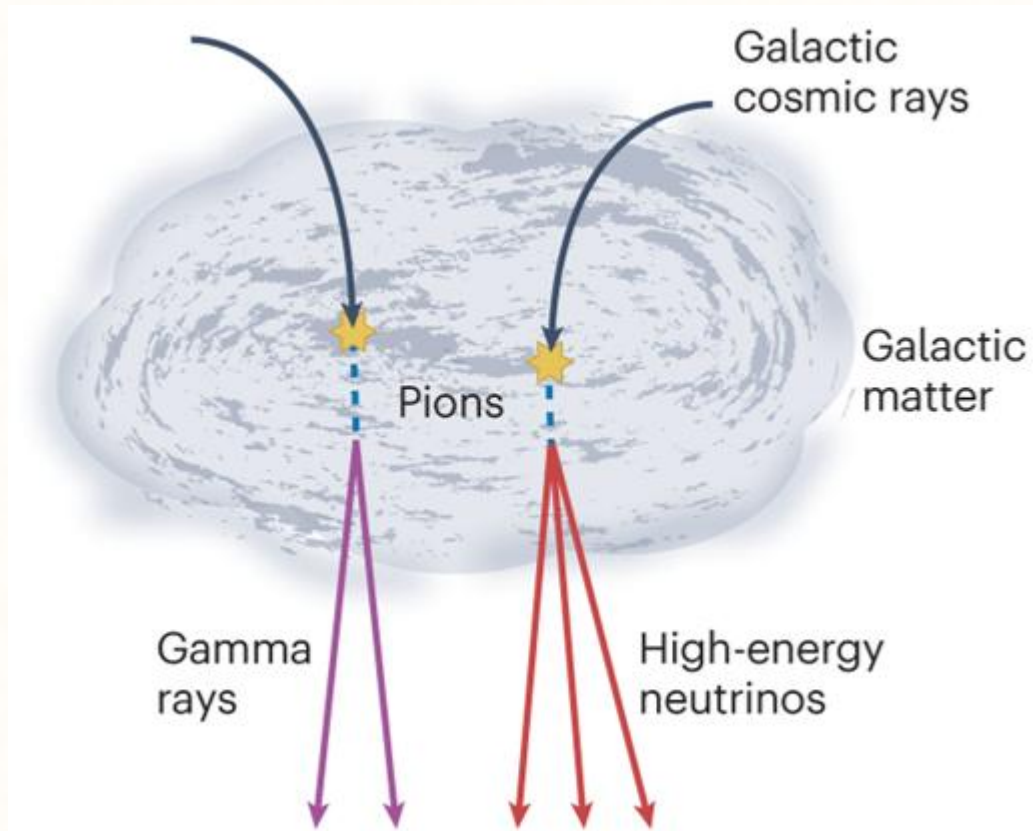
While the **first hardening is robustly observed in secondary cosmic rays**, we do not have statistics to verify this is the same for breaks at higher energies





# Other secondaries: Gamma rays and neutrinos

$$p_{\text{CR}} + p_{\text{ISM}} \rightarrow \pi^0 + \pi^\pm + X$$



$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

$$\pi^0 \rightarrow \gamma + \gamma$$

These interactions allow us to observe the “glow” of cosmic-ray interactions!

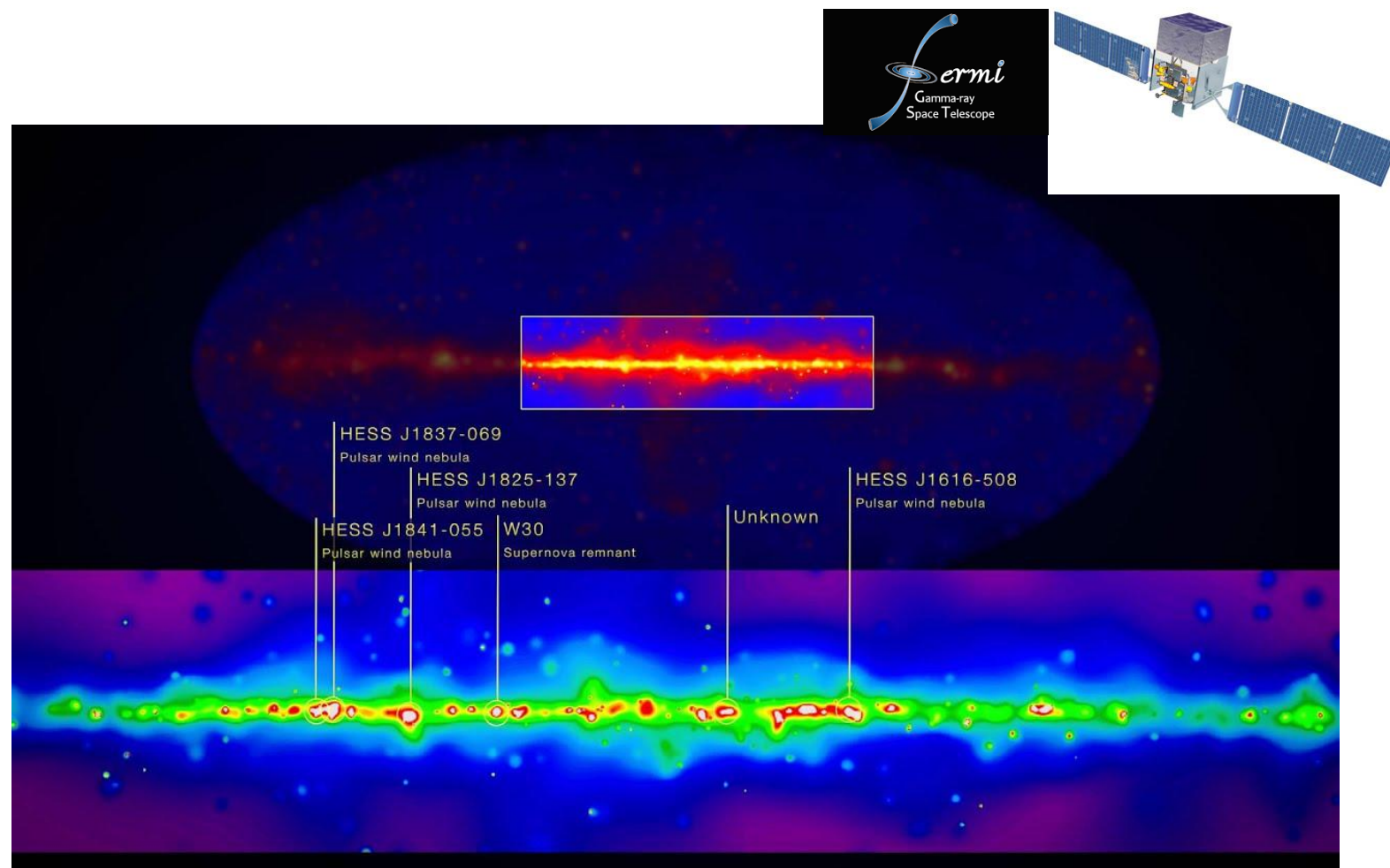
We need gamma ray telescopes and huge neutrino detectors to observe this... but we do see them!

# The Galactic gamma-ray sky

- > 5600 point-like sources have been observed by the Fermi satellite
- ~ 75 extended sources

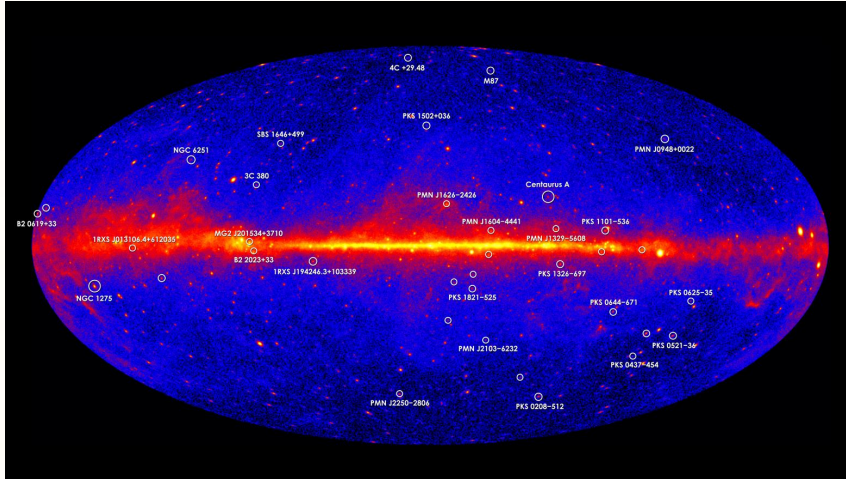
## Galactic diffuse gamma-ray background

Extragalactic background (EBL) is isotropic

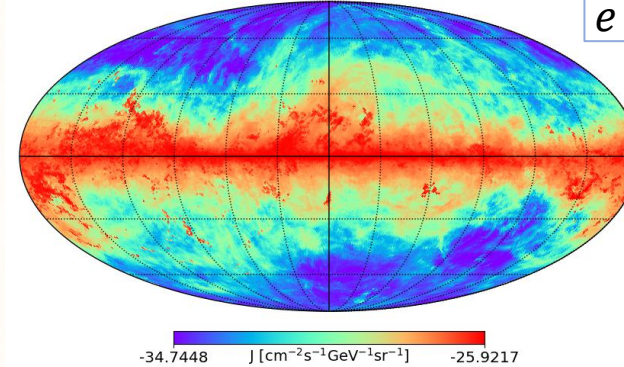


# The Galactic gamma-ray sky

Diffuse emission totally correlated  
with the propagation of cosmic rays  
Dominated by protons, He (and  $e^-$ )



Bremsstrahlung emission - 120 GeV



$$e + N \rightarrow e' + \gamma' + N$$

$$p + p \rightarrow \pi^0$$

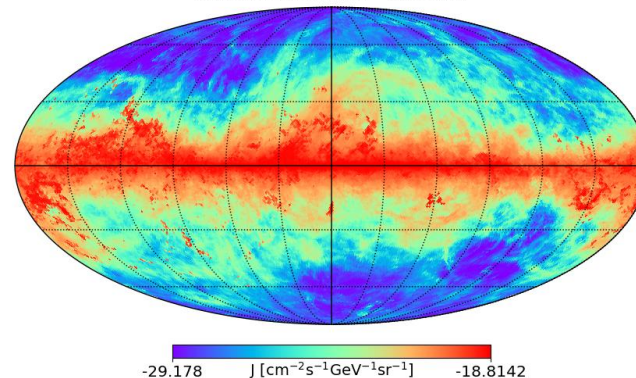
$$\pi^0 \rightarrow 2\gamma$$

$$e + \gamma \rightarrow e' + \gamma'$$

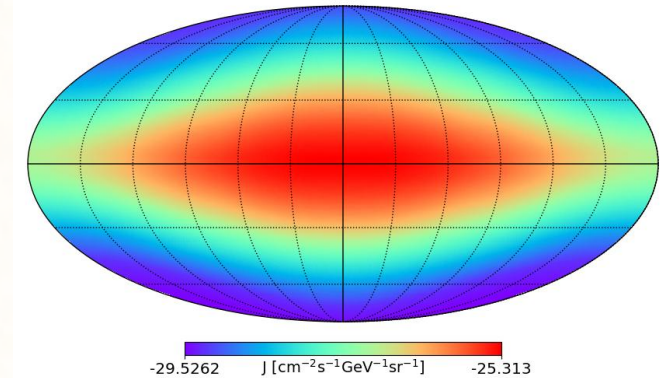
**Hadronic (and Bremss.) emission**  
follows the **ISM gas distribution**

**IC emission** depends on the  
energy density of the **ISRFs**

Hadronic emission - 120 GeV

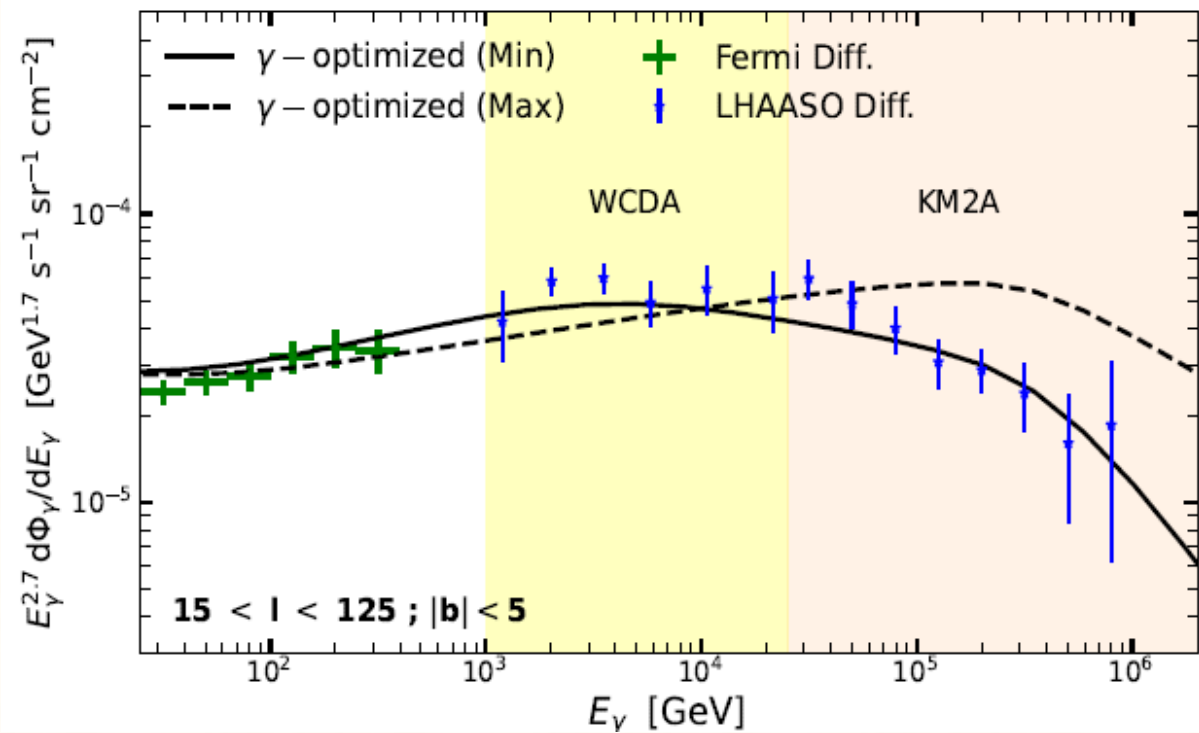
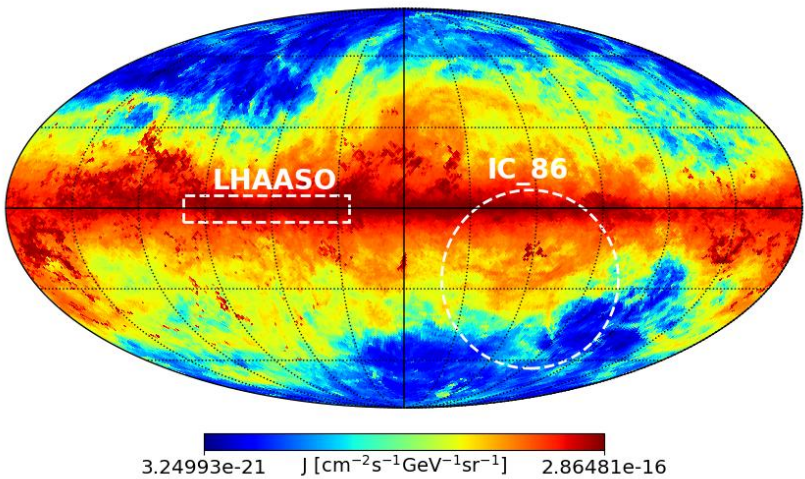


IC emission - 120 GeV

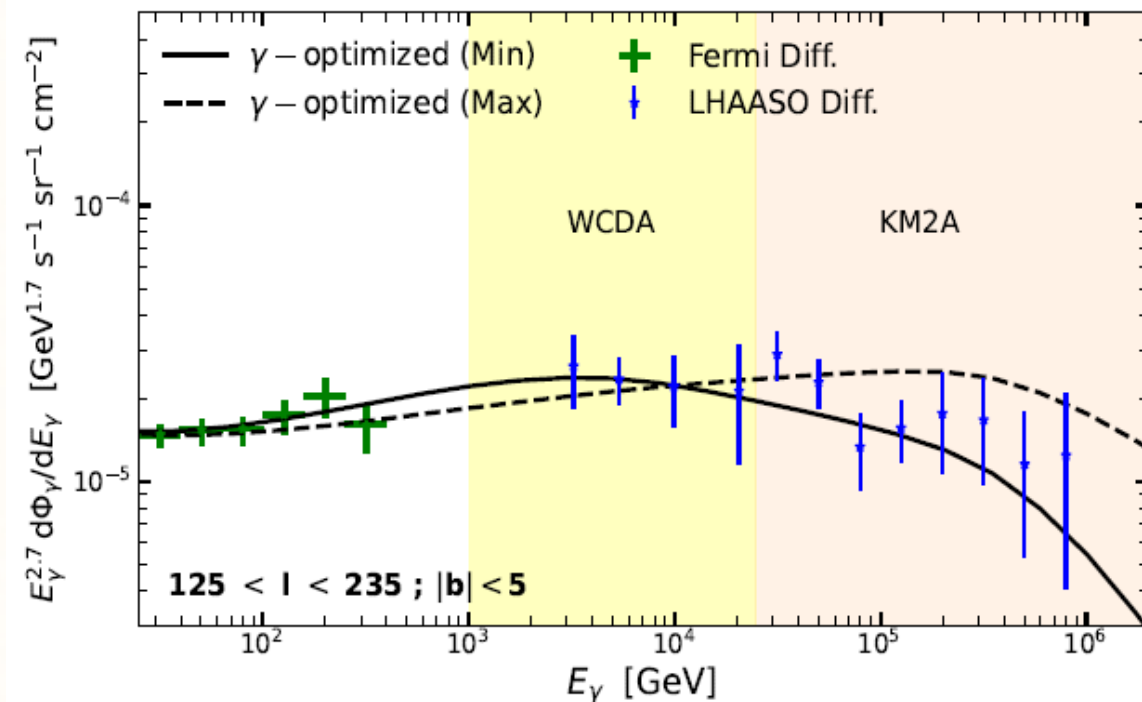




# The Galactic gamma-ray sky



| Region                              | $\phi_0$ at 10 TeV<br>( $10^{-13} \text{ TeV}^{-1} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ ) | $\alpha$                       |
|-------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------|
| $15^\circ < l < 125^\circ$ (inner)  | $8.50 \pm 0.58_{\text{stat}}$                                                                        | $-2.67 \pm 0.05_{\text{stat}}$ |
| $125^\circ < l < 235^\circ$ (outer) | $3.49 \pm 0.55_{\text{stat}}$                                                                        | $-2.83 \pm 0.19_{\text{stat}}$ |



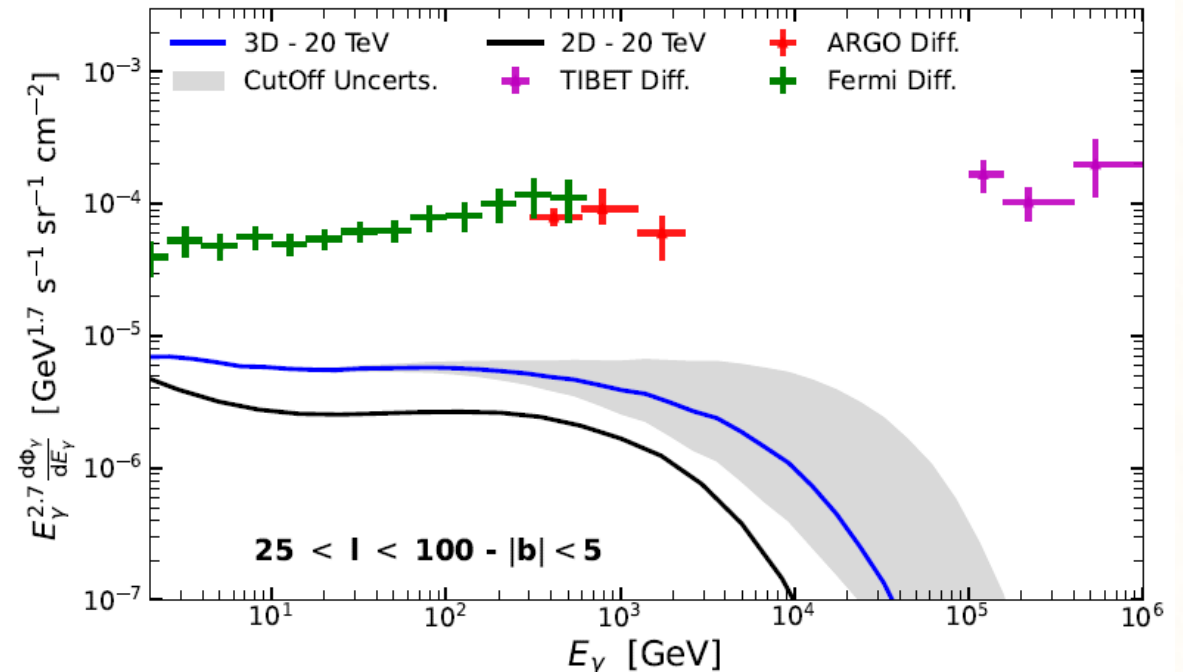
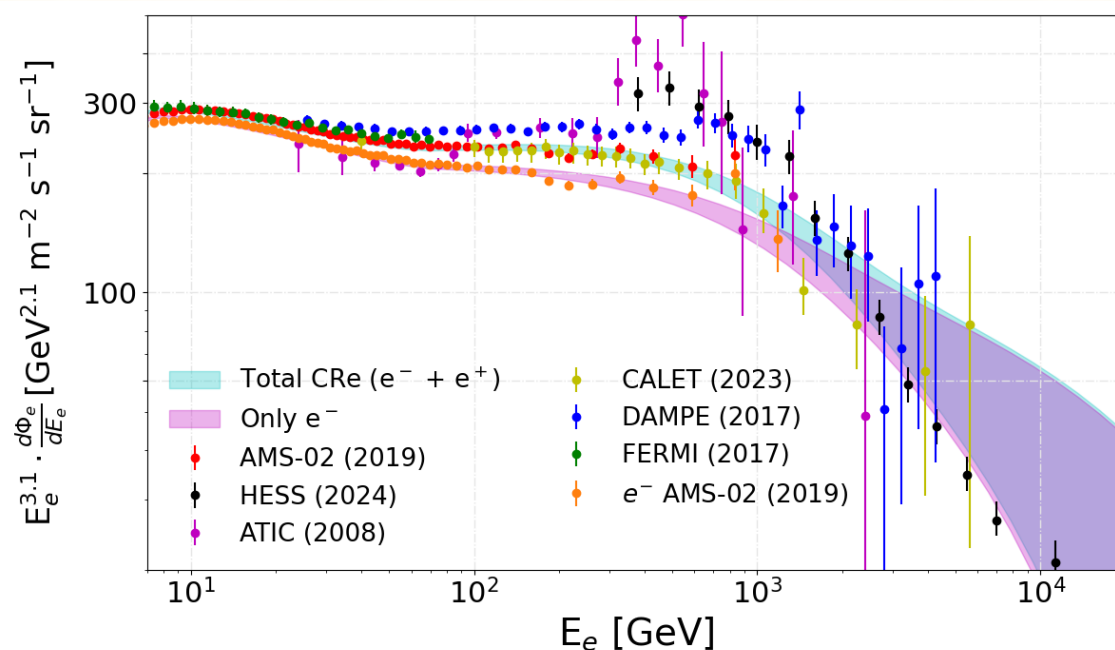
(Table II of ArXiv:2411.16021)

# The Galactic gamma-ray diffuse sky – electrons contribution

Electrons lose energy very quickly due to their interaction with the magnetic field (leading to synchrotron emission) and ISRFs (inverse Compton emission)

Therefore, the electron spectrum measured at Earth is very influenced by the presence of local sources → Difficult to estimate with precision the IC emission in different parts of the Galaxy

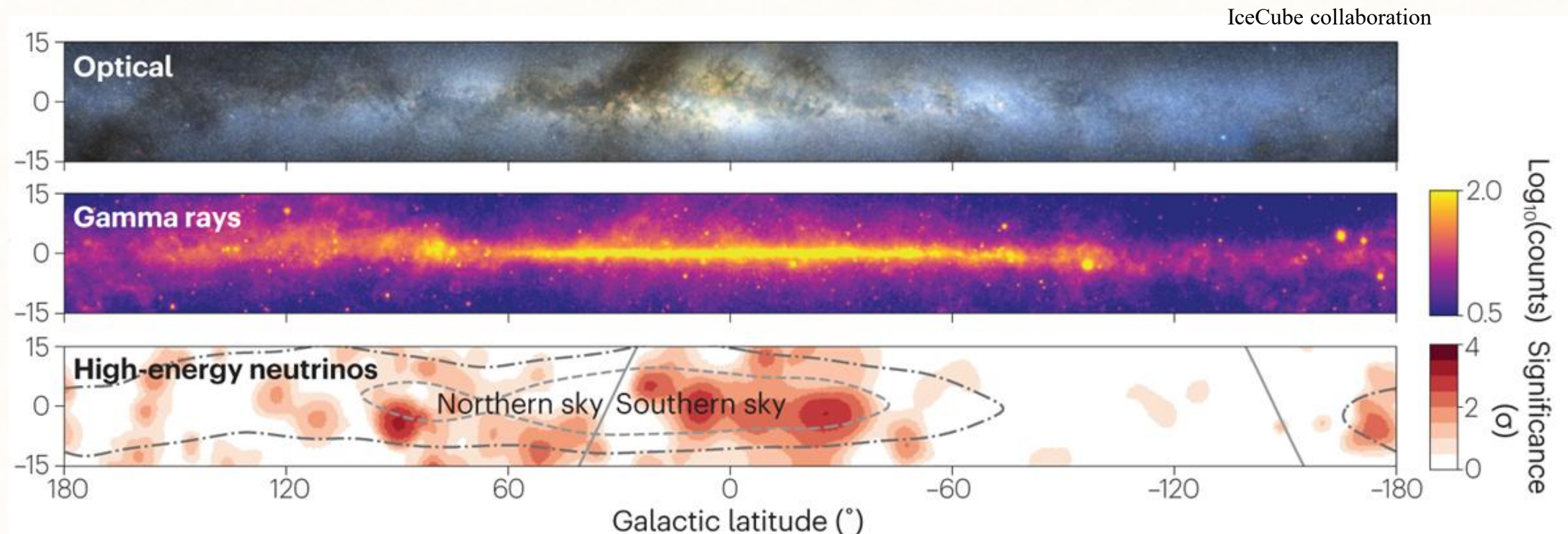
PDL. et al arXiv:2502.18268



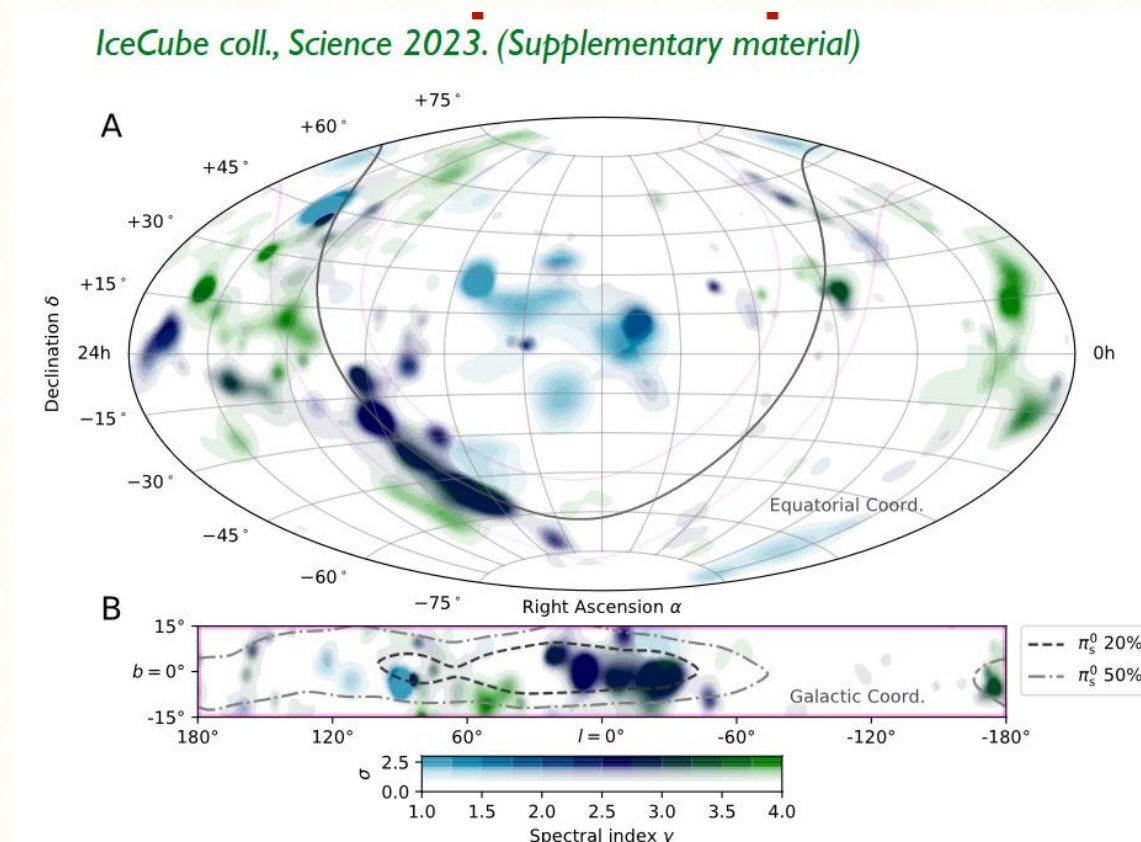
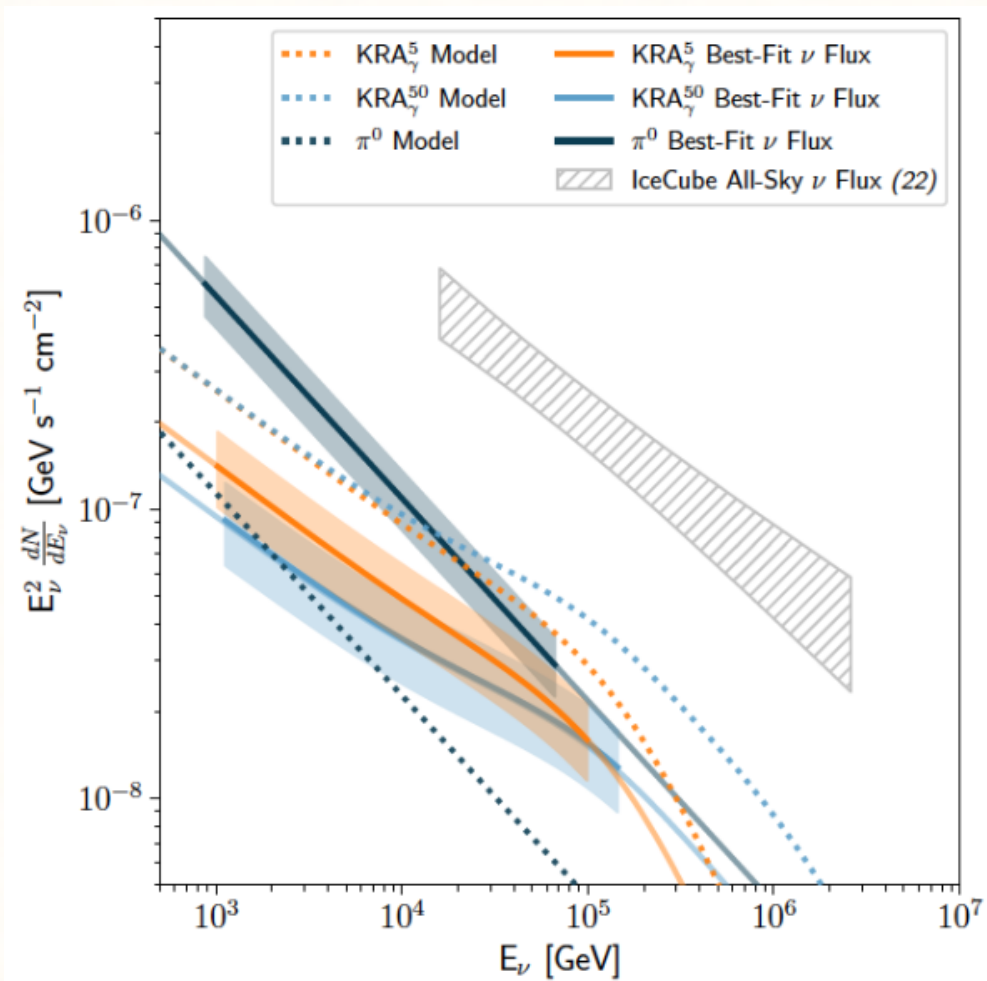


# The Galactic neutrino sky: Not clearly observed (Until recently!)

Although this analysis is much harder, IceCube recently reported a high significance of detection of neutrinos from the Galactic plane!



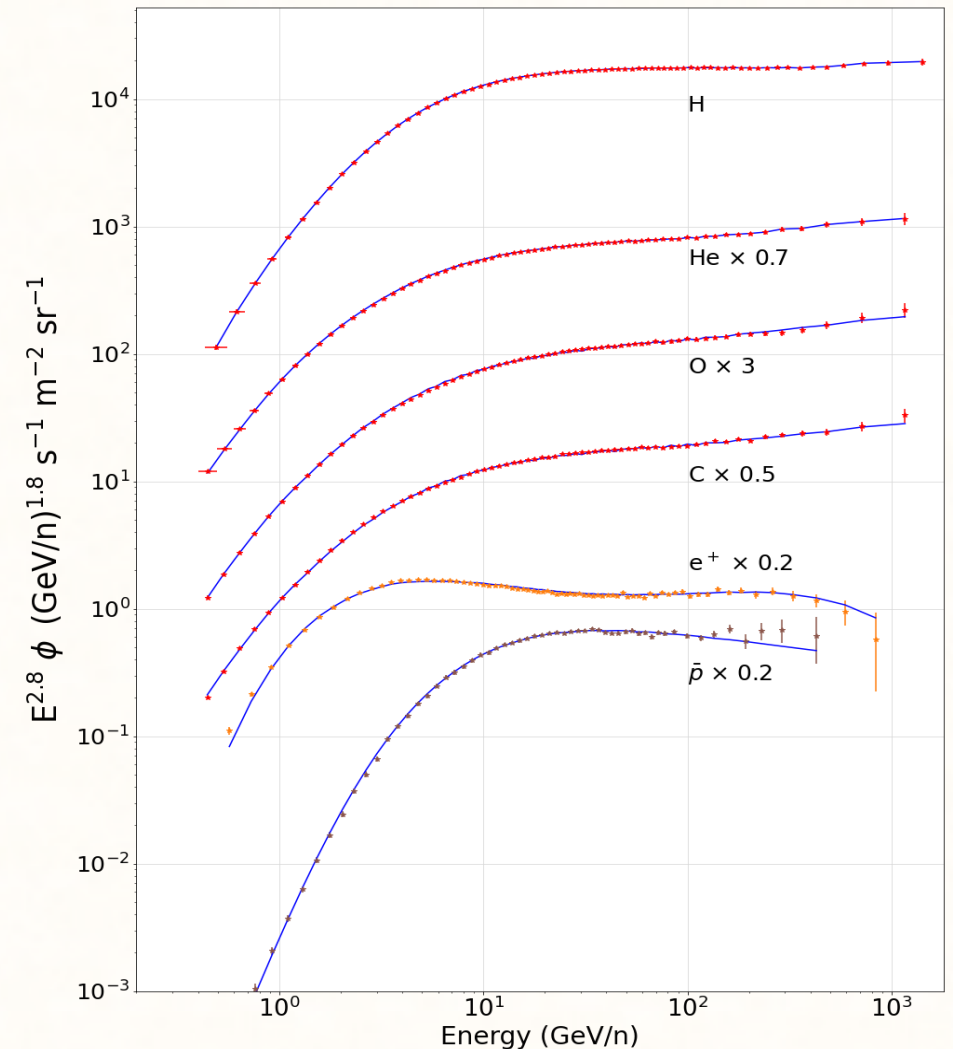
# IceCube measured the diffuse neutrino emission!



# Other secondaries: Antiparticles!

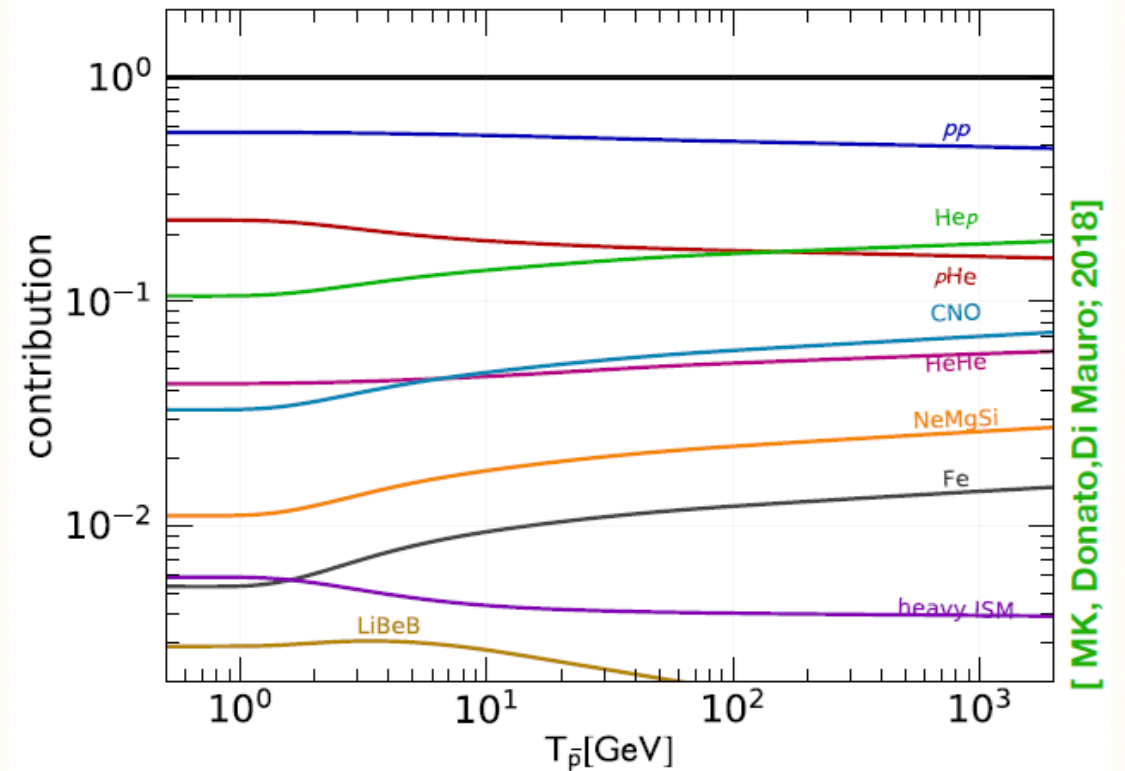
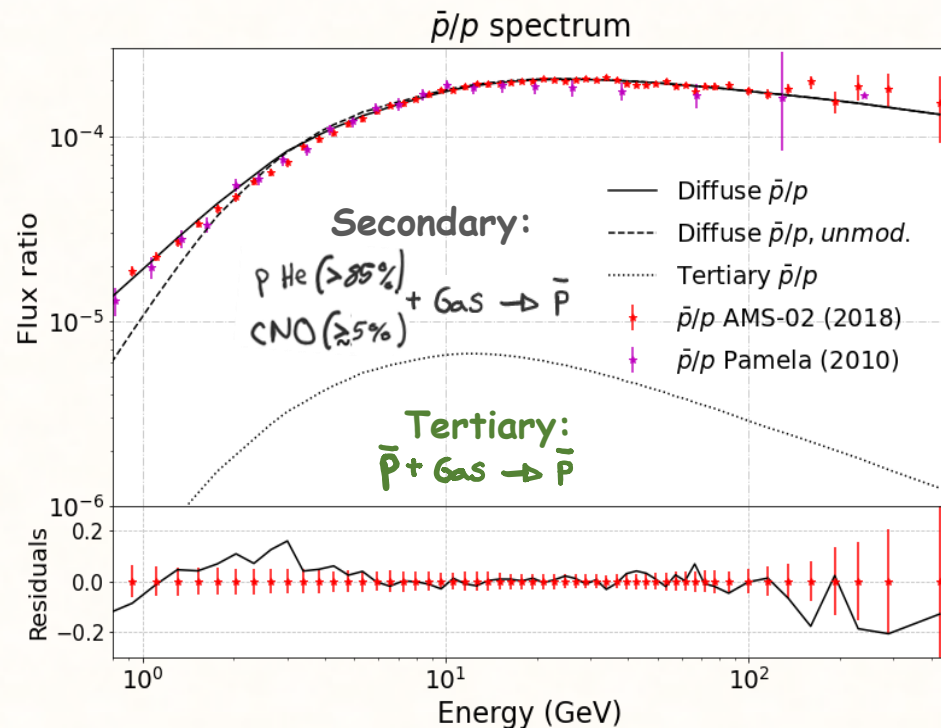
Positrons, antiprotons and even heavier antiparticles are also produced in CR interactions and **AMS-02 have detected them.**

- These particles are **produced similarly to secondary cosmic rays**, but can carry crucial information about other processes they are sensitive to, like nearby accelerators or the conditions of propagation
- In addition, they are used as **crucial observables to constrain the existence of exotic physics**, such as particle emission from dark matter or the existence of primordial black holes



# Channels of secondary antiproton production

$p + p \rightarrow p + p + p + \bar{p}$  (High energy protons produce lower energy antiprotons)



Different evaluations vary by tens of percent --  
parameterizations seem more precise than  
current event generators, with **uncerts ~ 12%**  
**Heavy CRs may have a contribution > 7%**

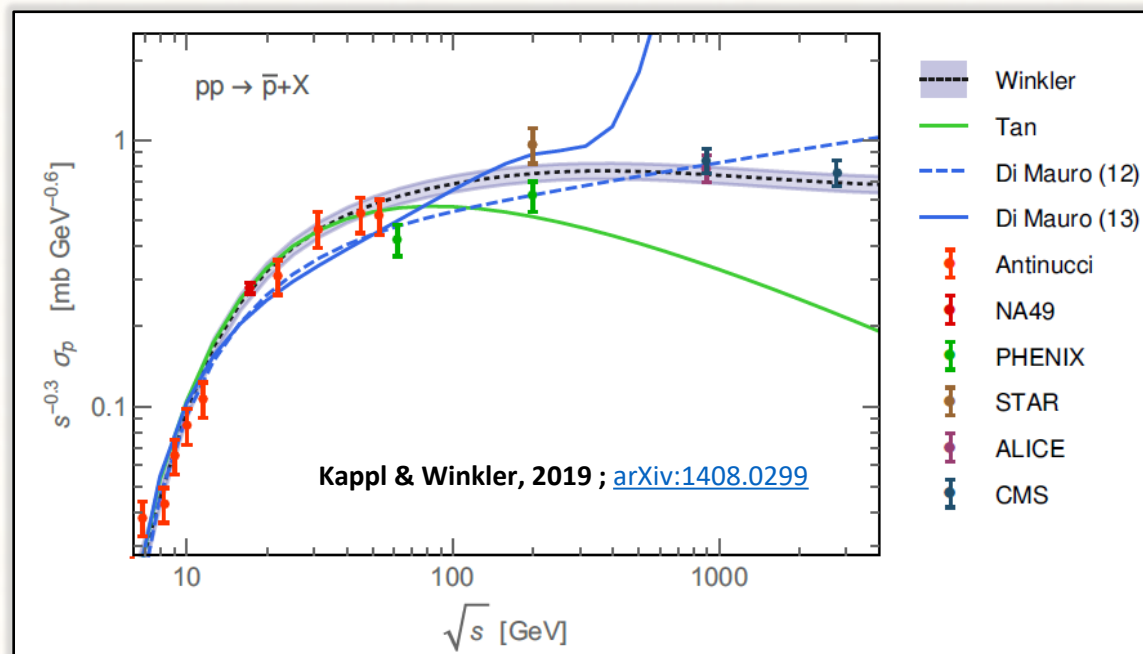


# Antiproton cross sections

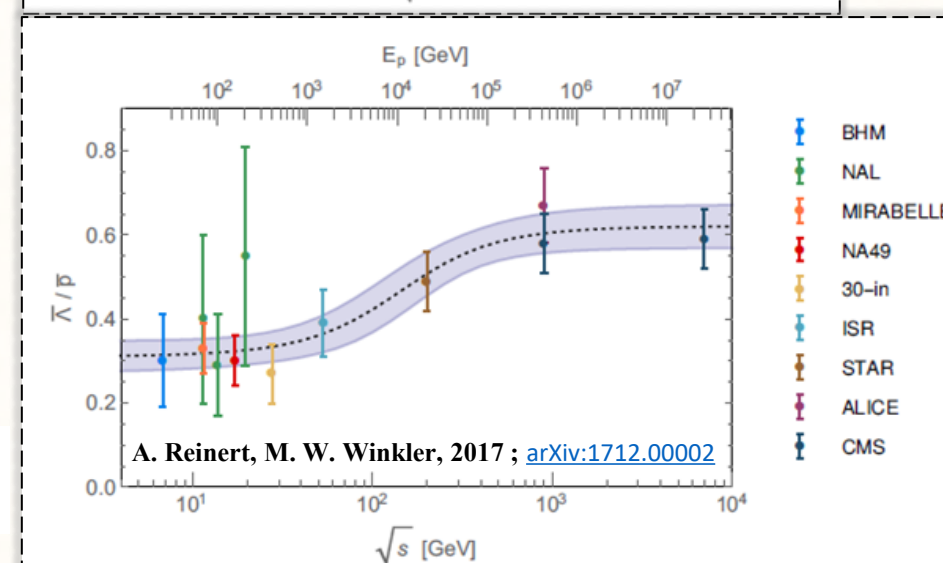
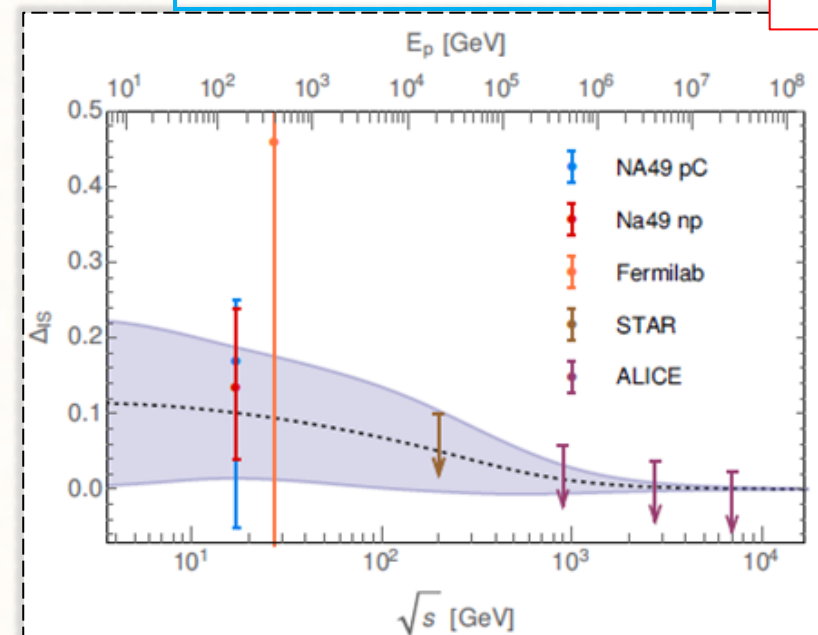
$$\left(E \frac{d^3\sigma}{dp^3}\right)_{pp \rightarrow \bar{p}} = \left(E \frac{d^3\sigma}{dp^3}\right)_{pp \rightarrow \bar{p}}^{\text{prompt}} \cdot (2 + \Delta_{IS} + 2 \Delta_{\Lambda})$$

$$p + p \longrightarrow \{\bar{n} \longrightarrow \bar{p}\} + X$$

$$\Delta_{IS} = \frac{\sigma_{pp \rightarrow \bar{n}}}{\sigma_{pp \rightarrow \bar{p}}} - 1$$



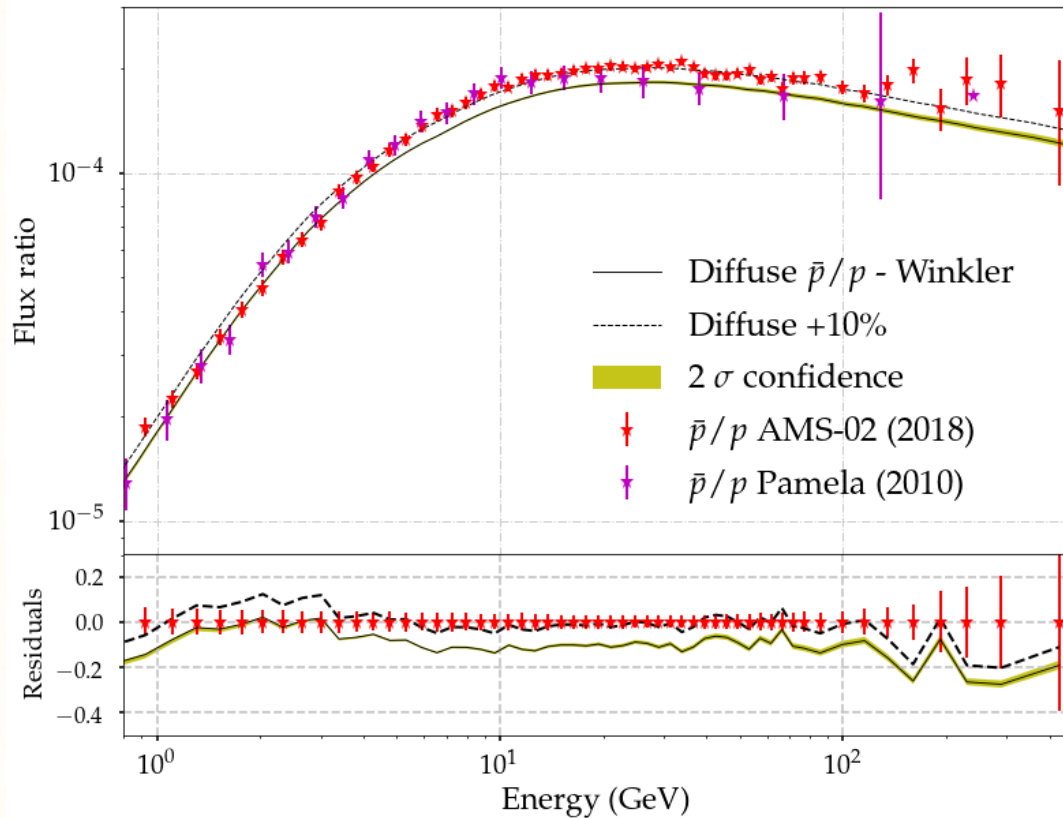
$$p + p \longrightarrow \{\bar{\Lambda}, \bar{\Sigma} \longrightarrow \bar{p}\} + X$$



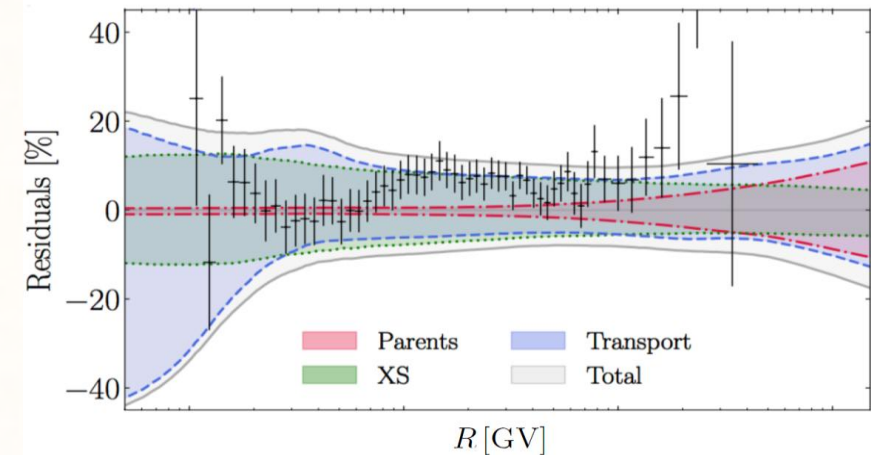
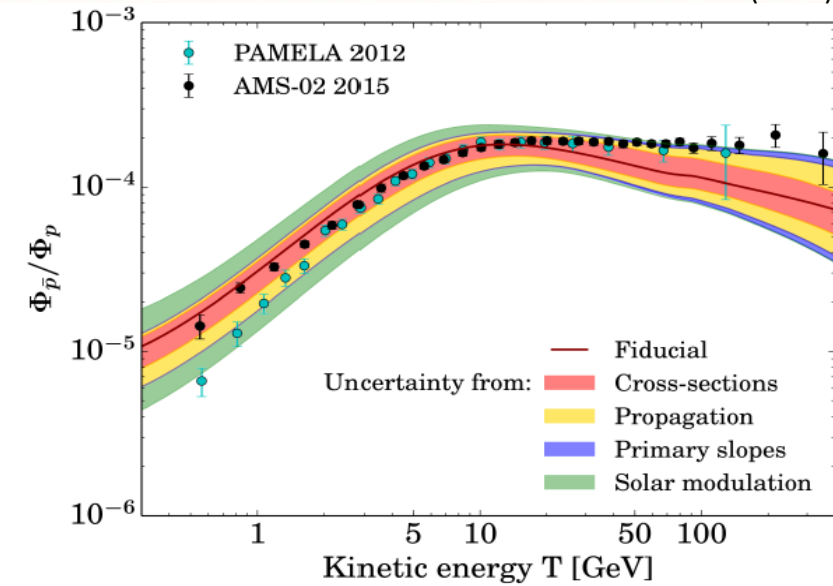


# An antiproton excess?

Tension in the propagation parameters predicted from the secondary CRs B, Be and Li and antiprotons

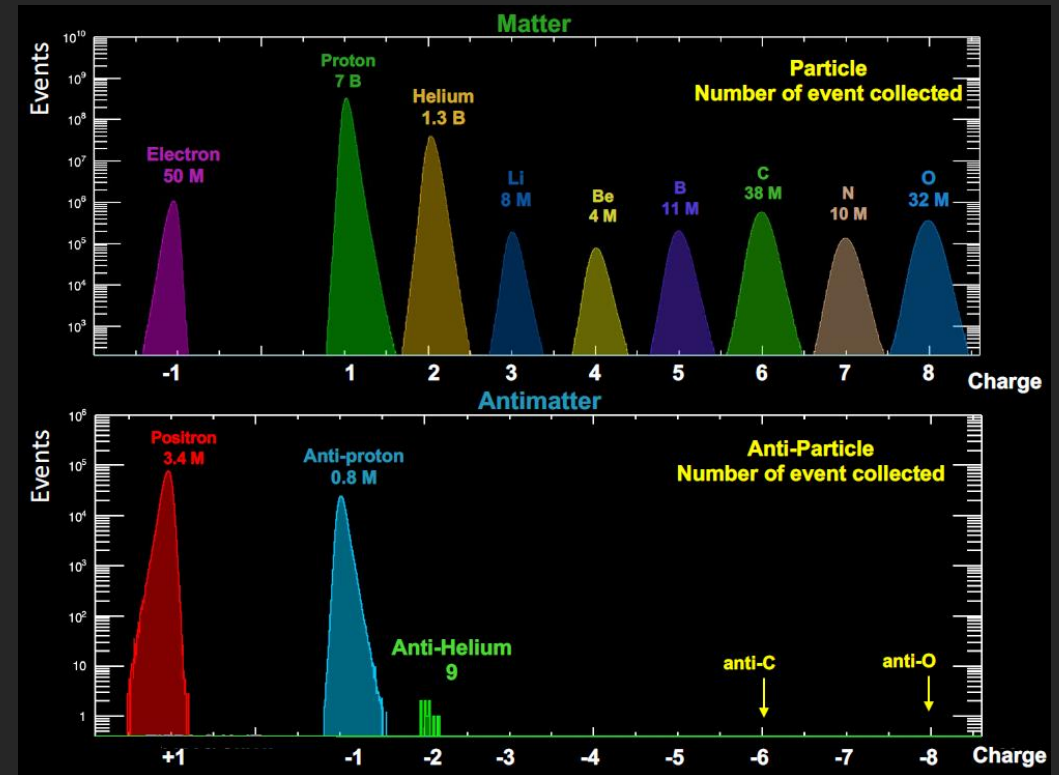
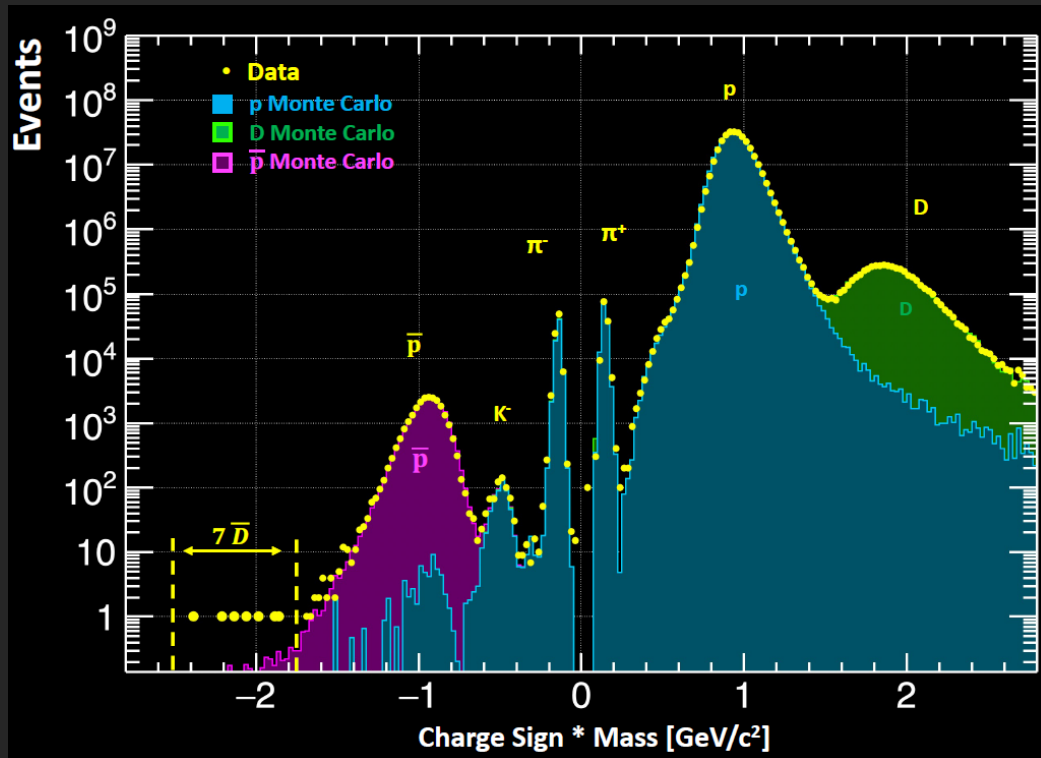


G. Giesen et al JCAP09(2015)023

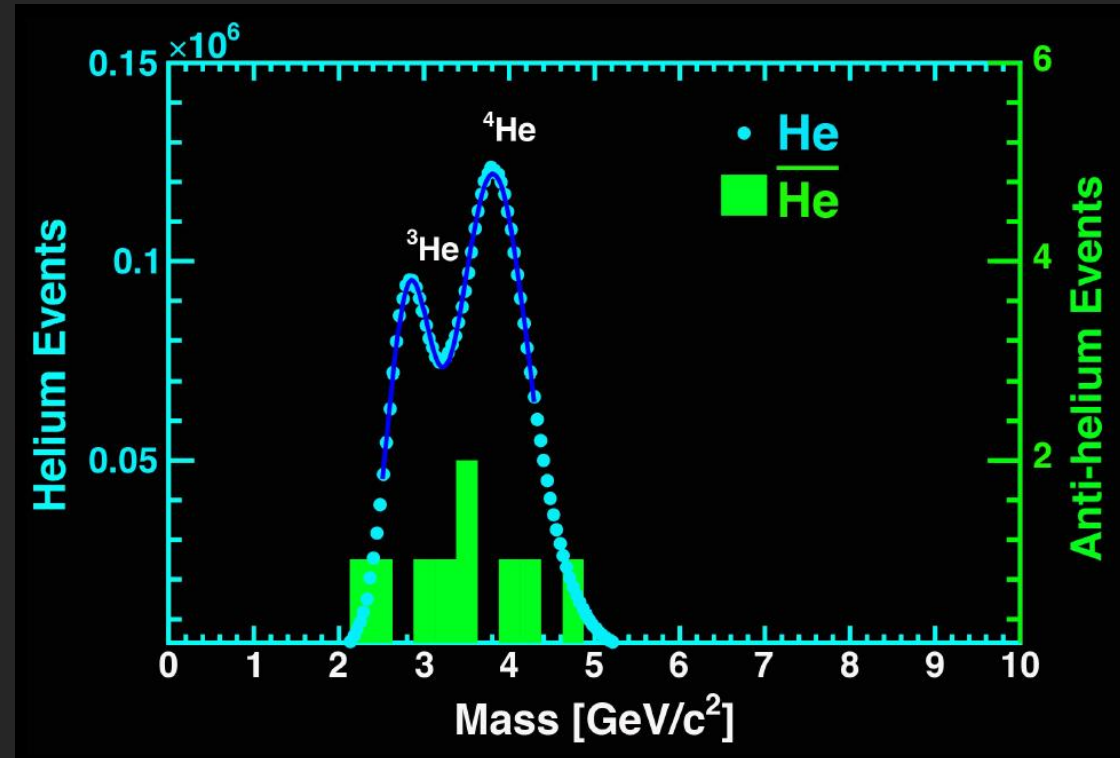
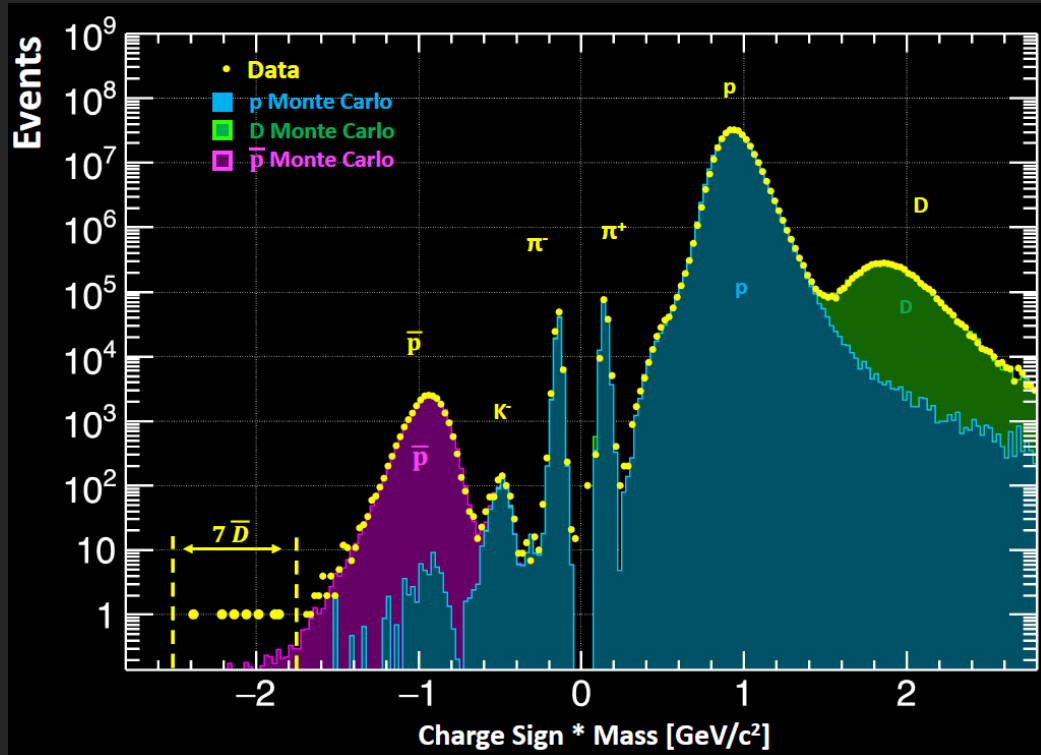


Boudaud et al Phys. Rev. Research 2, 023022

# ANTI-NUCLEI: AMS-02 mass-charge spectra



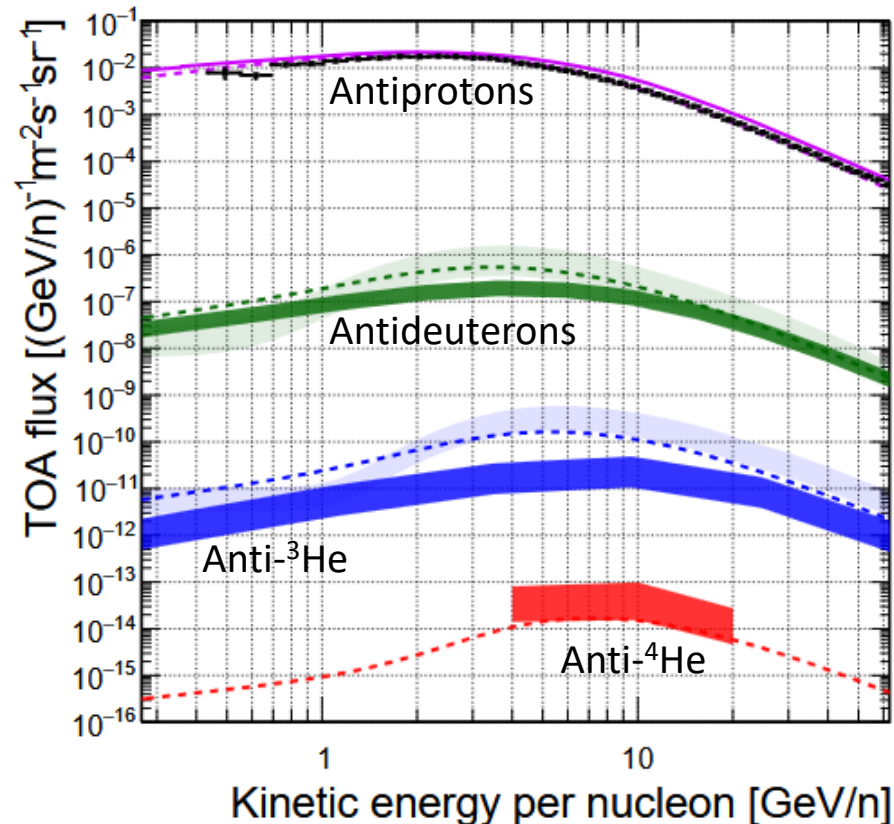
# ANTI-NUCLEI: AMS-02 mass-charge spectra



Paolo Zuccon MIAPP 2021

# Anti-nuclei as a smoking gun for new physics

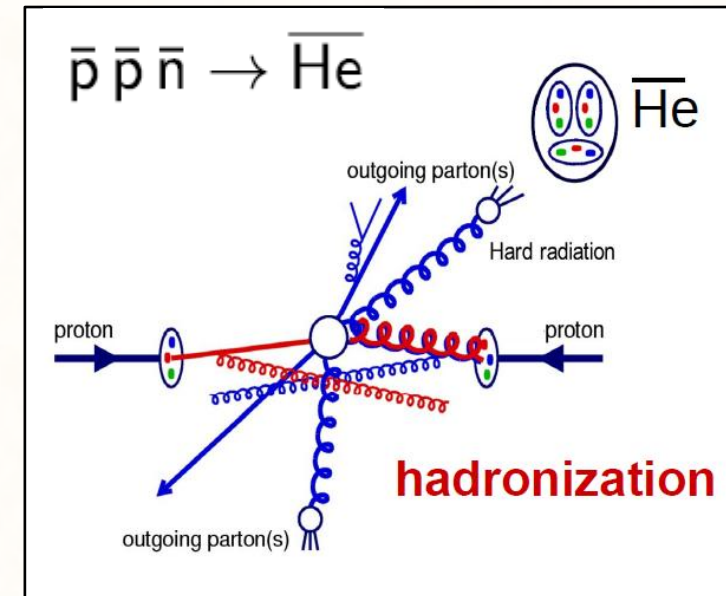
The window to prove (or disprove) many possible astrophysical excesses



Shukla *PRD* 102, 063004 (2020)

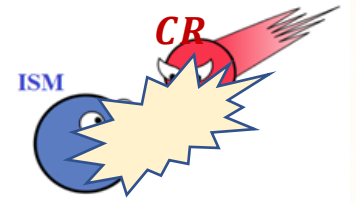
Secondary anti-nuclei produced from homologous interactions as for  $\bar{p}$ , but highly suppressed (due to coalescence)!

See Di Mauro+ ArXiv:2411.04815



# Astrophysical production

$$\text{CR} + \text{ISM} \rightarrow \bar{\text{He}}, \bar{d}$$

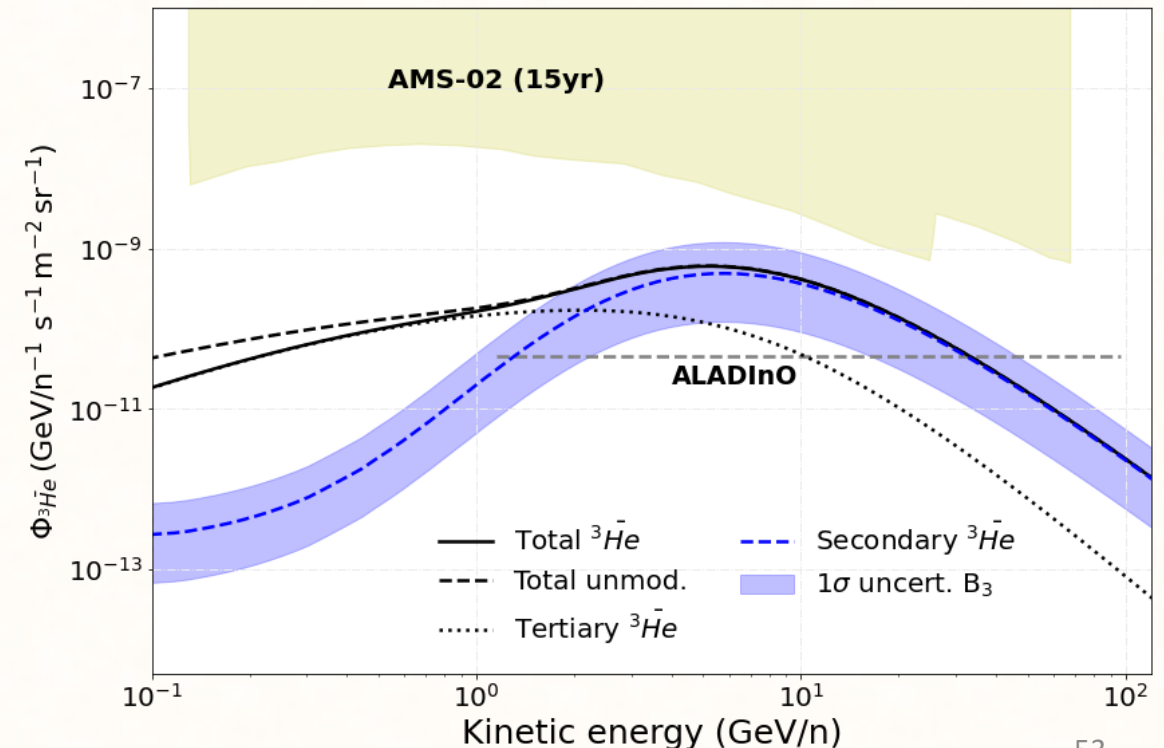
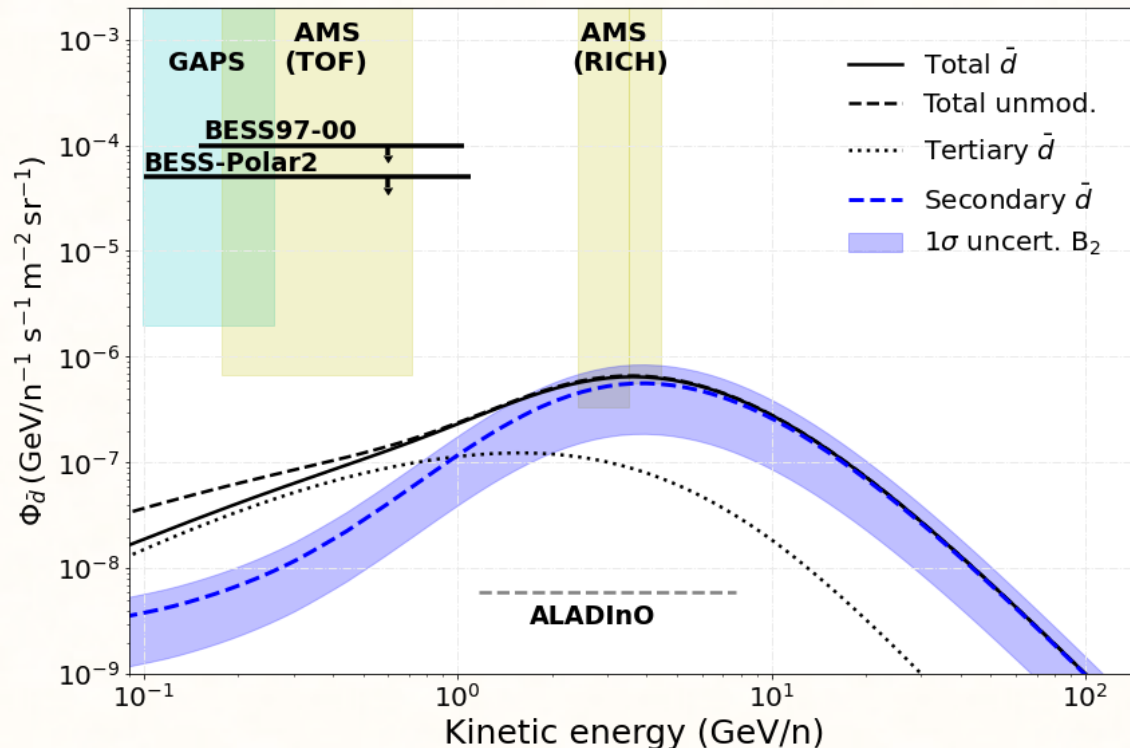


Can we explain the AMS-02 measurements without invoking any exotic source?

We expect to have measurements of the antideuteron flux in the next years!!

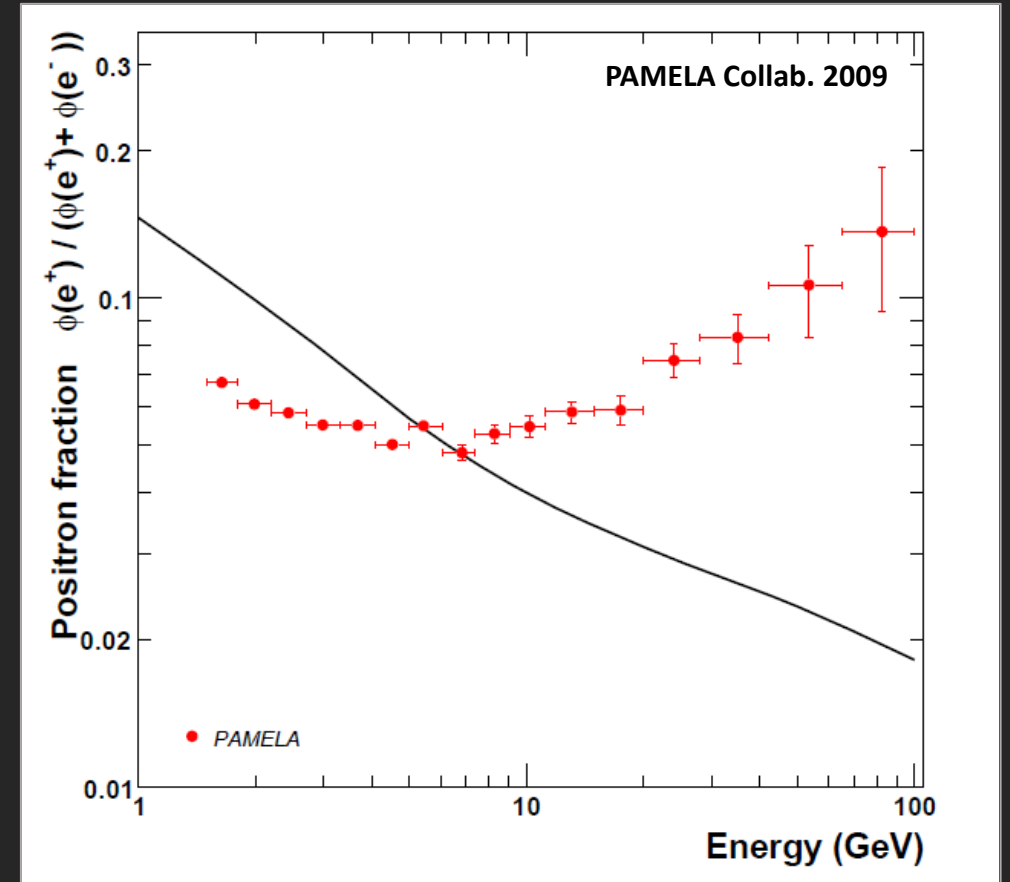
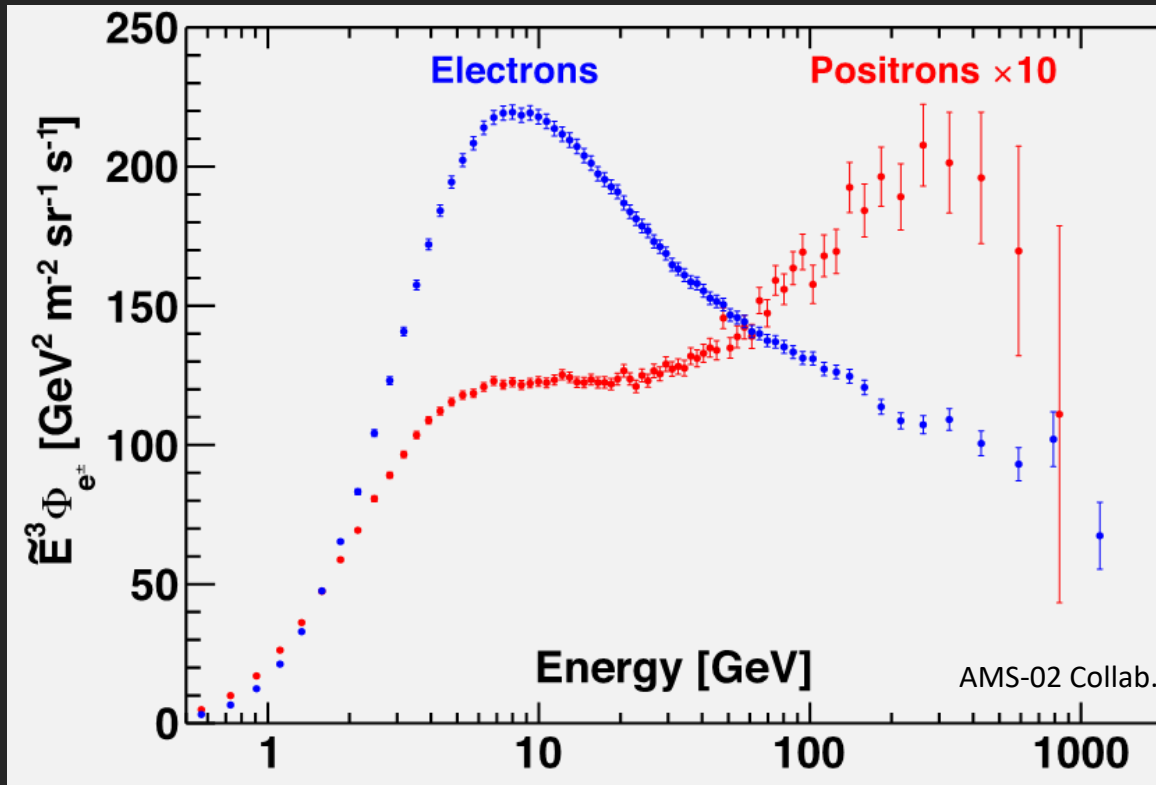
But no antihelium till future experiments: ALADInO or AMS-100 (foreseen to 2039)

PDL et al JCAP 10 (2024) 017





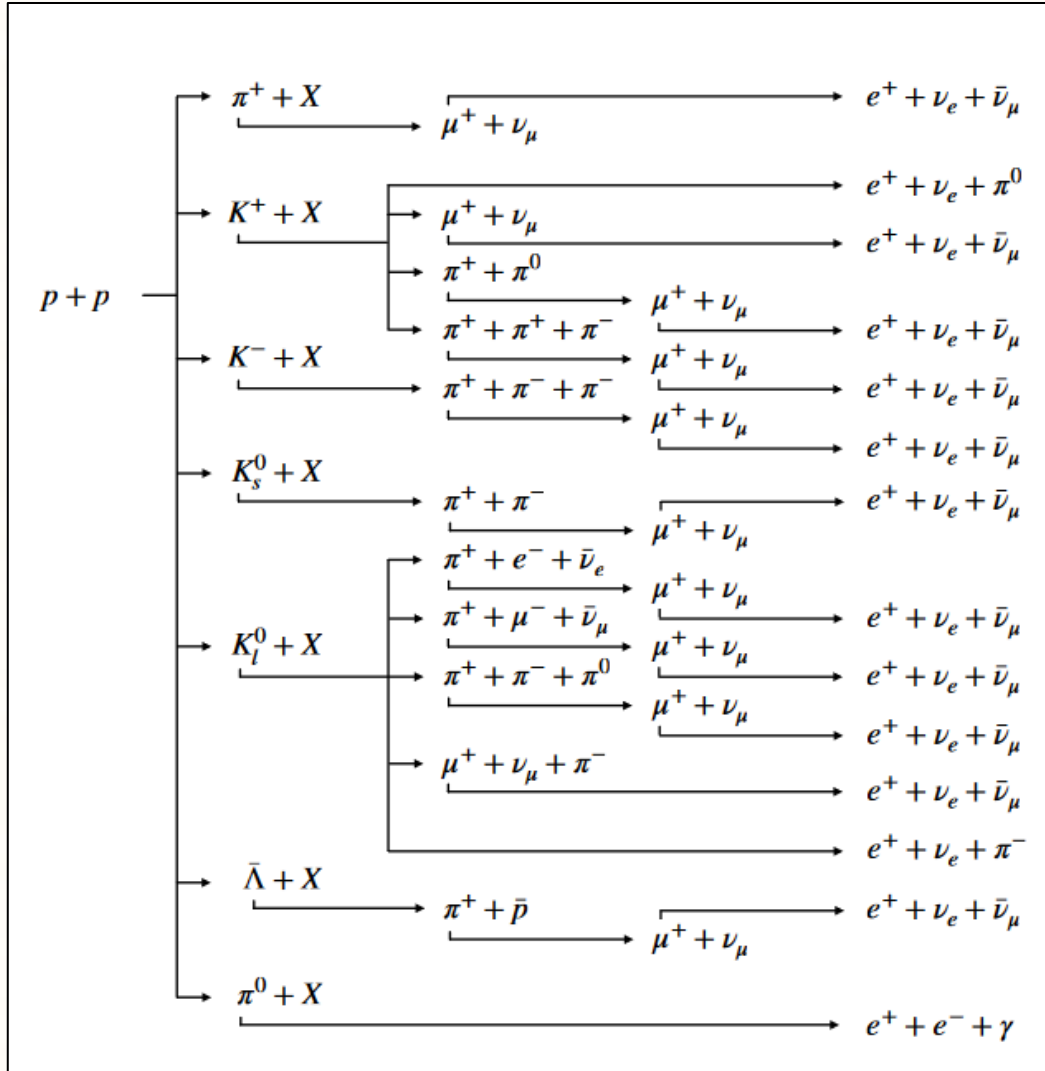
# The positron spectrum



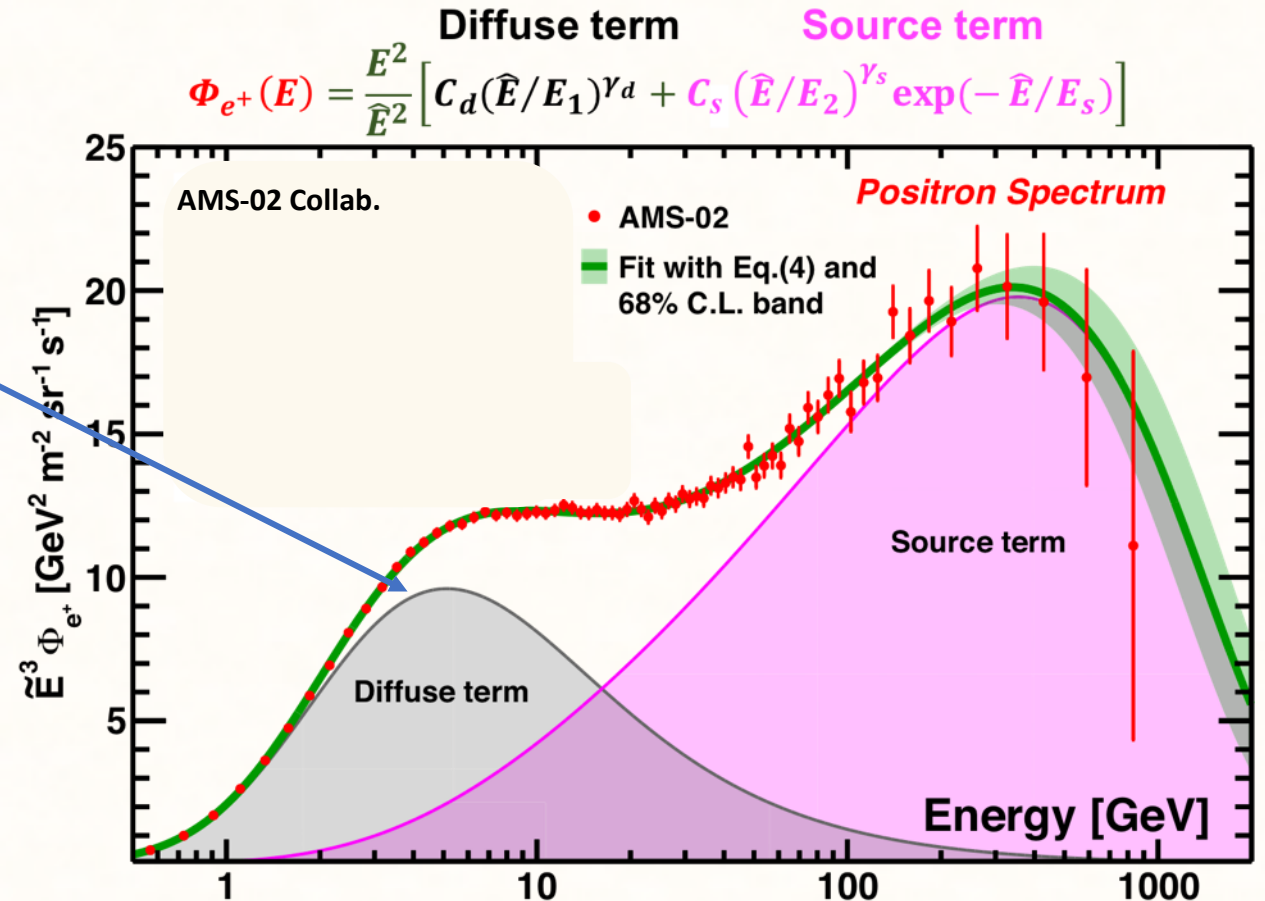
Unexpected and clear rise in the detected flux of positrons requires a new source of positrons. **What is this source of  $e^+$ ?**

# The positron spectrum

Orusa+ PRD 105 (2022) 12



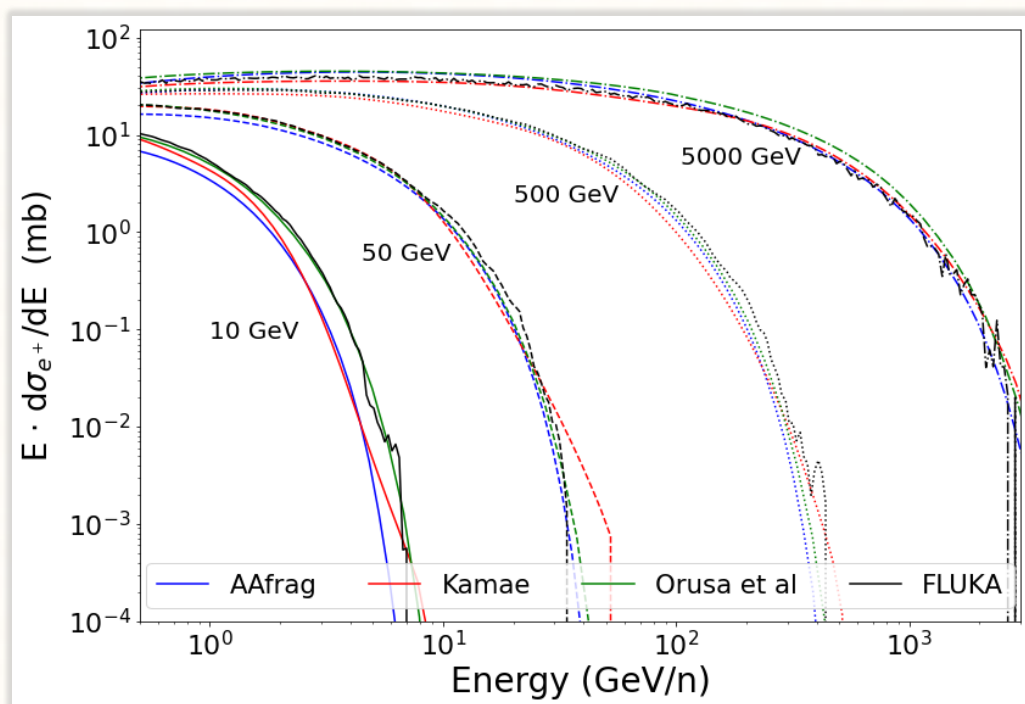
Positrons mainly produced from  $p+p$  interactions, but also, He and heavier CRs are involved and produce positrons!



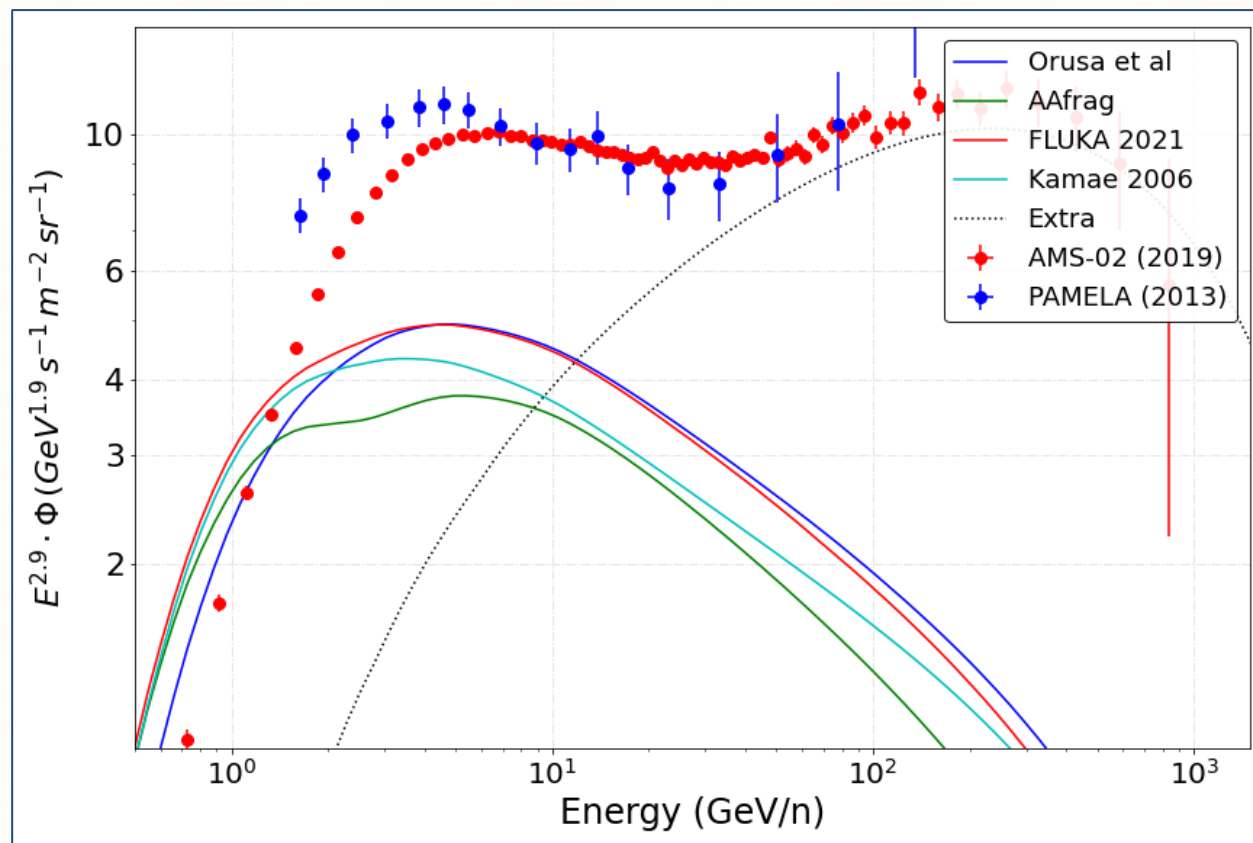
# The positron spectrum

Main cross sections data-sets available:

- Kamae 2006 APJ 647:692–708 (2006)
- AAfrag 2021 PRD 104, 123027 (2021)
- Orusa et al. PRD 105 (2022) 12, 123021
- Fluka 2021 JCAP10(2023)011



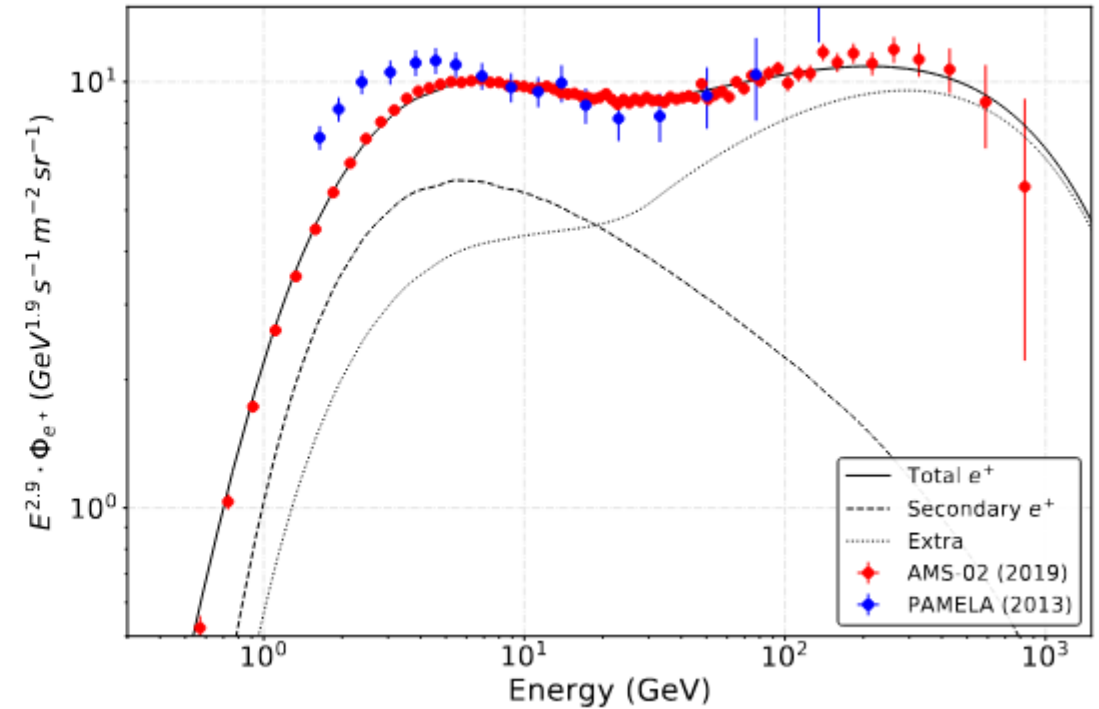
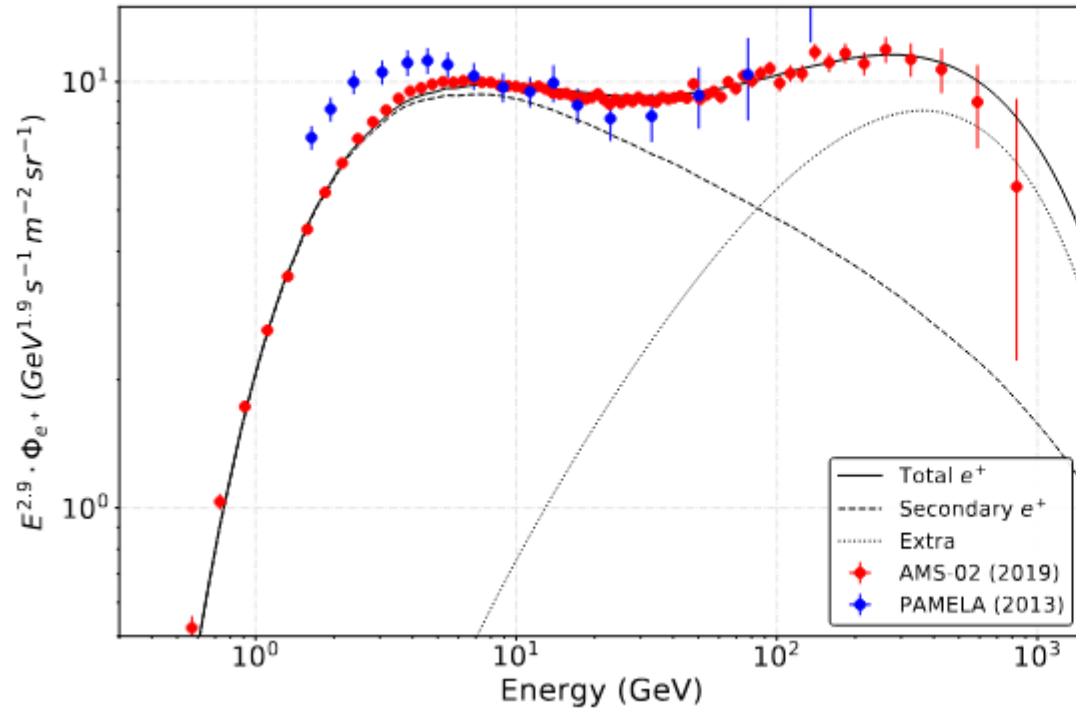
Different cross sections lead to a 25%-30% difference in the predicted flux below 30 GeV (depending on the exact diffusion parameters).



PDL+ ArXiv:2305.02958

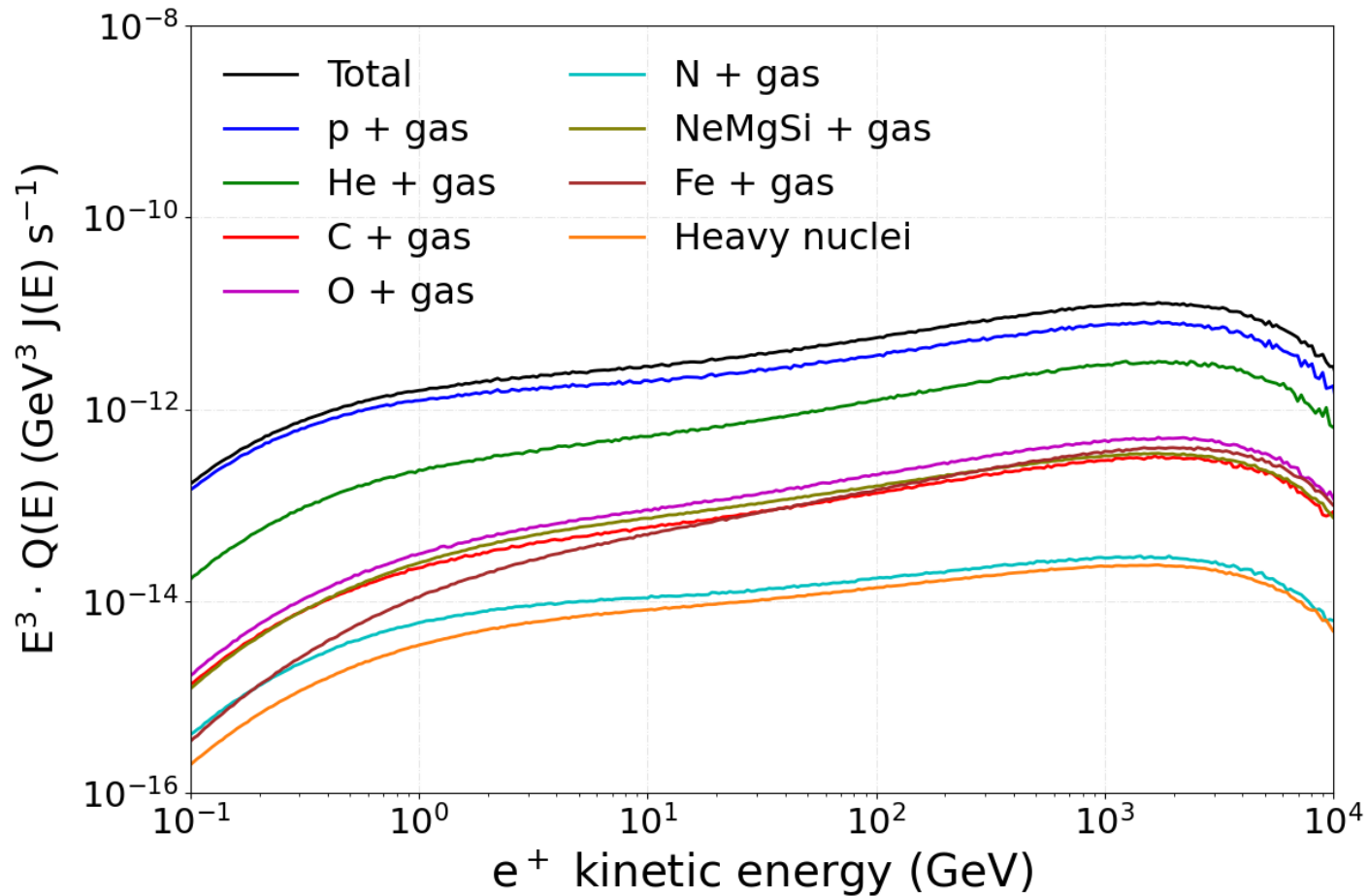
# The positron spectrum

Understanding the pulsar contribution is key to understand secondary positrons



# $e^+$ cross sections uncertainties

*Contribution of each channel to the total positron spectrum*



The contribution of elements with high mass number is as important as that from C or O above  $\sim 10$  GeV

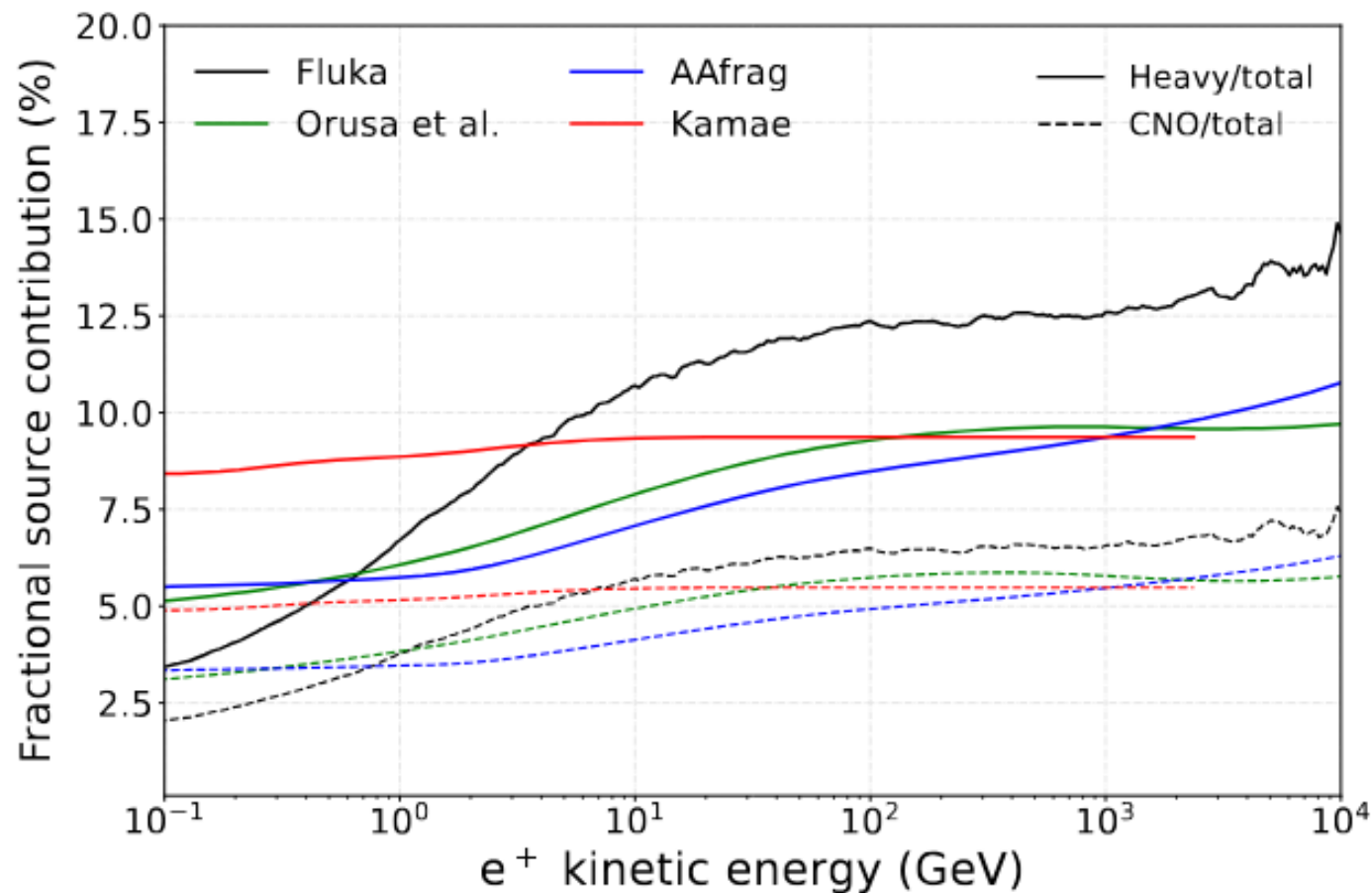
Contribution of heavy nuclei (above He and up to  $^{56}\text{Fe}$ ) constitutes between 7.5% and 10% of the total  $e^+$  flux at 10 GeV  $\rightarrow$  Overestimation due to the lack of data on sub-Fe elements

PDL+ ArXiv:2305.02958



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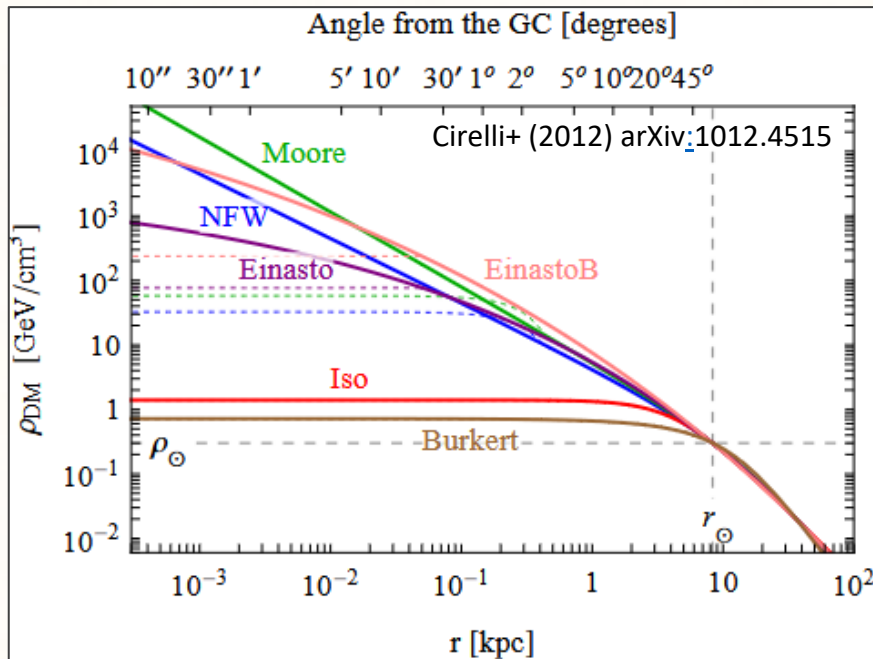
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PDL+ ArXiv:2305.02958

# Dark matter in the Galaxy

Extensive evidence for the existence of a “substance” that determines, through its gravitational pull, the dynamics and structure of Galaxies and larger objects: **Dark matter**

Expected to be concentrated around the Galactic Center and to interact with conventional matter very feebly



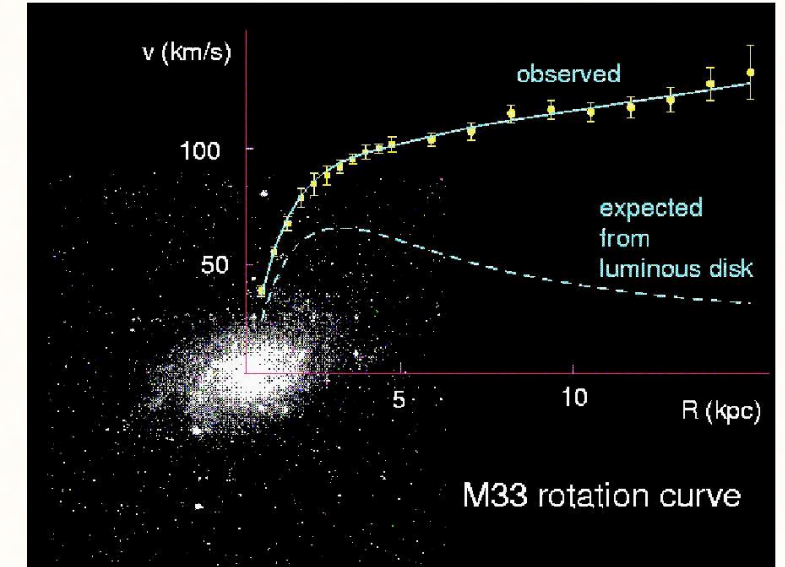
What's dark matter?

**A new particle?** (best motivated is the WIMP)

**Black holes?**

**MOND?**

Many other possibilities...



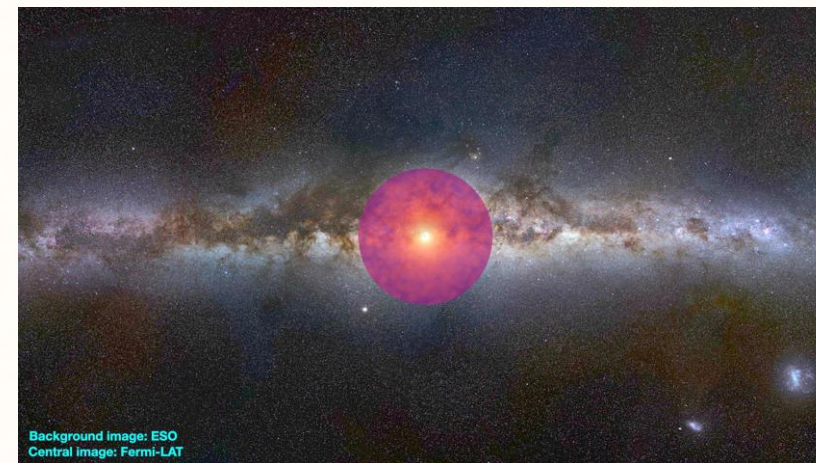
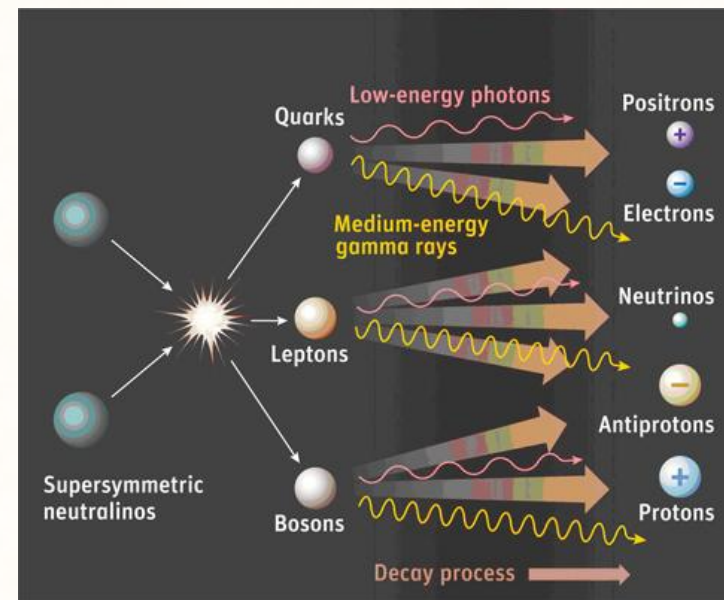
# Anti-particles from Dark matter

**WIMPs** must produce same number of particles and antiparticles in their decay/annihilation → Another potential source of antimatter

**Observing the direct products of dark matter would reveal its nature and origin**

Positrons and antiprotons **have long been considered critical channels for dark matter searches**: Many simple extensions of the SM lead to well-motivated dark matter candidates

$$Q_{\tilde{N}}^{\text{DM,ann}}(\vec{r}, T) = \frac{1}{2} \left( \frac{\rho(\vec{r})}{m_{\text{DM}}} \right)^2 \langle \sigma v \rangle_f \times \frac{dN_f^{\tilde{N}}}{dT}$$





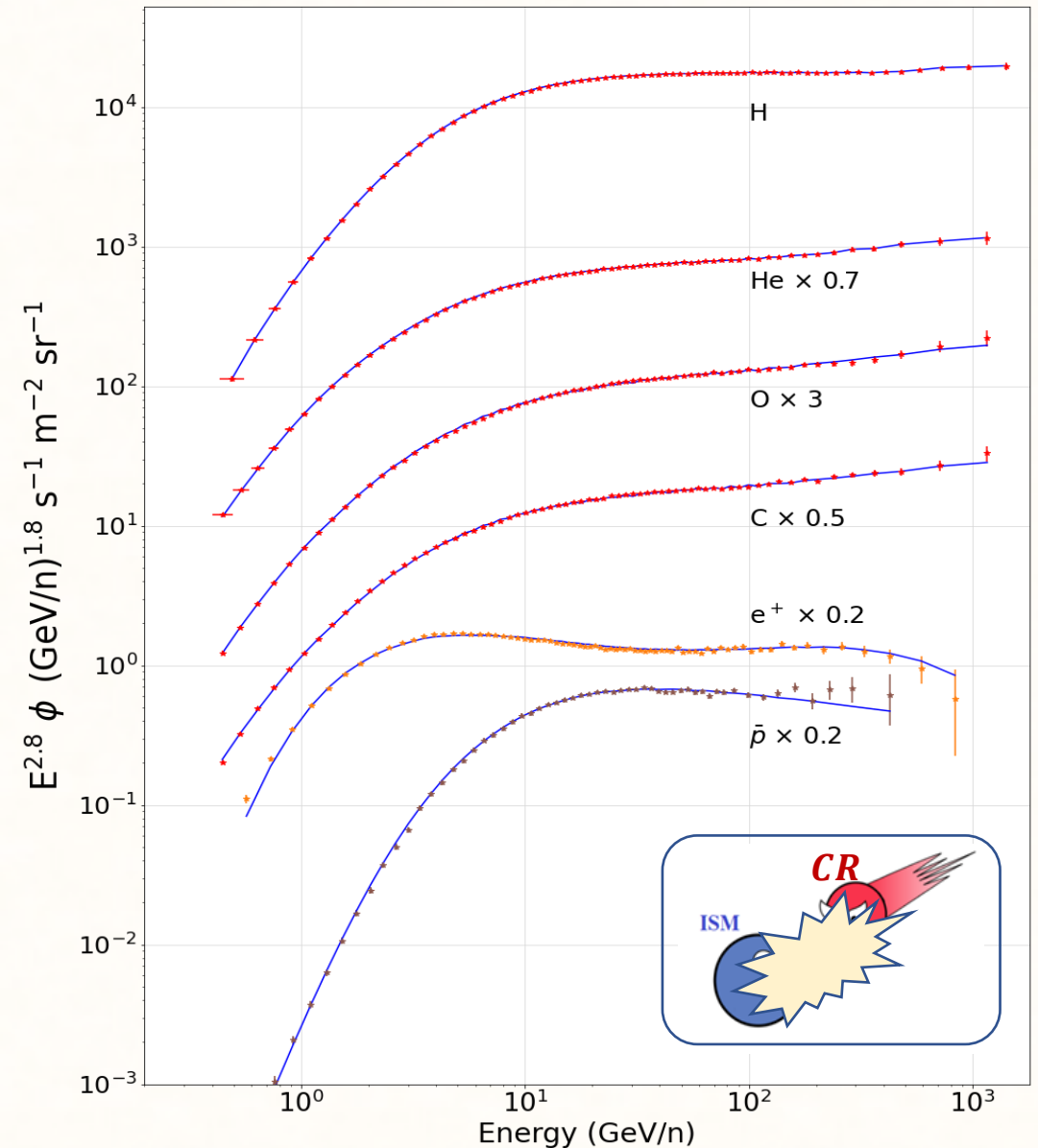
# Potential of antiparticles to reveal the existence of BSM physics

High precision data for the fluxes of CR nuclei allow us to accurately model the production of CR antiparticles and uncertainties related.

Antiproton and positron observations provide strong constraints on the existence of BSM physics due to the expected low production and uncertainty in their modelling.

Specially, well-motivated **WIMPs** ( $M_\chi \sim O(100 \text{ GeV})$ ) are expected to leave imprints in the GeV energy region.

Flux of CR nuclei and antiparticles (data from AMS-02)



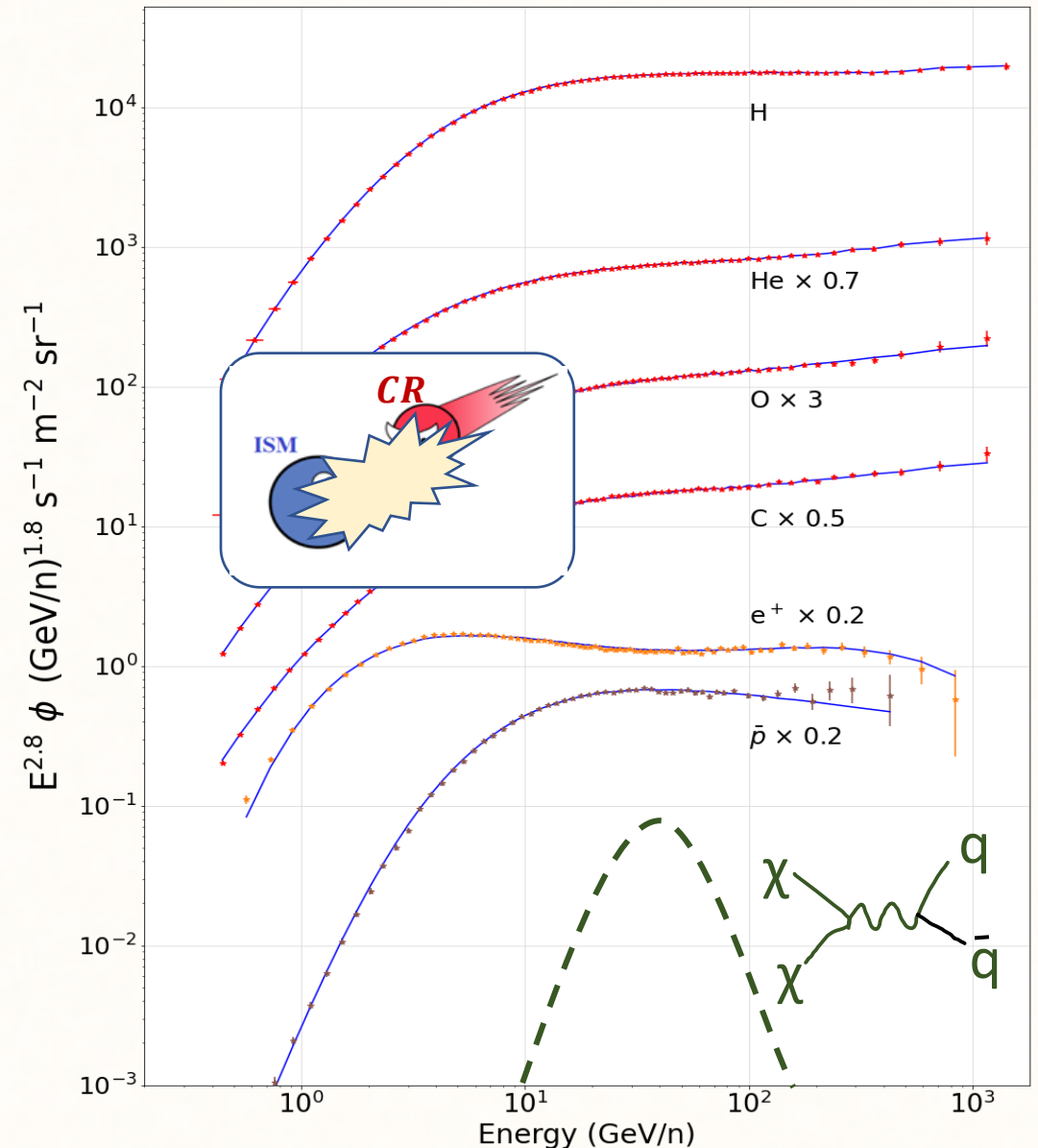
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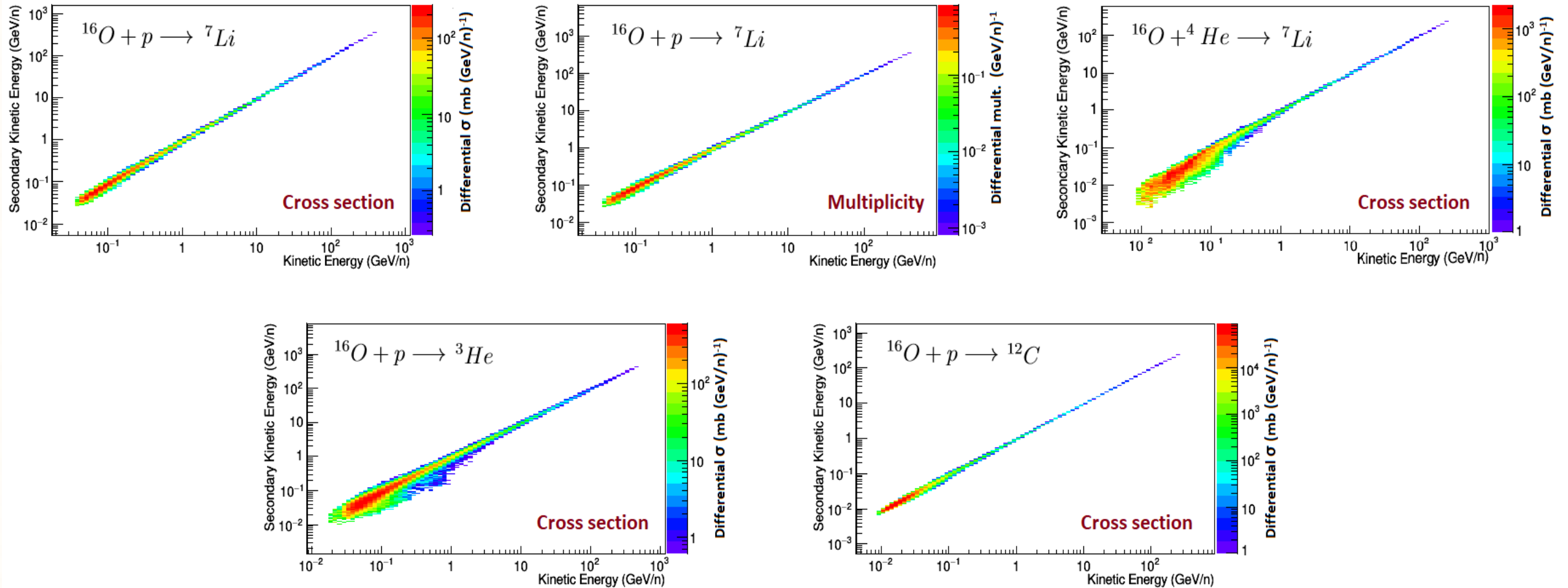


# TO SUMMARIZE...

- Secondary cosmic rays constitute the most precise tool to determine the diffusion coefficient
- Unstable isotopes like  $^{10}\text{Be}$  are necessary to constrain the volume where cosmic rays propagate
- Cosmic-ray interactions with the interstellar medium are **directly seen with gamma-ray telescopes**. These interactions allow us to understand better the Galactic environment
- **Antiparticles** are key to look for features of non-standard physics in astrophysical observations and represent an opportunity to **understand the nature of dark matter**

# BACK UP

# How valid is the head-on approximation below Li?



# Same story with neutrinos

See Orusa+ ArXiv:2509.16303

